

The Upside of Being Replaceable: AI, Market Structure, and Growth

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Abstract

I develop an endogenous-growth model in which a monopolistic AI developer supplies AI services and invests in recursive AI self-improvement. I characterize two long-run regimes under an upper bound on AI efficiency—the quantity of AI services produced per unit of compute. Complementarity between AI and labor, or a sufficiently low upper bound on AI efficiency, leaves labor as a production bottleneck. In this regime, output per worker and wages eventually grow at the exogenous rate of labor-augmenting technological progress. Instead, when AI and labor are substitutes and the upper bound is sufficiently high, an AI-dominated regime sustains permanently faster wage growth and a higher long-run return to capital. Output per worker grows faster than wages, so labor’s income share vanishes while AI-industry revenue increases as a share of output. Replacing monopoly with perfect competition eliminates private research incentives. Nevertheless, if AI is already sufficiently efficient, the competitive economy can sustain AI-dominated growth through capital accumulation and expanding AI use. By contrast, monopoly rewards research but restricts AI use through its markup. It may therefore produce more efficient AI yet slower long-run growth than competition. Numerical simulations show that early output-per-worker growth can remain moderate and similar across economies approaching different long-run regimes.

Keywords: Artificial Intelligence; Recursive AI self-improvement; Market Structure; Labor Share.

JEL codes: O41, O33, O31, D42, E25.

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1 Introduction

Advances in artificial intelligence raise both hopes and fears. One hope is that increasingly capable AI could generate unprecedented economic prosperity, with workers benefiting from higher wages. One fear is that the AI industry could capture a growing share of that prosperity while workers who depend on labor income receive a declining share. In this paper, I show how both outcomes can coexist: when AI and labor are substitutes, AI can permanently raise wage growth. Yet in the long run, output per worker grows even faster than wages, while AI-industry revenue grows faster than total labor income. Labor’s income share therefore converges to zero.

To show this, I extend the Ramsey–Cass–Koopmans model to include AI services as an additional production input. Competitive firms produce the final good using capital, effective labor, and AI services. A monopolistic AI developer supplies the AI services and invests in recursive AI self-improvement (RSI) to improve AI efficiency—the quantity of AI services produced per unit of compute. Through RSI, more efficient AI helps develop still more efficient AI. A representative household chooses consumption and saving, receiving income from labor, capital ownership, and the AI developer’s profits.

If labor and AI are substitutes and there are sufficiently strong research returns, the developer’s profit can become unbounded if AI efficiency can grow without bound. To rule this out, I first study an economy with an upper bound on AI efficiency and examine how equilibria change as the bound rises. Together, the elasticity of substitution between AI and labor and the upper bound on AI efficiency determine whether human labor remains a long-run production bottleneck or whether AI sustains permanently faster economic growth.

In the equilibria where labor remains a long-run production bottleneck, output per worker and wages eventually grow at the exogenous rate of labor-augmenting technological progress, and labor’s income share converges to a positive value. This is the result when there is complementarity between labor and AI production services or when the upper bound on AI efficiency is sufficiently low.

An AI-dominated long-run regime becomes possible when AI and labor are substitutes and the upper bound on AI efficiency is sufficiently high. As in [Jones and Manuelli \(1990\)](#), capital accumulation can sustain growth when its limiting return remains sufficiently high. In this regime, capital and AI production services eventually grow faster than effective labor. This does not require AI efficiency to keep improving: capital accumulation and expanding compute use sustain growth even as AI efficiency approaches its upper bound. The long-run growth rates of output per worker and wages, as well as the interest rate, exceed their labor-bottleneck values. Because output per worker grows faster than wages, labor’s income share converges to zero. Total labor income therefore grows faster than in the labor-bottleneck regime, but in the long run it grows more slowly than capital income and AI-industry revenue. Greater substitutability raises long-run output-per-worker growth, but long-run wage growth peaks at a finite elasticity of substitution.

I also characterize a competitive economy in which AI models and any improvements are freely available to all suppliers of AI production services. Without a private return to research, AI

efficiency remains at its initial level. When AI and labor are substitutes, a sufficiently high initial efficiency nevertheless sustains AI-dominated growth through capital accumulation and expanding compute use. Neither monopoly nor continuing RSI is therefore necessary to sustain this growth regime.

The growth comparison between monopoly and competition depends on whether monopoly-financed efficiency gains compensate for the reduction in AI use caused by monopoly pricing. A monopoly economy may reach AI-dominated growth from an initial efficiency too low to sustain it under competition. Conversely, competition can sustain AI-dominated growth from initial states for which monopoly-led research improves efficiency but leaves the economy labor-bottlenecked. Even when both economies approach AI-dominated growth, monopoly delivers faster long-run growth only if the research gains are large enough to offset its markup.

I also examine the effect of removing the upper bound on AI efficiency. With complementarity, long-run average output-per-worker growth cannot exceed the exogenous rate of labor-augmenting technological progress. At unit elasticity, I derive conditions for a balanced-growth equilibrium in which output per worker grows faster than this exogenous rate. Wages grow at the same rate as output per worker, preserving a constant labor income share. With substitution and sufficiently strong research returns, the developer can make discounted net profit arbitrarily large even over a fixed finite horizon, ruling out the existence of an equilibrium.

I use numerical simulations to illustrate the effects of introducing RSI under monopoly and replacing competition with monopoly. First, I simulate an unanticipated activation of RSI in economies that already use AI services under a monopoly. Depending on research productivity, output-per-worker growth can display a moderate increase or a noticeable jump in both long-run regimes. Eventually, output-per-worker growth converges to the rate of labor-augmenting technological progress in the labor-bottleneck regime and to a permanently higher rate in the AI-dominated regime. With lower research productivity, initial output-growth trajectories can remain similar across economies approaching different long-run regimes. Second, I simulate granting exclusive AI rights in economies that initially supply AI services competitively. Monopoly-financed research can move an economy from labor-bottleneck to AI-dominated growth. Conversely, monopoly pricing can leave an economy labor-bottlenecked even when competition would sustain AI-dominated growth with less efficient AI.

This paper has eight sections, including the present introduction. Section 2 relates the paper to the literature. Section 3 presents the model and defines the equilibrium. Section 4 establishes sufficient conditions for equilibrium existence with an upper bound on AI efficiency and characterizes the associated long-run growth and distributional outcomes. Section 5 characterizes competitive AI provision and compares its growth outcomes with those under monopoly. Section 6 examines the three substitution regimes without an upper bound on AI efficiency. Section 7 presents numerical simulations. Section 8 concludes. The appendix contains the proofs and describes the algorithm used for the simulations.

2 Related literature

My model shares the feedback between AI development and economic growth studied by [Davidson et al. \(2026\)](#), [Jones and Tonetti \(2026\)](#), and [Erdil et al. \(2025\)](#): better AI raises output, expanding the resources available for further AI development. My model also allows better AI to improve research itself through RSI, a technological feedback related to the automation of research in [Davidson et al. \(2026\)](#) and [Jones and Tonetti \(2026\)](#). The models differ in how saving and research expenditure are determined. [Davidson et al. \(2026\)](#) and [Jones and Tonetti \(2026\)](#) take saving and research allocation shares as given in their growth models. [Erdil et al. \(2025\)](#) instead let a social planner choose investment and allocate compute between developing and deploying AI. I instead study a market equilibrium.

I examine when AI can sustain faster growth rather than raise it only temporarily. [Aghion et al. \(2019\)](#) and [Jones and Tonetti \(2026\)](#) show how tasks that remain dependent on human labor can constrain growth despite automation elsewhere. [Palivos and Karagiannis \(2010\)](#) show how sufficiently high substitutability between capital and labor can allow capital accumulation to sustain growth without technological progress. [Restrepo \(2025\)](#) obtains sustained output growth from an exogenous expansion of compute after bottleneck tasks have been automated. In my model, RSI can improve AI efficiency without eliminating this labor bottleneck. I also identify conditions under which capital and AI services expand together and sustain permanently faster growth without further AI-efficiency gains.

I also study how AI-driven growth affects wages and labor’s share of income. [Acemoglu and Restrepo \(2018\)](#) show that capital accumulation can raise long-run wages even as automation displaces labor from existing tasks and reduces labor’s income share. Similarly, [Berg et al. \(2018\)](#) show that, when robots and labor are imperfect substitutes, higher robot productivity and capital accumulation can raise steady-state wages while lowering labor’s share. These comparative-static results concern wage levels, whereas I examine when AI can permanently increase wage growth. Like [Ray and Mookherjee \(2022\)](#), I show that wages can rise without bound even as labor’s income share vanishes. [Hémous and Olsen \(2022\)](#) show that low-skill wages grow more slowly than output in the long run, while high-skill wages keep pace with it. In this paper, I abstract from labor heterogeneity.

I also study how market structure affects AI development and growth. [Korinek and Ng \(2019\)](#) examine how digital innovation lowers production costs while generating monopoly rents and markups that restrict output. Holding AI capability fixed, [Gans \(2026\)](#) shows how competition and monopoly can affect long-run growth by changing the choice between incremental and exploratory research. [Liu and Wan \(2026\)](#) show how the prices charged by general-purpose AI suppliers affect incentives to improve both their models and the specialized applications that use them, shaping long-run growth. I compare monopoly and competition to examine whether monopoly-led research raises AI efficiency enough to compensate for the reduction in AI use caused by monopoly pricing.

3 Model

Time is continuous. The economy contains a representative household, competitive final-good firms, and an integrated AI developer. Labor-augmenting technology A grows exogenously. AI efficiency B improves through autonomous AI research: the only current input used to raise B is research compute. The model therefore describes an economy in which there are no longer human researchers developing AI and AI has achieved recursive self-improvement (RSI). As in [Trammell and Korinek \(2023\)](#), I distinguish automation of research from automation of production: human labor remains an input in final-good production.

3.1 Population and labor-augmenting technology

Population grows at the exogenous rate $n \geq 0$:

$$\dot{N} = nN, \quad n \geq 0. \quad (1)$$

Each person supplies one unit of homogeneous labor inelastically. The aggregate labor endowment is therefore N . Because autonomous research uses no human labor, all labor is employed in final production and equilibrium will impose $L = N$.

The variable A measures the level of labor-augmenting technology. One unit of labor supplies A efficiency units, so effective labor is AL . Labor-augmenting technology evolves according to

$$\dot{A} = \gamma A, \quad \gamma > 0. \quad (2)$$

3.2 Household

The representative household owns physical capital and the AI developer. Let Π denote the AI developer's net profit, which accrues to the household. The household takes the paths of the net interest rate r , the wage w in final-good units and Π as given. For a predetermined capital stock $K_0 > 0$, it chooses aggregate consumption C to solve

$$\max_{\{C_t\}_{t \geq 0}} \int_0^\infty e^{-\rho t} N_t \log(C_t/N_t) dt, \quad \rho > n, \quad (3)$$

subject to

$$\dot{K} = rK + wN + \Pi - C, \quad K \geq 0. \quad (4)$$

The variable r is the net real interest rate: depreciation has already been deducted from the rental return to capital. The household's Euler condition is

$$\frac{\dot{C}}{C} = n + r - \rho. \quad (5)$$

Its transversality condition is

$$\lim_{t \rightarrow \infty} e^{-\rho t} \frac{N_t K_t}{C_t} = 0. \quad (6)$$

3.3 Final-good production

Competitive firms produce the final good from physical capital K , effective labor AL , and effective AI production services X . Let $0 < \alpha < 1$ denote the capital share, $\delta \geq 0$ the depreciation rate, and $\sigma > 0$ the elasticity of substitution between effective labor and effective AI production services. The parameters ω_L and ω_X denote the CES weights on effective labor and effective AI production services, respectively. Both weights are positive and sum to one. Production is

$$Y = K^\alpha Z^{1-\alpha}, \quad (7)$$

$$Z = \left[\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma} \right]^{\sigma/(\sigma-1)}. \quad (8)$$

The composite input Z combines effective labor and AI production services.

I use the CES specification in equation (8) to represent how labor and AI services combine, either within a single task or as aggregate inputs across multiple tasks. The latter interpretation connects to [Jones and Tonetti \(2026\)](#), who show how a task-based production process can admit a CES-like aggregate representation.

Let p_X denote the price of one unit of effective AI production services. Taking the input prices $r + \delta$, w , and p_X as given, a final-good firm solves

$$\max_{K, L, X \geq 0} \left\{ K^\alpha Z(AL, X)^{1-\alpha} - (r + \delta)K - wL - p_X X \right\}. \quad (9)$$

Define s_X as the elasticity of Z with respect to X :

$$s_X = \frac{\omega_X X^{(\sigma-1)/\sigma}}{\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma}}. \quad (10)$$

The first-order conditions are

$$r = \alpha \frac{Y}{K} - \delta, \quad (11)$$

$$w = (1 - \alpha)(1 - s_X) \frac{Y}{L}, \quad (12)$$

$$p_X = (1 - \alpha)s_X \frac{Y}{X}. \quad (13)$$

Thus $(1 - \alpha)s_X$ is AI revenue as a share of output, while $(1 - \alpha)(1 - s_X)$ is labor's income share.

Constant returns and these conditions imply

$$Y = (r + \delta)K + wL + p_X X. \quad (14)$$

Solving the final-good firm's AI condition for its price yields the inverse demand schedule $p_X(X)$. The absolute elasticity of inverse demand, holding K , A , and L fixed, is

$$e_X \equiv -\frac{\partial \ln p_X(X)}{\partial \ln X} = \frac{1 - s_X}{\sigma} + \alpha s_X. \quad (15)$$

This demand schedule is an input to the developer's problem.

3.4 The AI developer

3.4.1 Effective AI production and research services

I use *compute* to denote the computational resources used by AI systems (Sastry et al., 2024). Compute can be measured in floating-point operations, with flows measured in operations per unit of time. I assume that each unit of compute requires one unit of the final good, which could otherwise be consumed or invested in physical capital.

AI efficiency B measures the services obtained per unit of compute. A rise in B represents algorithmic improvement rather than an expansion of physical computing capacity. This interpretation parallels measures of algorithmic efficiency based on the compute required to achieve a given level of performance (Hernandez and Brown, 2020; Erdil and Besiroglu, 2022).

Inference compute U supplies AI services for current final-good production, while research compute M is used to develop better AI. Korinek and McKelvey (2026) make a related distinction between compute used for inference and compute used for R&D and training. Research compute includes both training and running AI systems that conduct research (Davidson et al., 2026). I assume that the same B augments both activities, producing flows BU and BM of AI production and research services, respectively. AI production services therefore satisfy

$$X = BU. \quad (16)$$

Figure 1 summarizes the production block.

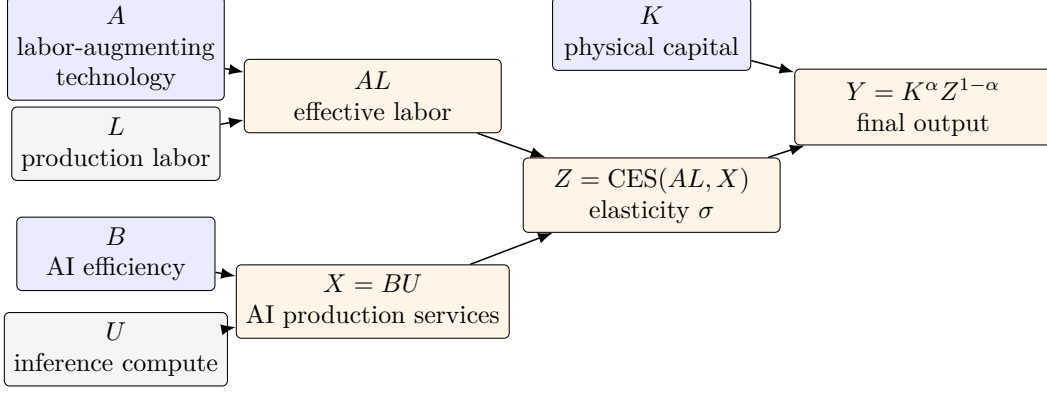


Figure 1: Production of final output and AI production services. AI efficiency B augments inference compute U to produce $X = BU$, the flow of AI production services. These services combine with effective labor AL to produce Z , which combines with capital K to produce Y .

3.4.2 RSI technology

Given initial AI efficiency $B_0 > 0$ and an upper bound $\bar{B} \in (B_0, \infty]$, AI efficiency evolves according to the RSI technology

$$\dot{B} = \chi(BM)^\eta \psi(B), \quad \psi(B) = 1 - \frac{B}{\bar{B}}, \quad \chi > 0, \quad 0 < \eta < 1. \quad (17)$$

Figure 2 summarizes the RSI mechanism.

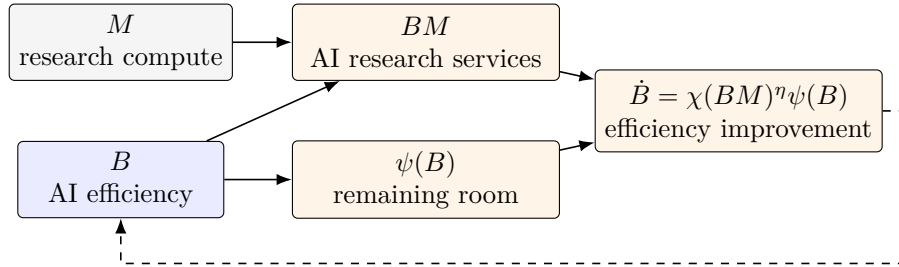


Figure 2: Recursive self-improvement in AI efficiency. AI efficiency B augments research compute M to produce BM , the flow of AI research services. These services and the remaining room for improvement $\psi(B)$ determine $\dot{B}$, which increases B .

The parameter χ shifts research productivity, while η is the elasticity of $\dot{B}$ with respect to research compute, holding B fixed. Equation (17) captures RSI: a higher B produces more AI research services BM from a given amount of research compute M , and these services help improve B further. This is the AI-research feedback studied by Davidson et al. (2026). The term $\psi(B)$ reduces the remaining room for improvement as AI efficiency approaches $\bar{B}$. A related approach appears in Erdil et al. (2025), who also reduce research productivity as hardware and software efficiency approach their respective upper bounds. The uncapped specification is recovered at $\bar{B} = \infty$.

Evidence that ideas become harder to find motivates allowing research productivity to decline as technology advances. Bloom et al. (2020) document that research effort has risen substantially while research productivity has fallen across several industries, products, and firms. This evidence motivates diminishing returns, but it does not establish a literal finite technological ceiling. The term $1 - B/\bar{B}$ is a deliberately stronger reduced-form specification. Without the upper bound, recursive improvement may lead to the absence of an equilibrium, as shown in Section 6.3.

3.4.3 Supply of effective AI production services

An integrated AI developer has exclusive ownership of B . It conducts research and sells effective AI production services. The developer sells X and purchases the $U = X/B$ units of inference compute needed to supply those services. Because X does not enter the RSI technology in equation (17) directly, its within-date choice can be solved conditional on B . Define optimized operating profit before research expenditure as

$$\pi(B) = \max_{X \geq 0} \left\{ p_X(X)X - \frac{X}{B} \right\}. \quad (18)$$

The notation $\pi(B)$ leaves its dependence on current K , A , and L implicit. The first-order condition equates marginal revenue to the compute cost of one additional service unit:

$$p_X(1 - e_X) = \frac{1}{B}. \quad (19)$$

Lemma 1 in Appendix A shows that problem (18) has a unique, strictly positive global maximizer characterized by condition (19). Thus, the developer charges the unit cost $1/B$ multiplied by the monopoly markup $1/(1 - e_X)$. The envelope theorem gives

$$\pi_B = \frac{X}{B^2} = \frac{U}{B}. \quad (20)$$

At a fixed X , a marginal increase in AI efficiency saves U/B units of the consumption good.

AI efficiency is nonrival in use. Once B has been developed, using it to augment one unit of inference or research compute does not reduce its ability to augment other units. This does not make effective AI services costless: BU and BM require the rival and costly compute inputs U and M . Nonrivalry also does not rule out excludability. An owner can restrict access to model weights, software, or AI services.

To see the resulting appropriability problem, suppose that competitive service providers had free access to B . Free entry would imply no operating rent. A developer that paid for M would improve a common AI-efficiency index B that competing providers could use without compensating it. The developer would bear the research cost but could not appropriate the resulting benefits. Exclusive ownership allows the developer to capture part of the value created by a higher B and makes the return to research appropriable, as in the standard endogenous-growth treatment of nonrival ideas (Romer, 1990).

That being said, monopoly is an institutional assumption, not a technological implication of nonrivalry. Licensing, oligopoly, public funding, or open-source provision with research subsidies could also support investment in B (Korinek and Vipra, 2025; Gans, 2024; Demirer et al., 2025). I also abstract from the conflict between general-purpose AI suppliers and application developers studied by Liu and Wan (2026).

3.4.4 The dynamic problem

After the within-date service choice, the developer's net profit is $\Pi_t = \pi(B_t) - M_t$: operating profit less research expenditure. Taking the paths of r, K, A , and L as given and substituting the static solution to problem (18), the developer chooses the path of research compute M to increase AI efficiency B :

$$\max_{\{M_t\}_{t \geq 0}} \int_0^\infty \exp \left\{ - \int_0^t r_s ds \right\} [\pi(B_t) - M_t] dt, \quad (21)$$

subject to the RSI technology given by (17) and taking B_0 as given.

Let q be the current-value shadow price of AI efficiency, measured in units of the current final good per unit of B . An interior choice of research compute M satisfies

$$q\chi\eta B(BM)^{\eta-1}\psi(B) = 1, \quad (22)$$

Equation (22) equates the marginal benefit of research compute to its unit cost. Holding current B fixed, an additional unit of M raises the rate of improvement in AI efficiency by $\chi\eta B(BM)^{\eta-1}\psi(B)$. Multiplication by q values the resulting efficiency gain, accounting for future profit gains and further improvements in AI efficiency.

The current-value costate equation is

$$\frac{\dot{q}}{q} = r - \frac{X}{qB^2} - \eta \frac{\dot{B}}{B} - \dot{B} \frac{\psi'(B)}{\psi(B)}. \quad (23)$$

Rearranging gives the return equation

$$rq = \frac{U}{B} + q\eta \frac{\dot{B}}{B} + q\dot{B} \frac{\psi'(B)}{\psi(B)} + \dot{q}. \quad (24)$$

The first term on the right-hand side of Equation (24) is the inference-compute saving from greater AI efficiency. The second is the research-productivity gain from RSI. With an upper bound, the third term is negative because a higher B leaves less room for further improvement. The final term is the capital gain on AI efficiency. Together these returns must equal the market return rq .

The developer's transversality condition is

$$\lim_{t \rightarrow \infty} \exp \left\{ - \int_0^t r_s ds \right\} q_t B_t = 0. \quad (25)$$

It rules out a path that leaves positive discounted value in the AI-efficiency state.

The preceding conditions are necessary for an interior optimum but do not establish global optimality. Lemma 2 in Appendix A gives sufficient conditions for a trajectory of research expenditure to maximize the present value of the developer's net profit in problem (21).

3.5 Equilibrium

Capital-market clearing identifies household capital with the capital rented by final-good firms. Labor-market clearing gives $L = N$. AI-service market clearing identifies the developer's sales with the final sector's purchases. Combining the household budget, final-good zero profit, developer profit, and market clearing yields the aggregate resource constraint

$$C + \dot{K} + \delta K + U + M = Y. \quad (26)$$

Definition 1 (Equilibrium). *For exogenous initial values $A_0, N_0, K_0, B_0 > 0$, an equilibrium is an infinite-horizon trajectory*

$$\{K_t, B_t, C_t, q_t, L_t, U_t, M_t, X_t, Z_t, Y_t, r_t, w_t, p_{X,t}, s_{X,t}, e_{X,t}, \Pi_t\}_{t \geq 0} \quad (27)$$

such that the developer solves problem (21) and the following conditions hold:

1. *The intratemporal allocation and prices satisfy*

$$L = N, \quad (28a)$$

$$Z = \left[\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma} \right]^{\sigma/(\sigma-1)}, \quad (28b)$$

$$Y = K^\alpha Z^{1-\alpha}, \quad (28c)$$

$$s_X = \frac{\omega_X X^{(\sigma-1)/\sigma}}{\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma}}, \quad (28d)$$

$$e_X = \frac{1 - s_X}{\sigma} + \alpha s_X, \quad (28e)$$

$$r = \alpha \frac{Y}{K} - \delta, \quad (28f)$$

$$w = (1 - \alpha)(1 - s_X) \frac{Y}{L}, \quad (28g)$$

$$p_X = (1 - \alpha) s_X \frac{Y}{X}, \quad (28h)$$

$$\frac{1}{B} = p_X (1 - e_X), \quad (28i)$$

$$X = BU, \quad (28j)$$

$$1 = q_X \eta B (BM)^{\eta-1} \psi(B), \quad (28k)$$

$$\Pi = p_X X - U - M. \quad (28l)$$

2. The four dynamic equations are

$$\dot{K} = Y - C - U - M - \delta K, \quad (29a)$$

$$\dot{B} = \chi(BM)^\eta \psi(B), \quad (29b)$$

$$\frac{\dot{C}}{C} = n + r - \rho, \quad (29c)$$

$$\frac{\dot{q}}{q} = r - \frac{X}{qB^2} - \eta \frac{\dot{B}}{B} - \dot{B} \frac{\psi'(B)}{\psi(B)}. \quad (29d)$$

3. The two transversality conditions are

$$\lim_{t \rightarrow \infty} e^{-\rho t} \frac{N_t K_t}{C_t} = 0, \quad (30a)$$

$$\lim_{t \rightarrow \infty} \exp \left\{ - \int_0^t r_s ds \right\} q_t B_t = 0. \quad (30b)$$

Remark 1 (The economy without AI). With $\omega_X = 0$ and $\omega_L = 1$, the model reduces to the Ramsey–Cass–Koopmans economy with logarithmic utility and exogenous labor-augmenting technological progress.

The system contains no separate household budget or final-good zero-profit equation because these are redundant after market clearing. The consolidated income decomposition follows from the equations in the definition:

$$1 = \frac{(r + \delta)K}{Y} + \frac{wL}{Y} + \frac{\Pi}{Y} + \frac{U}{Y} + \frac{M}{Y}. \quad (31)$$

It records how final output is divided among gross capital income, labor income, developer profit, inference compute, and research compute.

4 Equilibrium with an upper bound on AI efficiency

In this section, I characterize the long-run equilibrium behavior for various parameter configurations. The key finding is that long-run output-per-worker growth can exceed the rate of labor-augmenting technological change when the elasticity of substitution between AI production services and labor exceeds one and the upper bound on AI efficiency is sufficiently high.

To identify this threshold, I calculate the net return on capital implied by the production and pricing conditions as AI efficiency approaches its upper bound and labor's income share tends to zero. I compare this return with $\rho + \gamma$, the interest rate that sustains consumption-per-person growth at the exogenous rate γ . When the former exceeds the latter and AI and labor are substitutes, the model supports equilibrium paths along which capital and AI production services grow faster than effective labor, even though AI efficiency converges to a finite limit.

4.1 The return to capital and the efficiency threshold

An upper bound restricts AI efficiency, but not the compute used to supply AI production services. To identify the return when labor's income share vanishes, fix $B > 0$ and consider $s_X \rightarrow 1$. Since $wL/Y = (1 - \alpha)(1 - s_X)$, this is the zero-labor-share limit.

Combine the final-good firm's demand for AI services in equation (13) with the developer's pricing condition in equation (19). The result is:

$$\frac{X}{Y} = B(1 - \alpha)s_X(1 - e_X) \longrightarrow B(1 - \alpha)^2. \quad (32)$$

The limit uses $e_X \rightarrow \alpha$ from Equation (15). The CES aggregator in equation (8) and the definition of s_X in equation (10) imply

$$\frac{Z}{X} = \left(\frac{\omega_X}{s_X} \right)^{\sigma/(\sigma-1)} \longrightarrow \omega_X^{\sigma/(\sigma-1)}. \quad (33)$$

Using $Z/Y = (Z/X)(X/Y)$ and the limits in equations (32) and (33), I obtain

$$\frac{Y}{K} = \left(\frac{Z}{Y} \right)^{(1-\alpha)/\alpha} \longrightarrow \left[(1 - \alpha)^2 B \omega_X^{\sigma/(\sigma-1)} \right]^{(1-\alpha)/\alpha}. \quad (34)$$

Let $\mathcal{R}(B)$ denote the net return on capital in this zero-labor-share limit at a given B . Substituting the limiting output-capital ratio in equation (34) into $r = \alpha Y/K - \delta$ gives

$$\mathcal{R}(B) \equiv \alpha \left[(1 - \alpha)^2 B \omega_X^{\sigma/(\sigma-1)} \right]^{(1-\alpha)/\alpha} - \delta. \quad (35)$$

This is a return at a particular limit of the production and pricing conditions, not necessarily an equilibrium interest rate. For $\sigma > 1$, $s_X \rightarrow 1$ means that AI production services become arbitrarily large relative to effective labor. For $\sigma < 1$, $s_X \rightarrow 1$ means that effective labor services become arbitrarily large relative to AI production services. Whether the equilibrium interest rate converges to $\mathcal{R}(\bar{B})$ depends on both σ and $\bar{B}$, as shown below.

By the Euler equation (5), consumption per person grows at $r - \rho$. If this growth rate converges to γ , the interest rate must converge to $\rho + \gamma$. Define $\mathcal{B}(\sigma)$ as the AI-efficiency level at which the zero-labor-share return equals this benchmark:

$$\begin{aligned} \mathcal{R}(\mathcal{B}(\sigma)) &= \rho + \gamma, \\ \mathcal{B}(\sigma) &= \frac{[(\rho + \gamma + \delta)/\alpha]^{\alpha/(1-\alpha)}}{(1 - \alpha)^2 \omega_X^{\sigma/(\sigma-1)}}. \end{aligned} \quad (36)$$

The inequalities between $\bar{B}$ and $\mathcal{B}(\sigma)$ –whether AI can attain sufficient efficiency– and between σ and 1 –whether labor and AI are substitutes or complements– distinguish the long-run equilibrium regimes characterized below. As [Duffy and Papageorgiou \(2000\)](#) explain for CES models of capital

and labor, an elasticity above one permits sustained growth only if the limiting return on capital is sufficiently high.

4.2 Equilibrium existence and long-run regimes

Define $g_{Y/N}$ as the growth rate of output per worker and g_w as the wage growth rate. The next proposition characterizes the long-run behavior of equilibrium trajectories.

Proposition 1 (Equilibrium existence and long-run characterization). *Fix $\bar{B} < \infty$. For each parameter configuration covered by a case below, there exists a nonempty open set of initial conditions (A_0, N_0, K_0, B_0) such that every initial state in that set has an equilibrium trajectory with $B_t \rightarrow \bar{B}$ as $t \rightarrow \infty$ and the following properties:*

(i) **Complementarity.** *If $\sigma < 1$ and $\bar{B} > \mathcal{B}(\sigma)$, then*

$$g_{Y/N} \rightarrow \gamma, \quad g_w \rightarrow \gamma, \quad r \rightarrow \rho + \gamma, \quad \lim_{t \rightarrow \infty} s_X \in (0, 1). \quad (37)$$

(ii) **Unit elasticity.** *If $\sigma = 1$, then*

$$g_{Y/N} \rightarrow \gamma, \quad g_w \rightarrow \gamma, \quad r \rightarrow \rho + \gamma, \quad s_X = \omega_X \quad (38)$$

(iii) **Substitution with a labor bottleneck.** *If $\sigma > 1$ and $\bar{B} < \mathcal{B}(\sigma)$, then*

$$g_{Y/N} \rightarrow \gamma, \quad g_w \rightarrow \gamma, \quad r \rightarrow \rho + \gamma, \quad \lim_{t \rightarrow \infty} s_X \in (0, 1). \quad (39)$$

(iv) **AI-dominated growth.** *If $\sigma > 1$ and $\bar{B} > \mathcal{B}(\sigma)$, then*

$$g_{Y/N} \rightarrow \mathcal{R}(\bar{B}) - \rho > \gamma, \quad (40)$$

$$g_w \rightarrow \left(1 - \frac{1}{\sigma}\right) \gamma + \frac{1}{\sigma} (\mathcal{R}(\bar{B}) - \rho) > \gamma, \quad (41)$$

$$r \rightarrow \mathcal{R}(\bar{B}), \quad s_X \rightarrow 1.$$

Proof. See Appendix A, Proof of Proposition 1. □

I omit the case $0 < \sigma < 1$ with $\bar{B} < \mathcal{B}(\sigma)$ from Proposition 1 because it describes an AI production bottleneck induced by a sufficiently low upper bound on AI efficiency. Proposition 9 in Appendix A shows that, in any equilibrium in this region, AI services become scarce relative to effective labor and long-run average output-per-worker growth is below the exogenous rate γ of labor-augmenting technological progress.

The labor-bottleneck cases of Proposition 1 show that autonomous AI research need not eliminate the production bottleneck, a constraint also emphasized by Aghion et al. (2019). AI efficiency improves in each of these cases, including complementarity. In the complementary labor-bottleneck

equilibria, expanding AI services cannot substitute for the continued expansion of effective labor. As AI-efficiency growth tends to zero, output per worker and wages therefore return to growth at γ . A positive limiting labor share and the absence of a permanent growth acceleration are two consequences of this production bottleneck. Substitution is needed for a permanent increase in growth under an upper bound, not for research to improve AI efficiency. Moreover, $\sigma > 1$ is not sufficient to escape the labor bottleneck: if $\mathcal{R}(\bar{B}) < \rho + \gamma$, the return at a vanishing labor share is too low to sustain growth faster than effective labor.

The AI-dominated regime has an asymptotically linear production structure, as in the capital-accumulation mechanisms of [Rebelo \(1991\)](#) and [Jones and Manuelli \(1990\)](#). Equation (34) gives a positive limiting output–capital ratio even as B approaches the upper bound. Capital and inference compute can expand together, sustaining output-per-worker growth above γ even when B is arbitrarily close to $\bar{B}$ and further AI-efficiency gains vanish, as in the compute-expansion mechanism of [Restrepo \(2025\)](#). [Erdil et al. \(2025\)](#) also discuss how expanding digital-worker services alongside physical capital can generate AK -type dynamics. Here, the corresponding growth regime arises under fixed aggregate production weights and private research and pricing decisions. RSI can help the economy reach an efficiency at which this mechanism supports growth faster than γ ; continuing efficiency improvements are not necessary to sustain it. Section 5 shows that sufficiently high initial efficiency also allows this growth mechanism to operate under competition without RSI.

This joint expansion also explains the higher long-run interest rate. In the labor-bottleneck regime, the interest rate converges to $\rho + \gamma$. In the AI-dominated regime, however, AI services expand alongside capital, so production need not be constrained by the growth of effective labor. The output–capital ratio converges to a level at which $r = \alpha Y/K - \delta$ remains above $\rho + \gamma$: accumulating more capital does not restore the labor-bottleneck return.

The proof of Proposition 1 first removes the long-run trends from growing quantities and shrinking efficiency gaps. For each parameter configuration covered by the proposition, I identify candidate stationary values of these normalized variables. I then construct equilibrium trajectories from every sufficiently nearby initial state and verify convergence to those values. Local uniqueness results are given in the proof. These analytical results guide the algorithm’s boundary conditions (Appendix B).

Remark 2 (Wages and AI revenue in the AI-dominated regime). In the AI-dominated equilibria of Proposition 1(iv), output per worker grows asymptotically faster than wages, even though both grow faster than labor-augmenting technological progress:

$$\lim_{t \rightarrow \infty} g_{Y/N} > \lim_{t \rightarrow \infty} g_w > \gamma.$$

Labor’s income share therefore converges to zero, while the AI industry’s revenue as a share of output converges to $1 - \alpha$.

Figure 3 shows where the long-run regimes characterized in Proposition 1 lie relative to $\mathcal{B}(\sigma)$. For $0 < \sigma < 1$, the equilibria above the threshold have output-per-worker growth converging to γ

and a positive limiting labor income share. For $\sigma > 1$, these same long-run outcomes occur below the threshold. Above it, the regime is AI-dominated: output per worker grows asymptotically faster than γ and labor's income share converges to zero.

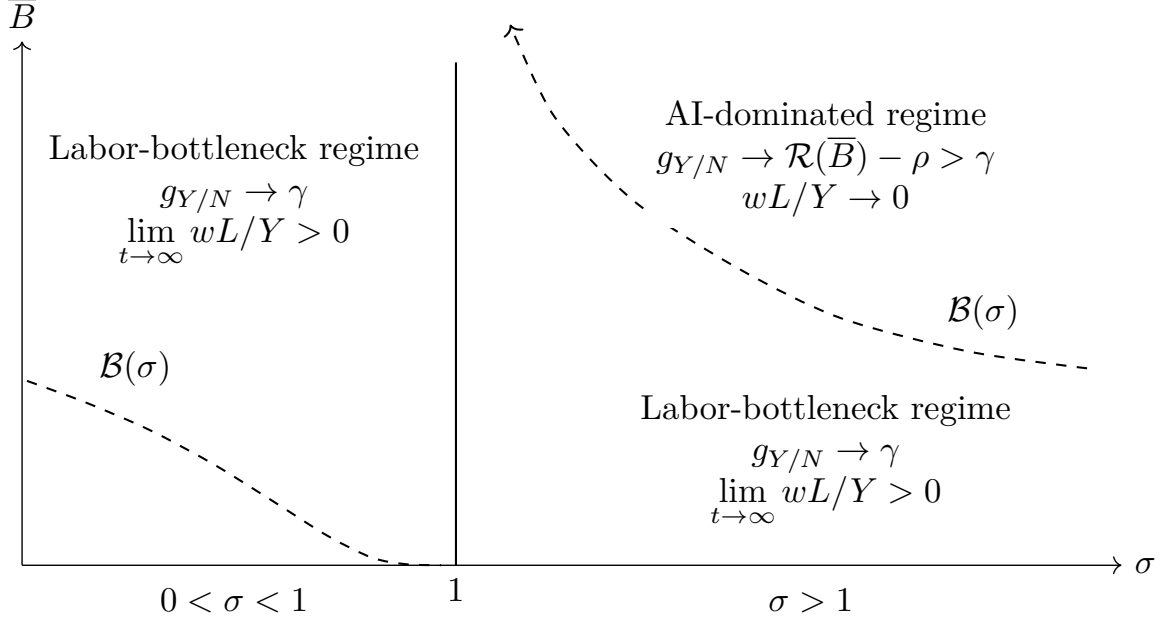


Figure 3: Growth regimes with an upper bound on AI efficiency.

Table 1 summarizes the four cases of Proposition 1. For $\sigma > 1$, an upper bound below $\mathcal{B}(\sigma)$ makes the return on capital at a vanishing labor share too low for capital and AI services to outgrow effective labor. Above the threshold, that return supports their faster growth along the AI-dominated equilibria. Across these equilibrium regimes, long-run wage growth exceeds γ if and only if labor's income share converges to zero.

Table 1: Long-run equilibrium characterization

Elasticity	AI efficiency bound	$\lim g_{Y/N}$	$\lim g_w$	$\lim wL/Y$	$\lim r$
$0 < \sigma < 1$	$\bar{B} > \mathcal{B}(\sigma)$	γ	γ	Positive	$\rho + \gamma$
$\sigma = 1$	Any finite $\bar{B}$	γ	γ	$(1 - \alpha)\omega_L$	$\rho + \gamma$
$\sigma > 1$	$\bar{B} < \mathcal{B}(\sigma)$	γ	γ	Positive	$\rho + \gamma$
$\sigma > 1$	$\bar{B} > \mathcal{B}(\sigma)$	$\mathcal{R}(\bar{B}) - \rho$	$(1 - \frac{1}{\sigma})\gamma + (\frac{1}{\sigma})(\mathcal{R}(\bar{B}) - \rho)$	Zero	$\mathcal{R}(\bar{B})$

4.3 Substitution and wage growth

When AI and labor are substitutes, cheaper AI reduces labor demand at a given wage and budget for labor and AI services. But this fixed-budget comparison misses the expansion of production in equilibrium. With sufficiently efficient AI, capital and AI services can grow faster than effective labor while maintaining a return high enough to sustain capital accumulation. Because substitution is imperfect, this joint expansion raises workers' marginal products and sustains wage growth

above the rate of labor-augmenting technological progress, even as labor's income share vanishes. With complementarity, by contrast, additional AI cannot compensate sufficiently for effective labor growing more slowly. The resulting decline in the return to capital prevents that faster expansion from continuing indefinitely. Wage growth in the labor-bottleneck regime therefore returns to the rate of labor-augmenting technological progress as AI-efficiency gains vanish.

To see why wages accelerate even as labor's share vanishes, I rewrite the wage condition (12) using the CES aggregator in equation (8):

$$w = (1 - \alpha)\omega_L A \left(\frac{K}{Z}\right)^\alpha \left(\frac{Z}{AL}\right)^{1/\sigma}. \quad (42)$$

In the AI-dominated regime, capital and AI services expand together, so K/Z converges to a positive constant while $Z/(AL)$ keeps growing. Although labor accounts for a declining share of output, its marginal product grows faster than A : workers participate in an expanding production process and remain imperfectly replaceable by AI.

Equations (40) and (41) make the distribution of this growth acceleration explicit:

$$\lim_{t \rightarrow \infty} g_w = \gamma + \frac{1}{\sigma} \left(\lim_{t \rightarrow \infty} g_{Y/N} - \gamma \right). \quad (43)$$

For $1 < \sigma < \infty$, only a fraction $1/\sigma$ of output-per-worker growth above γ translates into wage growth above γ . This wage-growth mechanism has a direct antecedent in the CES example of [Palivos and Karagiannis \(2010\)](#). For example, long-run output-per-worker growth of 4.0%, with $\gamma = 1.0\%$ and $\sigma = 1.5$, implies wage growth of 3.0%.

A higher elasticity has two opposing effects on long-run wage growth. On one hand, it raises the limiting return in equation (35) and therefore output-per-worker growth in the AI-dominated regime. On the other, it reduces the fraction $1/\sigma$ of the additional growth that reaches wages. This hints at the existence of an elasticity of substitution that maximizes long-run wage growth.

Proposition 2 (Substitution and wage growth). *If $\bar{B} > \mathcal{B}(\sigma)$ for some $\sigma > 1$, then long-run wage growth is maximized at a unique finite elasticity of substitution $\sigma^* > 1$.*

Proof. See Appendix A, [Proof of Proposition 2](#). □

5 Competitive Equilibrium

I now replace the monopolistic developer with competitive AI suppliers. This comparison separates the incentive to improve AI from the use of technology already developed. I focus on $\sigma > 1$, where both labor-bottleneck and AI-dominated growth are possible. I first characterize the competitive equilibrium and then compare it with the monopoly equilibria in Section 4.

5.1 Characterization

Consider now the case where multiple AI suppliers have free access to the same efficiency B . Entry is free, and any improvement becomes immediately available to every supplier, without royalties or other compensation to its developer. There are no research subsidies or publicly financed research. Suppliers and households take prices and the common technology as given. Preferences, final-good production, and the RSI technology remain unchanged.

Competition sets $p_X = 1/B$. Since $X = BU$, revenue then exactly covers inference compute: $p_X X = U$. A supplier that invests in research bears its cost but earns neither operating profit nor royalties on the improvement. It therefore chooses $M = 0$, leaving $B = B_0$. Final-good firms continue to solve problem (9), with production given by equations (7)–(8). Their demand for AI services in equation (13) then implies $U = (1 - \alpha)s_X Y$, since $X = B_0 U$. The household continues to choose consumption and saving according to the Euler equation (5) and the transversality condition (6). Capital accumulation follows the resource constraint (26) with $M = 0$. Competitive suppliers' optimality replaces the monopolist's pricing, research, and shadow-price conditions.

As labor's income share vanishes, the net return on capital under marginal-cost pricing approaches

$$\mathcal{R}^{\text{mc}}(B) = \alpha \left[(1 - \alpha) B \omega_X^{\sigma/(\sigma-1)} \right]^{(1-\alpha)/\alpha} - \delta. \quad (44)$$

Let $\mathcal{B}^{\text{mc}}(\sigma)$ be the unique efficiency at which $\mathcal{R}^{\text{mc}}(\mathcal{B}^{\text{mc}}(\sigma)) = \rho + \gamma$. Unlike the monopoly economy, the competitive economy cannot improve from B_0 toward $\bar{B}$. Its long-run regime depends on the efficiency inherited at the initial date.

Proposition 3 (Competitive equilibrium). *Suppose $\sigma > 1$. A unique competitive equilibrium exists from every admissible initial state (A_0, N_0, K_0, B_0) . At every date,*

$$M = 0, \quad B = B_0, \quad p_X = 1/B_0, \quad \Pi = 0. \quad (45)$$

Its long-run outcomes on either side of the threshold are:

(i) *If $B_0 < \mathcal{B}^{\text{mc}}(\sigma)$, labor remains a bottleneck:*

$$g_{Y/N} \rightarrow \gamma, \quad g_w \rightarrow \gamma, \quad r \rightarrow \rho + \gamma, \quad \lim_{t \rightarrow \infty} wL/Y > 0.$$

(ii) *If $B_0 > \mathcal{B}^{\text{mc}}(\sigma)$, growth is AI-dominated:*

$$\begin{aligned} g_{Y/N} &\rightarrow \mathcal{R}^{\text{mc}}(B_0) - \rho > \gamma, & g_w &\rightarrow \gamma + \frac{\mathcal{R}^{\text{mc}}(B_0) - \rho - \gamma}{\sigma} > \gamma, \\ r &\rightarrow \mathcal{R}^{\text{mc}}(B_0), & wL/Y &\rightarrow 0. \end{aligned} \quad (46)$$

Proof. See Appendix A, [Proof of Proposition 3](#). □

Neither monopoly nor RSI is necessary to sustain AI-dominated growth. At fixed efficiency,

$X = B_0 U$ can continue expanding through greater compute use. When B_0 exceeds the competitive threshold $\mathcal{B}^{\text{mc}}(\sigma)$, capital and AI services grow together faster than effective labor. This is the capital-accumulation mechanism identified in Section 4.2, now sustained without private research. Below the threshold, diminishing returns bring the interest rate back to $\rho + \gamma$. Since $B = B_0$, the competitive economy cannot escape this bottleneck.

5.2 Comparison with monopoly

I now compare the efficiency required under the two pricing rules, isolating the effect of pricing from the difference in research incentives.

Proposition 4 (Monopoly pricing and the AI-efficiency threshold). *Suppose AI and labor are substitutes ($\sigma > 1$). Let $\mathcal{B}(\sigma)$ and $\mathcal{B}^{\text{mc}}(\sigma)$ denote the AI-efficiency thresholds under monopoly pricing and marginal-cost pricing, respectively. These thresholds identify the efficiency needed for the return on capital to reach $\rho + \gamma$ in the limit where labor's income share vanishes. Monopoly pricing requires a higher AI efficiency:*

$$\mathcal{B}^{\text{mc}}(\sigma) = (1 - \alpha)\mathcal{B}(\sigma) < \mathcal{B}(\sigma). \quad (47)$$

Proof. See Appendix A, Proof of Proposition 4. □

To understand why monopoly pricing raises the threshold, compare the same AI technology with the same capital and effective labor under the two pricing rules. At marginal cost, AI services are cheaper, so final-good firms use more of them and produce more with the same capital. This raises the output-capital ratio and therefore the return on capital. For the threshold comparison, evaluate both returns as labor's income share tends to zero. In this limit, monopoly pricing requires a higher AI efficiency for the return on capital to reach the benchmark $\rho + \gamma$. With $\alpha = 0.33$, marginal-cost pricing reaches this benchmark at only 67.0% of the AI efficiency required under monopoly pricing.

Figure 4 adds the competitive threshold to the substitution region of Figure 3. Efficiencies between the curves exceed the competitive threshold but remain below the monopoly threshold.

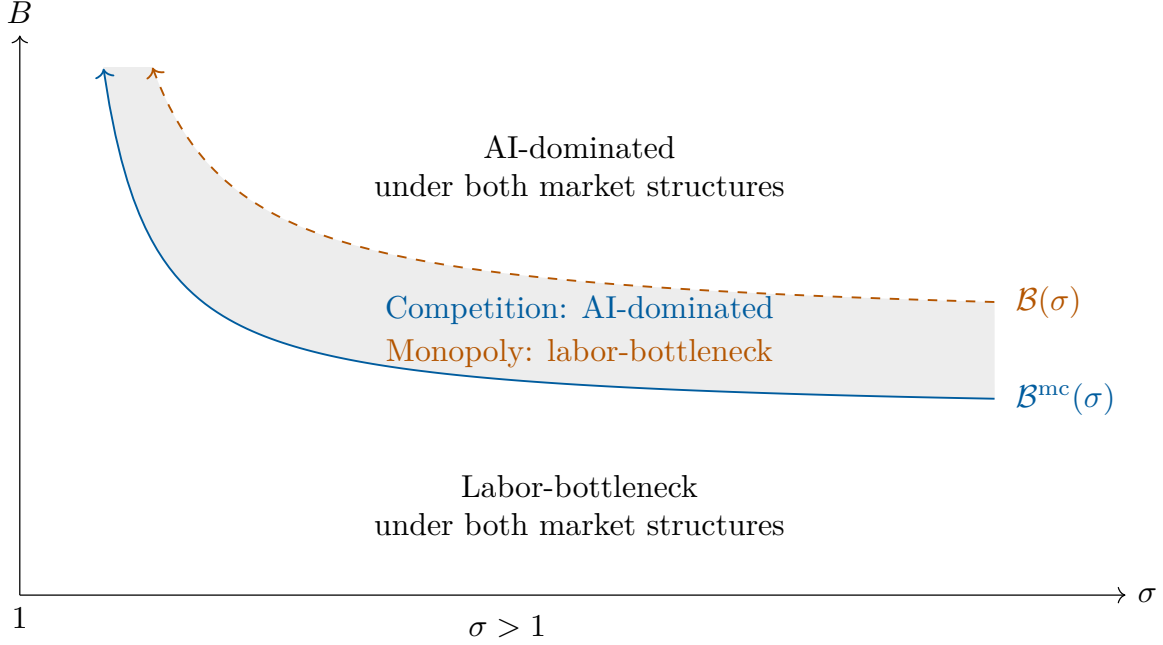


Figure 4: Long-run regimes under monopoly and competition ($\sigma > 1$). The blue solid curve is the competitive threshold; the orange dashed curve is the monopoly threshold. Their relationship is $\mathcal{B}^{\text{mc}}(\sigma) = (1 - \alpha)\mathcal{B}(\sigma)$ (Proposition 4). Read the vertical axis as B_0 under competition and as $\bar{B}$ under monopoly. The labels describe the characterized equilibria, not a comparison from common initial states.

Corollary 1 (Monopoly and the labor bottleneck). *Suppose $\sigma > 1$ and*

$$\mathcal{B}^{\text{mc}}(\sigma) < \bar{B} < \mathcal{B}(\sigma). \quad (48)$$

There is a set of initial states with $\mathcal{B}^{\text{mc}}(\sigma) < B_0 < \bar{B}$ from which the competitive equilibrium converges to AI-dominated growth, while a monopoly equilibrium converges to the labor-bottleneck regime. AI efficiency remains at B_0 under competition but approaches the higher level $\bar{B}$ under monopoly.

Proof. See Appendix A, Proof of Corollary 1. □

In this case, the efficiency gains generated by monopoly-led research do not compensate for the reduction in AI use caused by monopoly pricing. Competition sustains permanently faster output-per-worker and wage growth despite retaining a less efficient AI technology. AI services per unit of effective labor converge to a positive constant under monopoly but grow without bound under competition.

Corollary 2 (Growth under competition and monopoly). *Suppose $\sigma > 1$. Consider competitive and monopoly equilibria with the same parameters and initial stocks, both converging to AI-dominated growth, with AI efficiency approaching $\bar{B}$ under monopoly. Monopoly delivers a higher long-run return to capital and faster long-run growth in output per worker and wages if and only if*

$$B_0 < (1 - \alpha)\bar{B}. \quad (49)$$

Competition delivers higher limiting returns and growth rates when the inequality is reversed. At equality, these limiting outcomes coincide.

Proof. See Appendix A, [Proof of Corollary 2](#). □

The efficiency gains generated by monopoly-led research must be large enough to offset the reduction in AI use caused by its markup. With $\alpha = 0.33$, monopoly requires an eventual efficiency gain of more than 49.3% relative to B_0 to deliver faster long-run growth than competition, provided both equilibria are AI-dominated. This follows from rewriting equation (49) as $\bar{B}/B_0 > 1/(1 - \alpha)$.

There is therefore no universal growth ranking between competition and monopoly. Conditional on the existence of the compared equilibria, monopoly-led research can allow an escape from the labor bottleneck when $B_0 < \mathcal{B}^{\text{mc}}(\sigma)$ but $\bar{B} > \mathcal{B}(\sigma)$. Corollary 1 establishes the opposite regime ranking, while Corollary 2 shows what determines the ranking when both economies are AI-dominated. The relevant comparison is not efficiency alone, but whether monopoly's efficiency gains compensate for the reduction in AI use caused by its markup.

5.3 Research incentives and policy

At given AI efficiency, capital, and effective labor, marginal-cost pricing maximizes $Y - X/B$ and leaves more resources available for consumption or capital accumulation than monopoly pricing. However, the competitive arrangement studied here also removes the private return to research. This raises a classic time-inconsistency problem ([Kydlund and Prescott, 1977](#)). A government may promise monopoly protection to encourage research, but later wish to remove the markup once the technology has been developed. Anticipating this policy reversal can weaken the developer's incentive to invest in research.

An announced rule that ends monopoly protection at a specified efficiency level is different from renegeing on a promise. It nevertheless changes the developer's problem. Without compensation, reaching that level would remove future monopoly rents, potentially giving the developer an incentive to delay reaching it. The threshold $\mathcal{B}^{\text{mc}}(\sigma)$ identifies the AI efficiency at which the limiting return on capital under marginal-cost pricing reaches $\rho + \gamma$. It does not identify the optimal time to end monopoly protection. Choosing that time requires accounting for the research response and comparing consumption paths under alternative rules.

Patent buyouts offer a possible way to reward innovation without retaining the monopoly markup. [Kremer \(1998\)](#) studies government purchases of patents that are generally placed in

the public domain. Applied here, the idea would be to compensate the developer for the rights to an existing AI technology and allow competing suppliers to use it. Competitive supply at the same efficiency would require access to the algorithms and model weights. The buyout would also need a credible valuation and payment rule, a source of financing, and arrangements to reward subsequent improvements. Incentives for further development depend on whether developers can appropriate its benefits. These issues are particularly important for RSI, where today's efficiency is also an input into tomorrow's research. In this paper I leave aside the design of such a policy, including its effects on research and welfare.

6 Equilibrium without an upper bound on AI efficiency

I now return to the monopoly economy, remove the upper bound on AI efficiency, and compare the three substitution regimes. The RSI technology becomes

$$\dot{B} = \chi(BM)^\eta. \quad (50)$$

All other primitives remain as in Section 3. In Definition 1, I set $\psi(B) = 1$ and $\psi'(B) = 0$ and remove the upper restriction on B . Equilibrium still requires finite allocations at every finite date, global agent optimality, and both transversality conditions.

The results differ in scope across substitution regimes. Under complementarity, long-run average output-per-worker growth cannot exceed the rate of labor-augmenting technological progress, even without an upper bound on AI efficiency. At unit elasticity, I give sufficient parameter restrictions for an analytically tractable balanced-growth equilibrium. Under substitution and sufficiently strong returns to research, the developer can make discounted net profit arbitrarily large even over a fixed finite horizon, ruling out an equilibrium with a finite-valued developer optimum.

6.1 Complementarity

When $\sigma < 1$, effective labor remains necessary even if AI becomes arbitrarily efficient. Equation (8) gives an upper bound for Z and Y which does not depend on B :

$$Z \leq \omega_L^{\sigma/(\sigma-1)} AL, \quad (51)$$

$$Y \leq \omega_L^{\sigma(1-\alpha)/(\sigma-1)} K^\alpha (AL)^{1-\alpha}. \quad (52)$$

Proposition 5 (Production bounds under complementarity). *Let $\sigma < 1$ and $\bar{B} = \infty$. Along any feasible path satisfying the production equations (7) and (8), labor-market clearing $L = N$ in equation (28a), and the resource constraint (26), long-run average output-per-worker growth cannot exceed γ :*

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \log \left(\frac{Y_t/N_t}{Y_0/N_0} \right) \leq \gamma. \quad (53)$$

Proof. See Appendix A, Proof of Proposition 5. $\square$

Complementarity bounds long-run average output-per-worker growth despite RSI and the absence of an AI-efficiency upper bound. This does not prevent growth from temporarily exceeding γ .

Proposition 6 (Long-run outcomes under complementarity without an AI efficiency upper bound). *Suppose $\sigma < 1$, $\bar{B} = \infty$, and an equilibrium satisfies $B \rightarrow \infty$. Then*

$$\frac{K}{Y} \rightarrow \frac{\alpha}{\rho + \gamma + \delta} > 0, \quad \frac{C}{Y} \rightarrow 1 - \frac{\alpha(\delta + n + \gamma)}{\rho + \gamma + \delta} > 0, \quad (54)$$

$$\frac{U}{Y} \rightarrow 0, \quad \frac{M}{Y} \rightarrow 0, \quad (55)$$

$$g_{Y/N} \rightarrow \gamma, \quad g_w \rightarrow \gamma, \quad r \rightarrow \rho + \gamma, \quad (56)$$

$$s_X \rightarrow \frac{1 - \sigma}{1 - \alpha\sigma}, \quad (57)$$

$$\frac{wL}{Y} \rightarrow \frac{\sigma(1 - \alpha)^2}{1 - \alpha\sigma} > 0, \quad \frac{p_X X}{Y} \rightarrow \frac{(1 - \alpha)(1 - \sigma)}{1 - \alpha\sigma} > 0. \quad (58)$$

Proof. See Appendix A, Proof of Proposition 6. $\square$

Conditional on $B \rightarrow \infty$, inference and research compute vanish as shares of output, not necessarily in levels. The AI industry nevertheless retains a positive revenue share.

6.2 Unit elasticity

At unit elasticity, production becomes

$$Y = K^\alpha (AL)^{(1-\alpha)\omega_L} X^{(1-\alpha)\omega_X}. \quad (59)$$

Consider a balanced-growth path (BGP) with positive constant K/Y , C/Y , and M/Y . The AI-demand and monopoly conditions, equations (13) and (19), imply $U/Y = (1 - \alpha)^2 \omega_X^2$. Along the BGP, $g_K = g_U = g_M = g_Y$. Substituting $X = BU$ into production, equation (59), and log-differentiating this equation and the RSI technology in equation (50) yields

$$g_Y = n + \gamma + \frac{\omega_X}{\omega_L} g_B, \quad (60)$$

$$(1 - \eta) g_B = \eta g_Y. \quad (61)$$

The first relation describes how AI improvement raises production growth. The second describes how expanding research resources sustain AI improvement. Solving these relations gives candidate growth rates. The next proposition establishes that they belong to an equilibrium.

Proposition 7 (An uncapped balanced-growth equilibrium and local convergence). *Suppose $\sigma = 1$, $\bar{B} = \infty$, and*

$$\eta + \omega_X < 1. \quad (62)$$

(i) *There is an equilibrium trajectory such that*

$$g_{Y/N} = g_w = \gamma + \frac{\eta\omega_X}{1 - \eta - \omega_X}(n + \gamma), \quad (63)$$

$$r = \rho + \gamma + \frac{\eta\omega_X}{1 - \eta - \omega_X}(n + \gamma), \quad (64)$$

$$\frac{wL}{Y} = (1 - \alpha)\omega_L > 0, \quad (65)$$

$$g_B = \frac{\eta(1 - \omega_X)}{1 - \eta - \omega_X}(n + \gamma). \quad (66)$$

(ii) *There is an open neighborhood of the initial condition (A_0, N_0, K_0, B_0) of the BGP in part (i) such that, from every initial condition in that neighborhood, there exists an equilibrium trajectory converging to the corresponding BGP in normalized variables.*

Proof. See Appendix A, Proof of Proposition 7. □

Proposition 7 shows that RSI can sustain permanently faster wage growth without a vanishing labor income share. Equations (63) and (65) imply $g_w = g_{Y/N} > \gamma$ while $wL/Y = (1 - \alpha)\omega_L$ remains constant and positive. Wages therefore keep pace with output per worker. This contrasts with the regimes with an upper bound in Proposition 1, where a permanent wage-growth premium occurs only in the AI-dominated regime and accompanies a vanishing labor share. A declining labor share is therefore not a necessary consequence of recursive self-improvement.

Growth in effective labor expands the resources available for research, which raises B and further expands output. The common denominator $1 - \eta - \omega_X$ captures this reinforcement: a higher η strengthens the research feedback, while a higher ω_X makes output growth more responsive to AI improvement. A related dependence appears in Jones (1995), where long-run growth depends on an expanding research workforce. Here, effective labor supports research indirectly through production, rather than entering the RSI technology itself.

6.3 Substitution

With $\sigma > 1$, Proposition 1 establishes AI-dominated equilibria whose limiting interest rate is $\mathcal{R}(\bar{B})$, which grows without bound as $\bar{B} \rightarrow \infty$. The same function also provides a lower bound on the equilibrium interest rate at each date without an efficiency upper bound:

$$r_t \geq \mathcal{R}(B_t). \quad (67)$$

To see this, combine equations (32), (33), and (34) before taking the limit $s_X \rightarrow 1$:

$$\frac{Y}{K} = \left[B(1 - \alpha) \omega_X^{\sigma/(\sigma-1)} (1 - e_X) s_X^{-1/(\sigma-1)} \right]^{(1-\alpha)/\alpha}. \quad (68)$$

For fixed B and $\sigma > 1$, this expression decreases with s_X and approaches its lower limit as $s_X \rightarrow 1$. Substitution into $r = \alpha Y/K - \delta$ gives inequality (67). Thus $\mathcal{R}(B)$ is the lowest return compatible with the production and monopoly conditions at that efficiency level, attained in the zero-labor-share limit. If B grows without bound along a trajectory, inequality (67) requires $r \rightarrow \infty$ and $Y/K \rightarrow \infty$. Production also gives $Z/Y = (Y/K)^{\alpha/(1-\alpha)} \rightarrow \infty$, while the Euler equation (5) requires $g_{C/N} = r - \rho \rightarrow \infty$.

Davidson et al. (2026) show that technological and economic feedback loops can overcome diminishing returns to research and generate *hyperbolic growth*, with variables diverging at a finite date. This *finite-time singularity* is stronger than an indefinitely increasing growth rate. Their benchmark fixes saving and factor-allocation shares; I therefore cannot apply its explosion conditions directly to the optimal choices studied here.

Under Definition 1, a finite-time singularity cannot constitute an infinite-horizon equilibrium, which requires finite allocations at every finite date. The return bound in equation (67) does not determine whether divergence occurs in finite or infinite time. The following proposition instead shows that, when research returns are sufficiently strong, the developer can make discounted net profit arbitrarily large even over a fixed finite horizon. This is an unbounded optimization problem, not a trajectory that attains infinite profit in finite time.

Proposition 8 (Unbounded discounted net profit). *Suppose $\sigma > 1$, $\bar{B} = \infty$, and $\eta > \alpha$. Fix an initial efficiency B_0 , a finite horizon $T > 0$, continuous paths of K, A, L , and an integrable interest-rate path. The developer's discounted net profit in problem (21), restricted to $[0, T]$, is unbounded above over feasible research trajectories. Consequently, no equilibrium satisfying Definition 1 exists under these parameter restrictions.*

Proof. See Appendix A, Proof of Proposition 8. □

7 Numerical simulations

I use numerical simulations to illustrate the model's economic mechanisms, not to make calibrated predictions of the magnitude or timing of AI-driven growth. The AI-specific parameters are illustrative rather than jointly estimated from data. I therefore emphasize how research incentives, AI use, and capital accumulation shape the transition and the long-run regime; the reported magnitudes and dates are conditional on the chosen parameters and initial conditions.

The exercises in this section answer two questions. First, how does an economy adjust when RSI becomes available under an existing monopoly, and how does that adjustment depend on research productivity? Second, can research financed by monopoly rents compensate for the restriction on AI use caused by monopoly pricing and enable permanently faster growth? The [supplementary](#)

[simulations in the digital edition](#) also examine the opposite outcome: can monopoly pricing eliminate AI-dominated growth despite the efficiency gains from research? Appendix B describes the algorithm used to find the equilibrium and presents the links to replication files.

7.1 Parameters and initial conditions

In the next two subsections, I compare the transitional dynamics across four parameter configurations corresponding to the cases in Proposition 1. I study an unanticipated activation of RSI at date zero, retaining monopoly pricing before and after the event. Before date zero, the economy already uses AI production services, but AI efficiency is fixed at B_0 . For the chosen value of B_0 , each economy starts on a labor-bottleneck balanced growth path. I compute this path with the research productivity parameter χ equal to zero to obtain the corresponding initial capital stock K_0 for each parameter configuration. Setting $\chi = 0$ rules out improvements in AI efficiency, not the expansion of AI services through inference compute U . By treating activation as unanticipated, I exclude the advance adjustment of consumption and saving emphasized in the technology-news literature reviewed by [Beaudry and Portier \(2014\)](#).

I compare four economies with $\sigma = 0.9, 1.0, 1.1, 1.5$, holding all other parameters and the initial values A_0 , N_0 , and B_0 fixed. For each economy, I choose K_0 to place it on its pre-RSI balanced growth path. In particular, for each elasticity, I choose the pre-event capital stock to satisfy

$$r_0 = \rho + \gamma, \quad \frac{K_0}{Y_0} = \frac{\alpha}{\rho + \gamma + \delta} = 3.30. \quad (69)$$

This setup guarantees that Y , X , U , w , r , and p_X behave continuously rather than jumping at date zero. Consumption, research expenditure, and growth rates can nevertheless jump when RSI becomes available. Within each elasticity, the two research-productivity scenarios inherit exactly the same stocks; consumption and research are chosen anew after activation.

Table 2 reports the parameter values and initial stocks used for the simulations. Time is measured in years, and the four values of K_0 follow the order of the elasticities in the table. I use standard values for α , δ , and ρ , while n and γ are guided by long-run population and productivity projections. The AI production weight $\omega_X = 0.10$ and research exponent $\eta = 0.20$ are illustrative choices, not empirical estimates. I choose them to satisfy $\omega_X + \eta < 1$ and $\eta < \alpha$.

The common upper bound on AI efficiency satisfies $\mathcal{B}(1.5) < \bar{B} < \mathcal{B}(1.1)$. Thus, AI and labor are substitutes in both the $\sigma = 1.1$ and $\sigma = 1.5$ cases, but only the latter converges to the AI-dominated regime. Holding this upper bound fixed, I study two levels of research productivity for each of the four elasticities of substitution: $\chi = 7.5$ in the higher-productivity scenario and $\chi = 1.5$ in the lower-productivity scenario. For a given elasticity, both lead to the same long-run regime, output and wage growth rates, and interest rate, but produce different initial adjustments and transition speeds. These are illustrative choices, intended to show how research productivity affects the short-run impact of RSI and the timing of subsequent growth.

Table 2: Simulation parameters and initial stocks

Symbol	Value	Interpretation	Source
α	0.33	Capital share	Standard calibration value.
δ	0.05	Depreciation rate	Standard calibration value.
ρ	0.04	Discount rate	Standard calibration value.
n	0.003	Population growth rate	Rounded constant-rate approximation to the UN 2024–2100 world projection (United Nations, 2024).
γ	0.01	Growth of labor-augmenting technology	OECD labor-productivity projections (OECD, 2025).
η	0.20	Research exponent	Illustrative.
ω_X	0.10	AI production weight	Illustrative.
ω_L	0.90	Effective-labor weight	$1 - \omega_X$.
χ	7.5 (higher); 1.5 (lower)	Research productivity after activation	Illustrative.
σ	0.90; 1.00; 1.10; 1.50.	Elasticity of substitution	Illustrative.
$\bar{B}$	1,361	AI-efficiency upper bound	$1.10 \times \mathcal{B}(1.50)$.
A_0	1	Initial labor-efficiency units	Normalization.
N_0	1	Initial population	Normalization.
B_0	13.6	Initial AI efficiency	$0.01 \times \bar{B}$.
K_0	3.43; 4.65; 5.04; 5.50.	Initial capital	Adjusted to match $K_0/Y_0 = 3.30$, the pre-RSI balanced growth path ratio.

I do not model a separate stock of computing hardware or the costs of rapidly expanding it. [Erdil et al. \(2025\)](#) incorporate adjustment costs for compute capacity and other physical capital. Introducing such frictions could change the initial reallocation of resources and the timing of the transition shown here.

7.2 Fast takeoff

I first ask how research, consumption, and capital accumulation respond when RSI becomes available under an existing monopoly, and how this adjustment differs across economies approaching the two long-run regimes. I compare the four elasticities under relatively high research productivity, $\chi = 7.5$. Research productivity affects the transition, but not the limiting growth rates characterized in Section 4. All four economies inherit the pre-RSI balanced growth paths described in Section 7.1. Table 2 reports their common parameters and elasticity-specific initial capital stocks. I examine quantities first, then wages and prices, and finally income shares.

I use Figure 5 to present the annual growth rates of output, AI production services, and capital per unit of effective labor: $Y/(AL)$, $X/(AL)$, and $K/(AL)$. Zero therefore means that the

corresponding aggregate grows at the same rate as effective labor, $n + \gamma$. The upper row shows two pre-event years and the first ten years after RSI activation. The lower row follows the same variables from year 10 to year 500. Throughout the three figures, green dashed, black solid, orange dotted, and blue dash-dotted lines denote $\sigma = 0.90, 1, 1.10, 1.50$, respectively. The vertical line marks RSI activation, and black dotted horizontal lines mark the analytical limits for $\sigma = 1.50$.

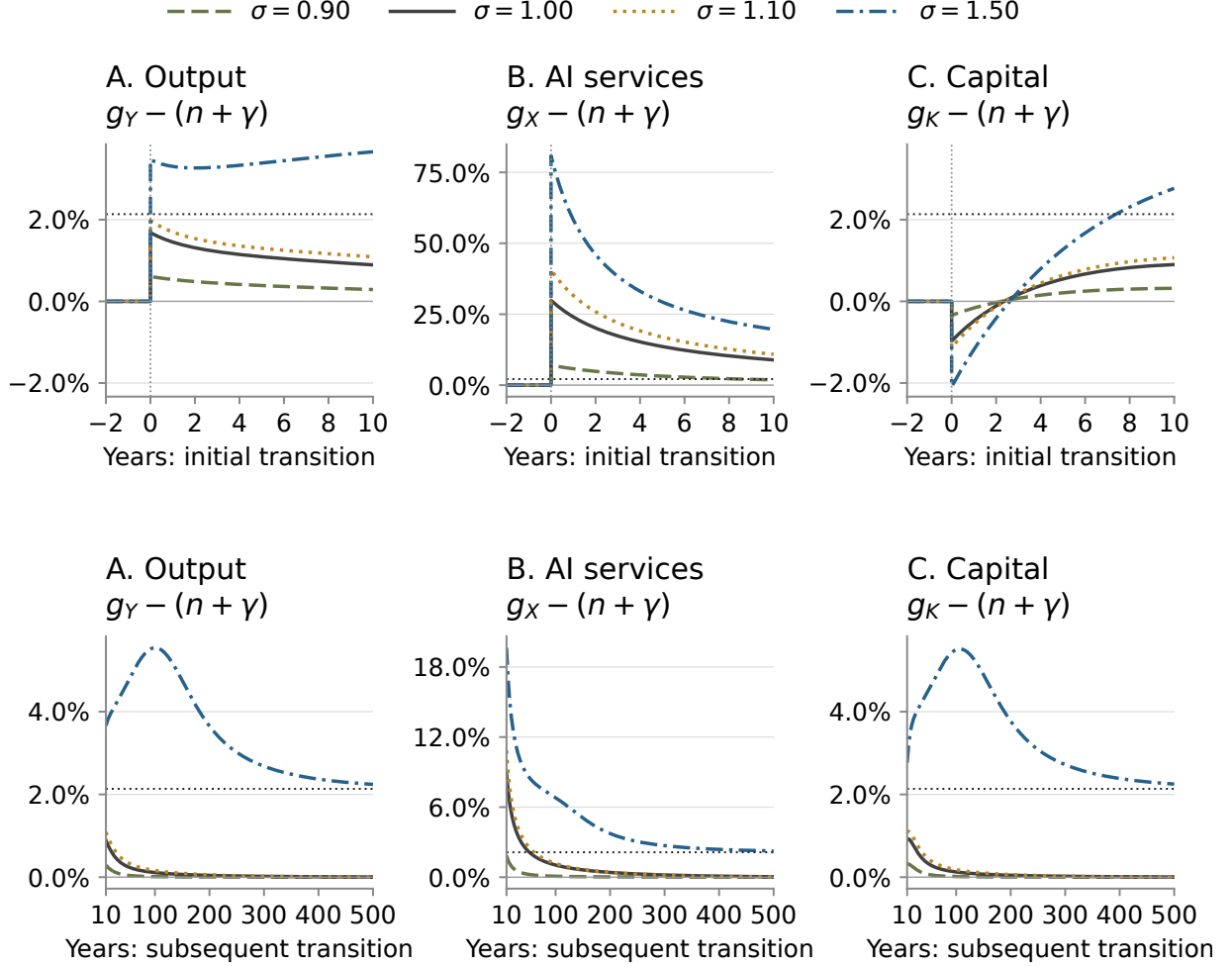


Figure 5: Higher research productivity ($\chi = 7.5$): growth of output, AI production services, and capital per unit of effective labor. Rates are instantaneous annual rates. The upper row shows two pre-event years and years 0–10; the lower row shows years 10–500. Panels A and C share a scale within each row. The vertical line marks RSI activation.

The top row of Figure 5 shows the short-run effects of RSI activation. At $\sigma = 1.50$, $X/(AL)$ initially grows at 80.8% per year, whereas $Y/(AL)$ grows at 3.5%. Research begins to raise B , allowing more services to be obtained from compute through $X = BU$. Initial growth of $Y/(AL)$ is smaller at the other elasticities, reaching only 0.6% at $\sigma = 0.90$. The third panel shows the capital response: capital per unit of effective labor, $K/(AL)$, falls in every case. Households increase consumption at RSI activation while the developer starts spending on research. The resource

constraint thus requires lower investment in physical capital. Capital accumulation subsequently recovers as output expands and as the marginal product of capital increases.

The bottom row of Figure 5 separates temporary growth gains from permanent ones. At $\sigma = 0.90, 1, 1.10$, all three normalized growth rates decline toward zero. This is the labor-bottleneck outcome in Proposition 1: output per worker eventually grows at $\gamma = 1.0\%$. At $\sigma = 1.50$, output and capital growth instead rise further before declining toward their long-run values. As explained in Section 4.2, sufficient substitution and a sufficiently high efficiency bound allow capital and inference compute to expand together without being limited by effective-labor growth. Their limiting growth rates exceed effective-labor growth by 2.1 percentage points; effective labor itself grows at 1.3% per year.

I next examine wages and prices in Figure 6. The figure shows annual wage growth g_w , the annual net interest rate r , and the level of p_X in final-good units on a logarithmic scale. The top panels present the short-run dynamics and the bottom panels present the long-run dynamics.

Wage growth increases in all four economies, but the initial response is not ordered monotonically by substitutability: the immediate increase is larger at $\sigma = 1$ than at $\sigma = 1.50$. Wage growth in the highest-elasticity case soon surpasses the other cases. After the initial jump, wage growth declines in the labor-bottleneck cases but continues to rise in the AI-dominated case.

The interest rate starts at 5.0% in every case and then rises as output initially outpaces capital. The higher return accompanies faster consumption-per-person growth, as required by the Euler condition in Equation (5). A related mechanism underlies Chow et al. (2026), who link faster expected consumption growth to higher long-term real interest rates.

The bottom row of Figure 6 shows the long-run adjustment of wage growth, the interest rate, and the AI-service price. The three labor-bottleneck economies return toward wage growth of 1.0% and an interest rate of 5.0%. At $\sigma = 1.50$, both the wage growth and the interest rate rise above their long-run values before declining. Equation (41) gives limiting wage growth of 2.4%, below output-per-worker growth of 3.1% but above labor-augmenting technological progress of 1.0%. Meanwhile, the interest rate tends to $\mathcal{R}(\bar{B}) = 7.1\%$, above its pre-RSI level.

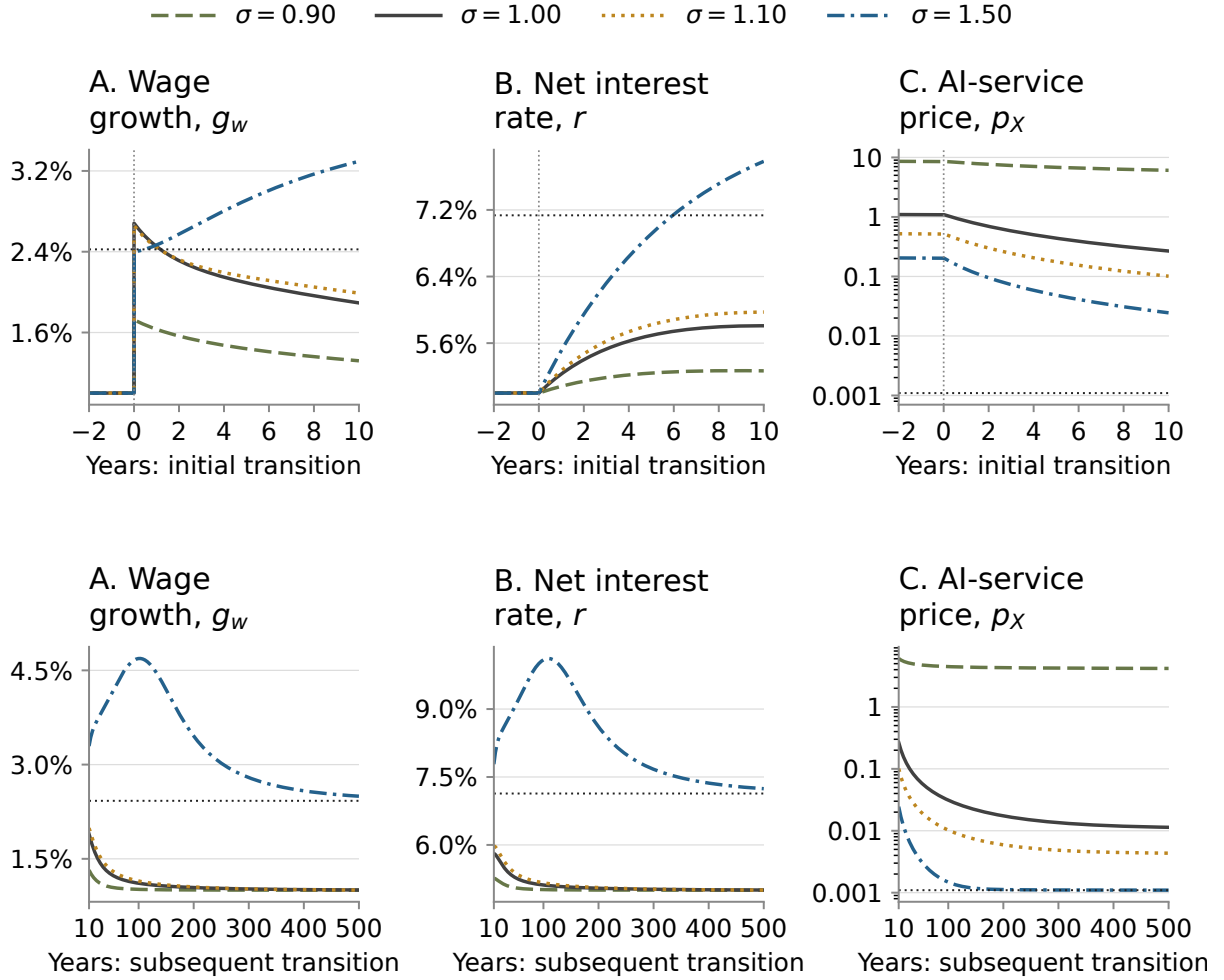


Figure 6: Higher research productivity ($\chi = 7.5$): wage growth, the net interest rate, and the AI-service price. Rates are annual; the price is in final-good units on a logarithmic scale. The vertical line marks RSI activation.

Finally, I use Figure 7 to show labor income wL/Y , developer net profit Π/Y , inference expenditure U/Y , and research expenditure M/Y , all as a share of output. The top four panels show the short run and the bottom four panels show the long run. Line styles again identify the four elasticities. The omitted gross capital income share remains $\alpha = 33.0\%$.

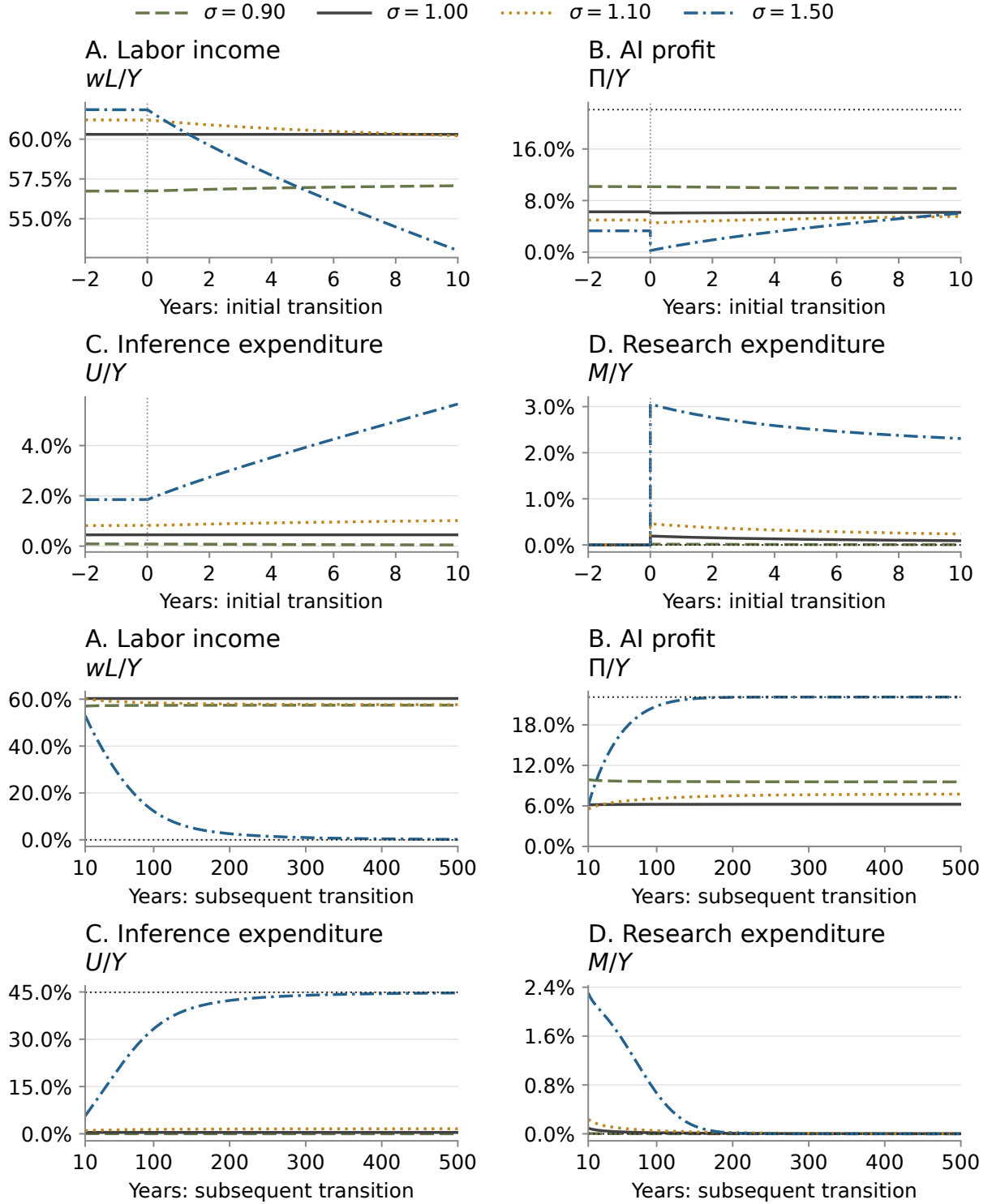


Figure 7: Higher research productivity ($\chi = 7.5$): labor income, AI net profit, inference expenditure, and research expenditure relative to output. The first two rows show the pre-event BGP and years 0–10; the last two show years 10–500. The omitted gross capital share remains $\alpha = 0.33$. Short-run panels A and C use scales fitted to the initial trajectories and omit the long-run reference lines.

In the short run, research spending as a share of output jumps in all four economies. The adjustment is strongest at $\sigma = 1.50$: immediately after activation, research absorbs 59.4% of AI revenue, compared with 35.9% for inference compute. The developer’s net profit therefore falls sharply. Profit subsequently recovers as research absorbs a smaller share of revenue and AI sales expand. Labor’s income share also falls most strongly in this scenario, while changes are much smaller in the other three economies.

Over the longer horizon, the AI-dominated economy combines permanently faster wage growth with an even faster expansion of output per worker, so labor’s income share tends to zero. Rising wages alongside a declining labor share also appear in [Autor and Kausik \(2026\)](#) and [Ray and Mookherjee \(2022\)](#). As labor’s share falls, AI revenue rises relative to output. Inference compute eventually absorbs 67.0% of AI revenue, while research expenditure vanishes relative to that revenue.

7.3 Slow takeoff

I next ask whether lower research productivity makes the early transition less informative about the eventual growth regime, and how it changes the initial allocation of resources. I set $\chi = 1.5$, one fifth of its value in Section 7.2. All other parameters and initial stocks remain unchanged, including the elasticity-specific pre-RSI capital stocks in Table 2. Consumption and research are solved again for each economy. The new exercise changes both the initial adjustment and the timing of the transition, but not the limiting growth rates, interest rates, and income shares.

I use Figure 8 to present the growth of $Y/(AL)$, $X/(AL)$, and $K/(AL)$, in percent per year. As before, zero means growth at the rate of effective labor, $n + \gamma$. The upper row shows two pre-event years and the first ten years after RSI activation. The lower row follows the same trajectories from year 10 to year 500. Throughout the three figures, line styles identify the same four elasticities as in Figure 5, and black dotted lines mark the analytical limits for $\sigma = 1.50$.

The upper row shows a much smaller initial increase in growth. Growth of $Y/(AL)$ ranges from 0.1% to 0.5% across the four economies. AI services still grow faster than output, but their initial normalized growth rate at $\sigma = 1.50$ is 11.6%, compared with 80.8% in the preceding exercise. Lower research productivity and a smaller research allocation produce slower efficiency improvements. Consumption still rises at activation, and research uses resources that could otherwise finance capital accumulation. Both adjustments are smaller, however, so $K/(AL)$ falls less. As capital accumulation recovers, output growth rises during the first decade even though AI-service growth is already slowing. The four economies therefore begin with more moderate growth and less pronounced differences.

The lower row reveals the eventual separation between regimes. At $\sigma = 1.50$, output and capital growth rise for several centuries before declining. AI-service growth first slows, then rises again as capital and inference use expand. At year 500, $Y/(AL)$ still grows at 3.2%, above its 2.1% long-run limit, showing a significantly slower transition. A similar pattern of delayed acceleration appears in [Jones \(2026\)](#), drawing on [Jones and Tonetti \(2026\)](#).

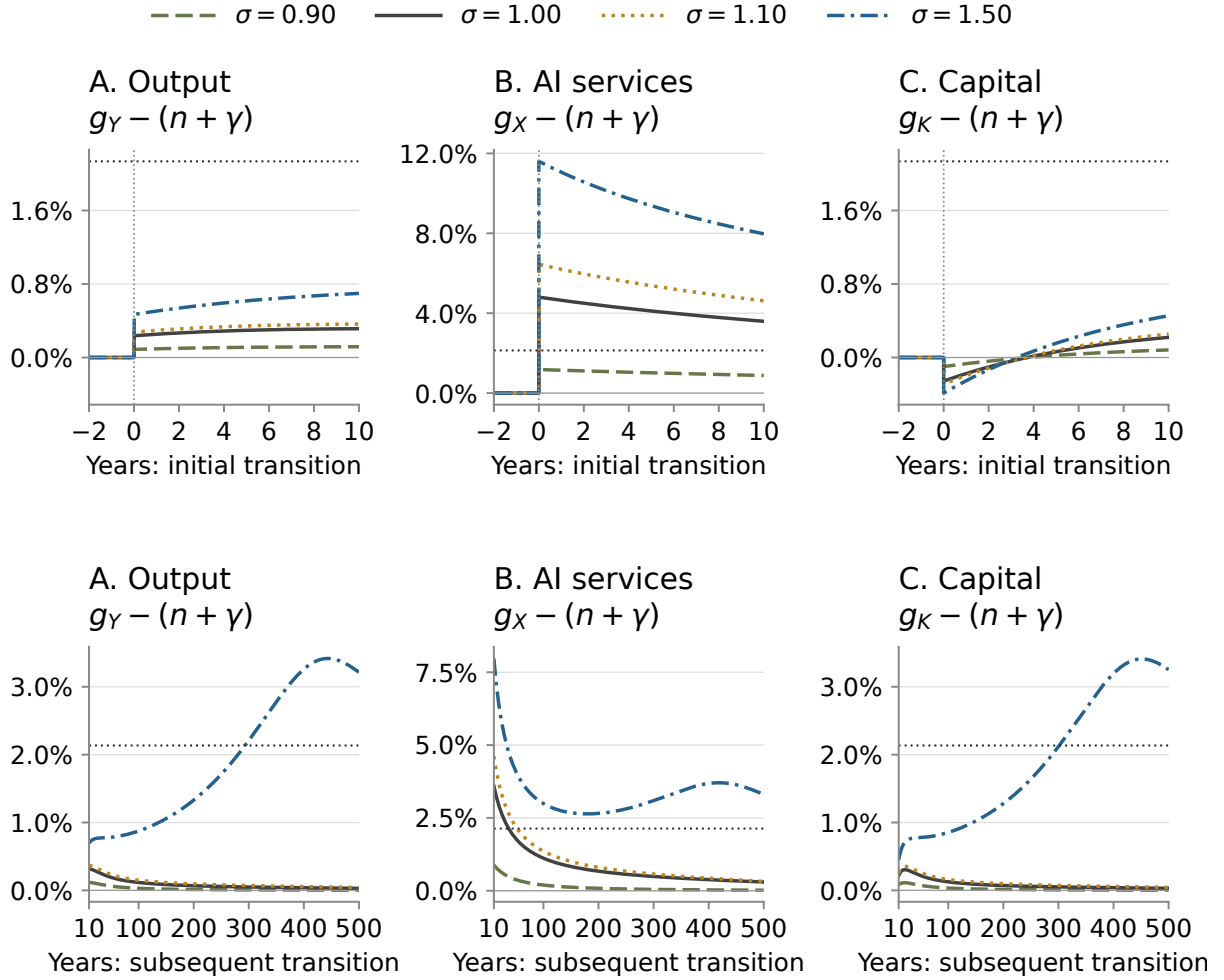


Figure 8: Lower research productivity ($\chi = 1.5$): growth of output, AI production services, and capital per unit of effective labor. Rates are instantaneous annual rates. The upper row shows two pre-event years and years 0–10; the lower row shows years 10–500. Panels A and C share a scale within each row. The vertical line marks RSI activation.

I next examine wages and prices in Figure 9. The panels show annual wage growth g_w , the annual net interest rate r , and the level of p_X in final-good units on a logarithmic scale. The top panels show the short run and the bottom panels the long run. Initially, wage growth rises modestly in all four economies. Unlike the preceding exercise, it continues to increase during the first years even in the labor-bottleneck cases: capital accumulation helps sustain the increase in labor's marginal product. The interest rate also rises more gradually as output outpaces capital. At $\sigma = 1.50$, it reaches 5.5% after ten years, compared with 7.8% under higher research productivity. Meanwhile, slower efficiency improvements make the decline in p_X more gradual.

In the bottom row, wage growth and the interest rate eventually diverge between regimes. The three labor-bottleneck cases approach 1.0% and 5.0%, respectively. At $\sigma = 1.50$, the rise and subsequent decline occur much later, and adjustment remains incomplete at year 500: the

interest rate is still 8.2%, above its 7.1% long-run value. The limiting wage-growth rate remains 2.4%, below output-per-worker growth of 3.1%. Lower research productivity therefore delays the transition without eliminating the long-run difference between these growth rates.

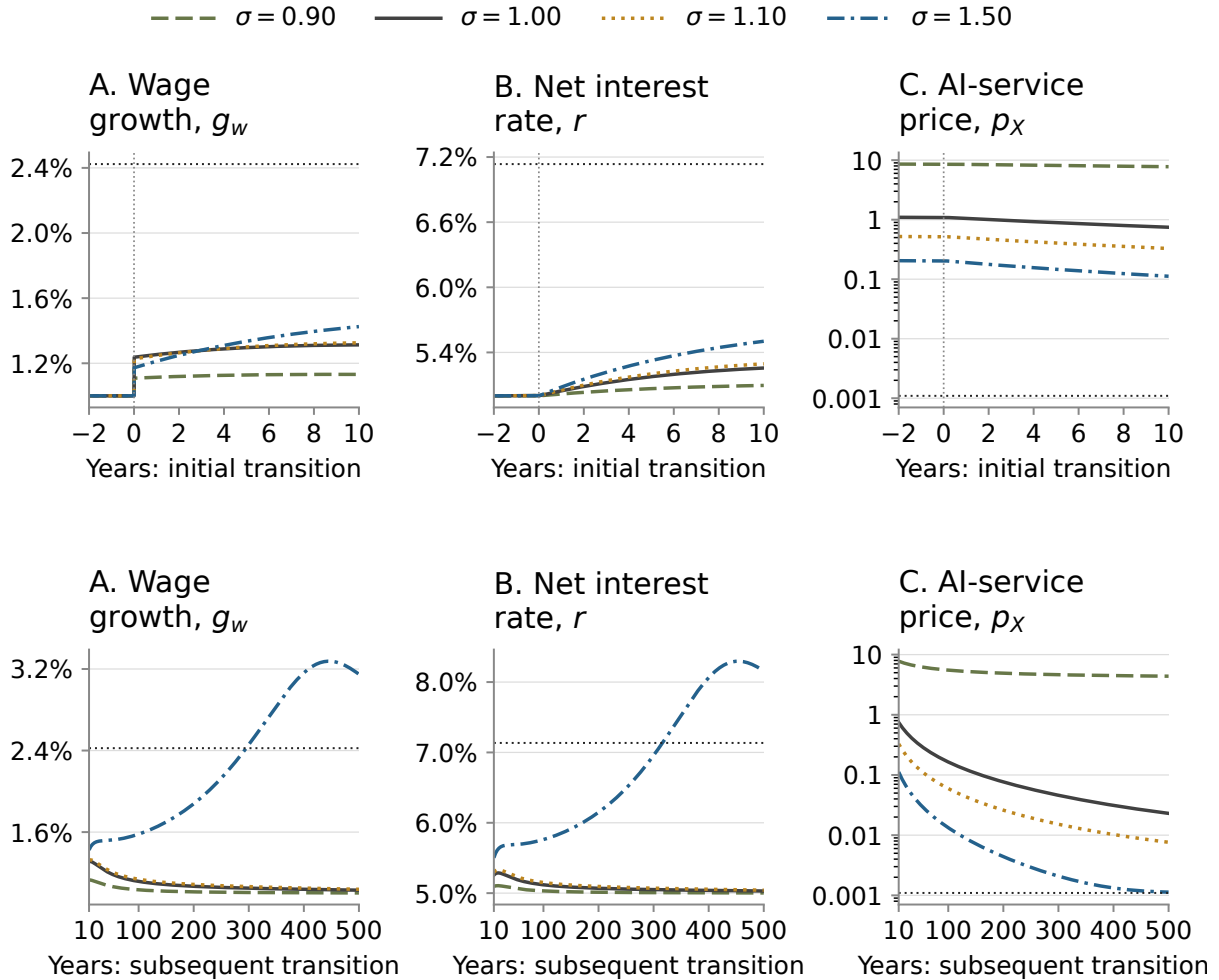


Figure 9: Lower research productivity ($\chi = 1.5$): wage growth, the net interest rate, and the AI-service price. Rates are annual; the price is in final-good units on a logarithmic scale. The vertical line marks RSI activation.

I complete the comparison with Figure 10. The panels report wL/Y , Π/Y , U/Y , and M/Y , all as percentages of output. The upper two rows cover the initial transition and the lower two cover years 10–500, with the same scenario lines. The omitted gross capital share remains 33.0%.

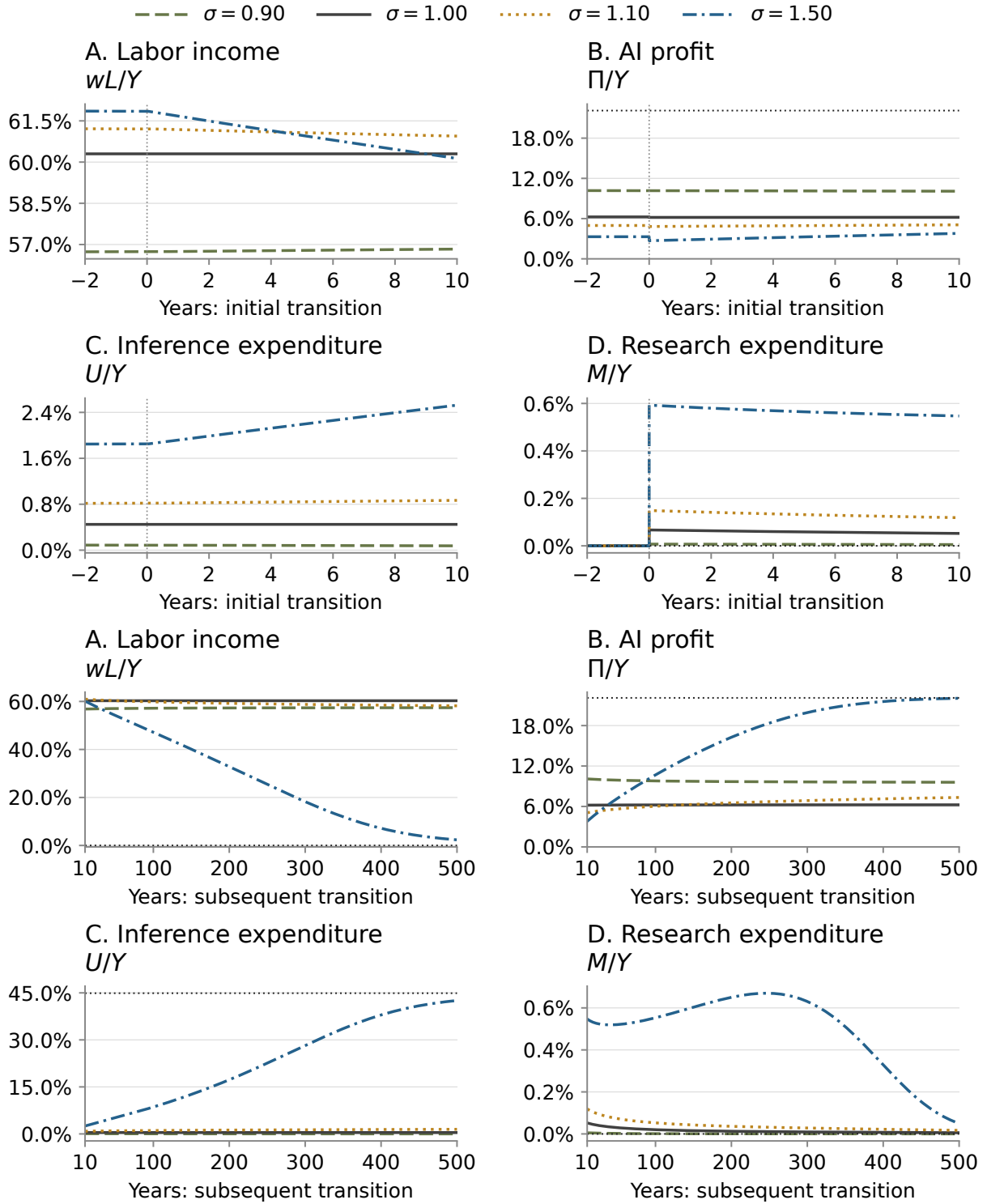


Figure 10: Lower research productivity ($\chi = 1.5$): labor income, AI net profit, inference expenditure, and research expenditure relative to output. The first two rows show the pre-event BGP and years 0–10; the last two show years 10–500. The omitted gross capital share remains $\alpha = 0.33$. Short-run panels A and C use scales fitted to the initial trajectories and omit the long-run reference lines.

In the short run, research absorbs a smaller share of output, leaving more of AI revenue as net profit. At $\sigma = 1.50$, initial research expenditure is 0.6% of output rather than 3.1%, and net profit is 2.7% rather than 0.2%. Research spending reduces net profit at activation, but less sharply than in the preceding exercise. Inference expenditure and labor’s income share change little in the labor-bottleneck cases. In the AI-dominated case, inference’s share increases and labor’s share declines, but both movements are initially gradual, consistent with the smaller output–wage growth gap.

The longer horizon shows the same eventual redistribution as in Figure 7: labor’s share tends to zero at $\sigma = 1.50$, while inference expenditure and developer profit approach the same respective limiting shares of output as in the higher-productivity exercise. Research expenditure relative to output first declines, then rises again before falling toward zero. Expanding AI sales provide an incentive for further research, while proximity to the upper bound eventually limits the efficiency gains it can deliver. The other three economies retain a positive labor share. Lower research productivity changes the adjustment and delays the divergence between regimes without changing their limiting growth rates and income shares.

7.4 Monopoly boosts growth

I now ask whether research financed by monopoly rents can compensate for the restriction on AI use caused by monopoly pricing and allow an economy to escape the labor bottleneck. The pre-event economy is a competitive economy in a labor-bottleneck regime with stagnant AI efficiency. I compare continuing with competition against introducing a monopoly. I set $\sigma = 1.5$ and study $\chi = 7.5$ and $\chi = 1.5$. Unlike the preceding exercises, RSI is available before date zero, with the same positive χ as after the change. Research is nevertheless zero under the competitive arrangement in Section 5, because improvements cannot be privately appropriated.

I retain the technology parameters in Table 2. Before the change, $p_X = 1/B_0$, $r = \rho + \gamma$, and $K_0/Y_0 = 3.3$. Initial efficiency lies below $\mathcal{B}^{\text{mc}}(1.5)$, so continued competition preserves the labor-bottleneck. The upper bound instead exceeds $\mathcal{B}(1.5)$, allowing the monopoly paths considered here to approach AI-dominated growth.

At date zero, an unexpected permanent exclusive right covers both the existing AI technology and its subsequent improvements. In particular, competitive suppliers can no longer use the old technology. This experiment changes research incentives and pricing, not the RSI technology. It abstracts from the cost of enforcing exclusivity and is not an optimal policy exercise.

Figure 11 compares both research productivities with continued competition from the same initial stocks. The panels show output per worker, wages, and consumption per person. The gray dashed line denotes continued competition; the solid blue and dash-dotted gold lines denote monopoly with $\chi = 7.5$ and $\chi = 1.5$. The upper row includes two pre-event years and years 0–10, measured as percentages of the competitive levels. The middle and lower rows cover years 10–100 and 10–500, measured as multiples on logarithmic scales.

Levels relative to continued competition | $\sigma = 1.5$

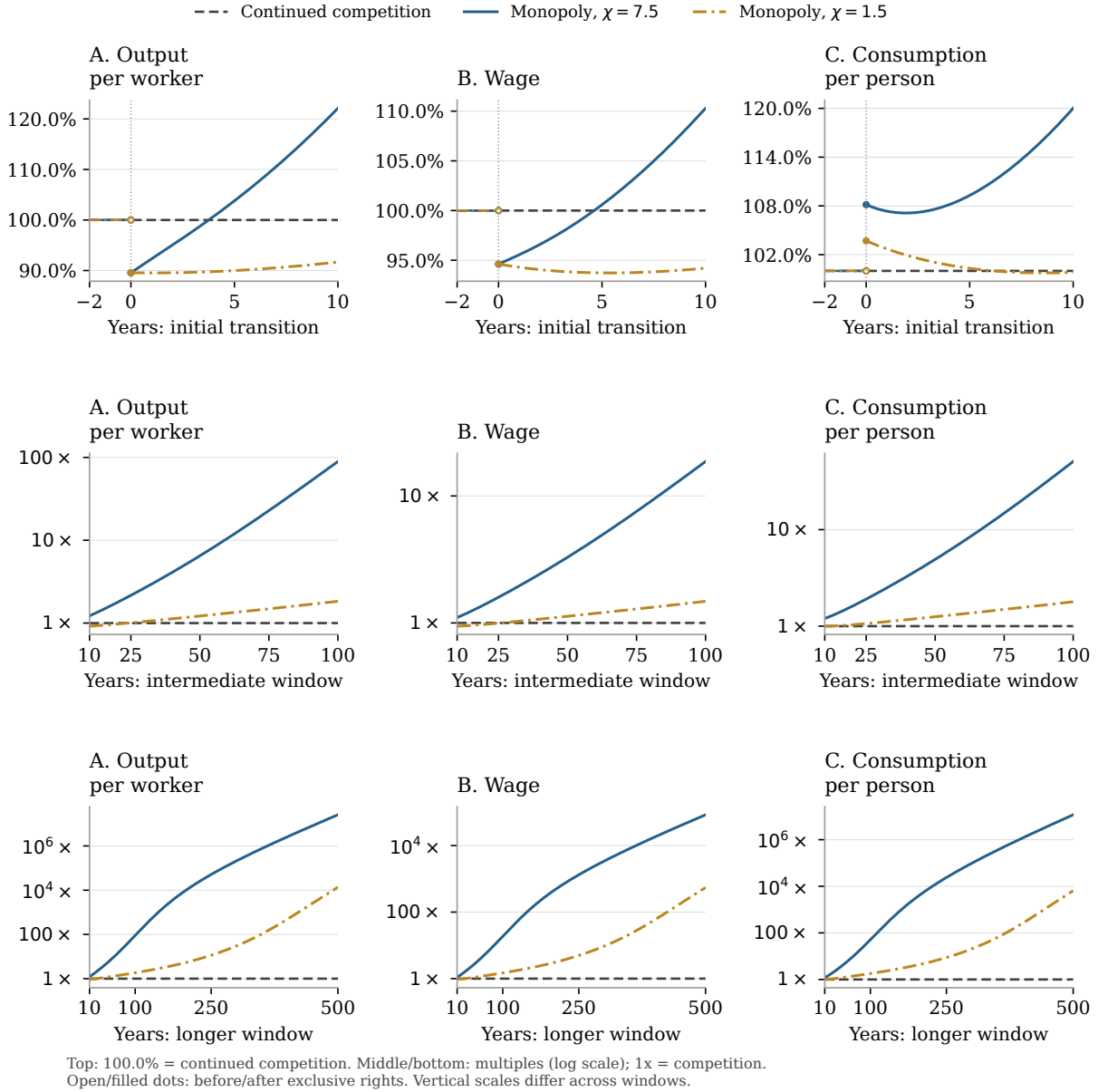


Figure 11: From competition to monopoly with RSI at $\sigma = 1.5$: output per worker, wages, and consumption per person relative to continued competition. The two monopoly paths differ only in χ . Rows cover years $-2-10$, $10-100$, and $10-500$. The upper row uses percentages; the lower rows use multiples on logarithmic scales. Open and filled dots mark levels before and after the grant. Vertical scales differ across windows.

The upper row shows the initial restriction in AI use before efficiency can improve: output falls by 10.4% and wages by 5.4% under either χ . With $\chi = 7.5$, research subsequently raises efficiency enough for output to regain its competitive counterpart around year 3.8 and wages around year 4.6. With $\chi = 1.5$, both remain below their competitive counterparts throughout the first decade.

Consumption responds differently. It rises by 8.2% at the grant under high research productivity and by 3.7% under low productivity, even though output falls. The household chooses less capital accumulation, while reduced inference use releases resources. Research initially absorbs 3.3% of output with $\chi = 7.5$, compared with 0.6% with $\chi = 1.5$.

The middle row makes the slower recovery visible: output and wages under $\chi = 1.5$ regain their competitive counterparts around years 24.0 and 26.0. The lower row shows an expanding gap under both monopoly paths. Research allows efficiency to rise beyond the monopoly threshold, whereas continued competition retains B_0 and remains labor-bottlenecked. Limiting output-per-worker growth, wage growth, and the interest rate are 3.1%, 2.4%, and 7.1% under either χ , as characterized in Proposition 1. The simulated horizon does not establish completed convergence: the interest rate at year 500 is still 7.2% with high research productivity and 8.2% with low research productivity.

The growth advantage of monopoly in this exercise is not general. As Corollary 1 shows, initial AI efficiency can exceed the competitive threshold while even its upper bound remains below the monopoly threshold. Competition then sustains AI-dominated growth, whereas monopoly-led research produces more efficient AI but leaves the economy labor-bottlenecked. I illustrate this opposite outcome in the [supplementary simulations available in the digital edition](#).

8 Conclusion

This paper endogenizes investment in AI research, which in turn affects the pace of economic growth. Even under RSI, computing requires resources that compete with consumption and accumulation of physical capital. Algorithms and model weights, by contrast, are non-rival: their use in one computation does not reduce their availability for others. Following [Romer \(1990\)](#), I use monopoly rents to make research investment profitable.

Paradoxically, when AI efficiency is bounded, permanently faster wage growth requires AI to substitute for labor, not complement it. If the upper bound on AI efficiency is sufficiently high, substitution allows capital and AI services to expand faster than effective labor, raising workers' marginal products and sustaining wage growth above the rate of labor-augmenting technological progress. Output per worker grows even faster, so labor's income share vanishes. In the labor-bottleneck regime, by contrast, labor retains a positive income share, but wages and output per worker eventually grow at the rate of labor-augmenting technological progress. Without the upper bound, continued AI efficiency improvements at unit elasticity can sustain a common growth rate for wages and output per worker above the rate of labor-augmenting technological progress, while preserving a constant positive labor share.

The comparison between monopoly and perfect competition in the supply of AI services shows how research incentives and AI pricing jointly determine long-run growth. Under free access to AI models and their improvements, competitive suppliers do not finance RSI. Nevertheless, sufficiently high initial efficiency sustains AI-dominated growth without monopoly or further research. The

simulations show that, from an initial efficiency too low to sustain AI-dominated growth under competition, monopoly rents can finance improvements that allow an escape from the labor bottleneck. But markups restrict AI use and raise the efficiency required for AI-dominated growth relative to marginal-cost pricing. Competition can therefore sustain AI-dominated growth where monopoly produces more efficient AI but leaves the economy labor-bottlenecked. Even when both economies are AI-dominated, monopoly delivers faster long-run growth only if its efficiency gains offset the reduction in AI use caused by its markup.

My comparison between monopoly and competition raises the question of how to reward AI research without restricting the use of the technology it produces. Monopoly rents encourage research, but monopoly pricing can leave the economy labor-bottlenecked even when competition would sustain AI-dominated growth. Removing those rents after innovation, however, can undermine incentives to invest beforehand. Patent buyouts could address this tension by compensating the developer for opening access to existing AI technology (Kremer, 1998).

The numerical simulations show that early output-per-worker growth can remain moderate and similar across economies approaching different long-run regimes. The question remains whether human labor will be essential enough for wages to keep pace with output per worker. The answer depends on both technology and the market conditions governing AI development and use.

A Proofs

A.1 Developer optimization

Lemma 1 (Optimal AI-service supply). *The developer's operating profit in problem (18) has a unique global maximizer $X > 0$. This optimal supply of AI services is characterized by condition (19) and varies smoothly with K, A, L , and B , for each fixed σ .*

Proof of Lemma 1. Fix $K, A, L, B > 0$ and let $R(X) = p_X(X)X$ denote AI-service revenue. To establish uniqueness, I first show that marginal revenue falls as AI-service supply increases wherever $R'(X) > 0$. The inverse-demand elasticity in (15) and differentiation of (10) give

$$R'(X) = p_X(X)(1 - e_X), \quad \frac{ds_X}{d \ln X} = \left(1 - \frac{1}{\sigma}\right) s_X(1 - s_X).$$

For supply levels with $R'(X) > 0$, differentiation with respect to X gives

$$-\frac{XR''(X)}{p_X(X)} = e_X(1 - e_X) + \left(\alpha - \frac{1}{\sigma}\right) \left(1 - \frac{1}{\sigma}\right) s_X(1 - s_X). \quad (70)$$

Since $R'(X) > 0$ implies $0 < e_X < 1$, the term $e_X(1 - e_X)$ is positive. If $(\alpha - 1/\sigma)(1 - 1/\sigma) \geq 0$, the remaining term is nonnegative, so the right-hand side is positive. If $(\alpha - 1/\sigma)(1 - 1/\sigma) < 0$,

the parameter restrictions imply $0 < \alpha < 1/\sigma < 1$, and

$$-\frac{XR''(X)}{p_X(X)} = \frac{1}{\sigma} \left(1 - \frac{1}{\sigma}\right) (1 - s_X)^2 + \left[\frac{1}{\sigma^2} + 2\alpha - \frac{3\alpha}{\sigma}\right] s_X(1 - s_X) + \alpha(1 - \alpha)s_X^2.$$

The coefficient of $s_X(1 - s_X)$ is positive because

$$\frac{1}{\sigma^2} + 2\alpha - \frac{3\alpha}{\sigma} = \left(\frac{1}{\sigma} - \alpha\right)^2 + \alpha\left(2 - \frac{1}{\sigma} - \alpha\right) > 0.$$

Thus, wherever $R'(X) > 0$, marginal revenue decreases strictly as AI-service supply X increases: $R''(X) < 0$.

As $X \downarrow 0$, revenue is asymptotically proportional to $X^{1-1/\sigma}$ when $\sigma > 1$, to $X^{(1-\alpha)\omega_X}$ when $\sigma = 1$, and to $X^{1-\alpha}$ when $\sigma < 1$. Every exponent lies between zero and one, so $R'(X) \rightarrow \infty$. As $X \rightarrow \infty$, marginal revenue converges to zero from above if $\sigma \geq 1$; if $\sigma < 1$, marginal revenue becomes negative at sufficiently large X . Hence $R'(X) = 1/B$ has exactly one solution. As X increases, operating profit $R(X) - X/B$ rises below this supply level and falls at higher supply levels. The linear cost X/B dominates revenue as $X \rightarrow \infty$, so this supply level is the unique global maximizer. Since $R''(X) < 0$ at the optimum, the implicit-function theorem makes the maximizing service choice smooth in K, A, L, B . $\square$

Lemma 2 (Optimal research expenditure). *Suppose $\bar{B} < \infty$. Consider a feasible research trajectory $\{B_t, M_t\}_{t \geq 0}$ for problem (21), starting from B_0 and satisfying the RSI technology (17). Assume $M_t > 0$ at every date and a finite present value of net profit. This trajectory maximizes the developer's discounted net profit if:*

- (i) *There exists a continuously differentiable shadow-price trajectory $\{q_t\}_{t \geq 0}$ satisfying the research condition (22), the costate equation (23), and the transversality condition (25).*
- (ii) *For every $t \geq 0$ and every $b \in [B_0, \bar{B})$, the following inequality holds. It compares the candidate with all alternative efficiencies reachable from B_0 , not only nearby values:*

$$\begin{aligned} \pi_t(b) - \pi_t(B_t) + \frac{1-\eta}{\eta} M_t \left[\left(\frac{b}{B_t}\right)^{\eta/(1-\eta)} - 1 \right] \\ \leq \frac{\bar{B} - B_t}{B_t} (U_t + M_t) \log \left(\frac{\bar{B} - B_t}{\bar{B} - b} \right), \end{aligned} \tag{71}$$

where $\pi_t(b)$ is operating profit as defined in equation (18).

Condition (71) requires the maximized Hamiltonian, expressed in the transformed efficiency variable used below, to lie below its tangent at the candidate state.

Proof of Lemma 2. Feasible research expenditure is nonnegative and integrable on every finite interval, with a locally absolutely continuous state satisfying the RSI technology almost everywhere. Integrating that technology gives

$$\bar{B} - B_t = (\bar{B} - B_0) \exp \left\{ -\frac{\chi}{\bar{B}} \int_0^t (B_s M_s)^\eta ds \right\} > 0. \quad (72)$$

Local integrability of M and $0 < \eta < 1$ make the exponent finite at every finite date. Thus every feasible trajectory remains in $[B_0, \bar{B})$, where the following transformation is well defined.

To express condition (71) as a supporting-tangent inequality, I transform AI efficiency. Define

$$R_B(B) = -\bar{B} \log \left(1 - \frac{B}{\bar{B}} \right), \quad B(R_B) = \bar{B} \left(1 - e^{-R_B/\bar{B}} \right). \quad (73)$$

The transformation $R_B(B)$ maps $B \in [B_0, \bar{B})$ into $R_B \in [R_{B,0}, \infty)$, where $R_{B,t} = R_B(B_t)$ and $R_{B,0} = R_B(B_0)$. Since $R'_B(B) = 1/\psi(B)$, this transformation removes the factor $\psi(B)$ from the RSI technology (17):

$$\dot{R}_B = \chi[B(R_B)M]^\eta. \quad (74)$$

The corresponding current-value shadow price is $q_{R,t} = q_t \psi(B_t)$, which is positive by the research condition (22). At each date t , hold $q_{R,t}$ at its value along the candidate trajectory and define the maximized Hamiltonian for an alternative state $\tilde{R}_B$:

$$\hat{\mathcal{H}}_t(\tilde{R}_B) \equiv \max_{\tilde{M} \geq 0} \left\{ \pi_t(B(\tilde{R}_B)) - \tilde{M} + q_{R,t} \chi[B(\tilde{R}_B)\tilde{M}]^\eta \right\}.$$

Because $0 < \eta < 1$, the expression being maximized is strictly concave in research expenditure. Using the candidate's research condition (22), the maximizing expenditure at $\tilde{R}_B$ is

$$M_t \left[\frac{B(\tilde{R}_B)}{B_t} \right]^{\eta/(1-\eta)},$$

where M_t is the candidate's research expenditure. Substitution gives

$$\hat{\mathcal{H}}_t(\tilde{R}_B) = \pi_t(B(\tilde{R}_B)) + \frac{1-\eta}{\eta} M_t \left[\frac{B(\tilde{R}_B)}{B_t} \right]^{\eta/(1-\eta)}. \quad (75)$$

Differentiating the maximized Hamiltonian with respect to transformed efficiency R_B and using the envelope condition (20) gives

$$\hat{\mathcal{H}}'_t(R_{B,t}) = \frac{\psi(B_t)}{B_t} (U_t + M_t). \quad (76)$$

Because

$$R_B(b) - R_{B,t} = \bar{B} \log \left(\frac{\bar{B} - B_t}{\bar{B} - b} \right),$$

condition (71) is equivalent to the following inequality for every $\tilde{R}_B \geq R_{B,0}$:

$$\hat{\mathcal{H}}_t(\tilde{R}_B) \leq \hat{\mathcal{H}}_t(R_{B,t}) + \hat{\mathcal{H}}'_t(R_{B,t})(\tilde{R}_B - R_{B,t}).$$

To compare discounted profits, I use the costate equation to bound the alternative's discounted profit gain at each date by a total derivative. Integrating this bound will leave only the shadow value of the difference between the terminal states. Let $D_t = \exp(-\int_0^t r_s ds)$ be the discount factor and $\mu_t = D_t q_{R,t}$ the present-value shadow price of the transformed state R_B . The research condition (22) and the RSI technology (17) imply $q_t \eta \dot{B}_t = M_t$. Differentiating $q_{R,t} = q_t \psi(B_t)$ and using the costate equation (23) therefore gives

$$\dot{q}_{R,t} = r_t q_{R,t} - \frac{\psi(B_t)}{B_t}(U_t + M_t), \quad \dot{\mu}_t = -D_t \hat{\mathcal{H}}'_t(R_{B,t}).$$

Consider any feasible alternative $\{\tilde{B}_t, \tilde{M}_t\}_{t \geq 0}$ starting from the same B_0 , and write $\tilde{R}_{B,t} = R_B(\tilde{B}_t)$. Let

$$J_T = \int_0^T D_t [\pi_t(B_t) - M_t] dt$$

denote discounted net profit through date T under the candidate, and define $\tilde{J}_T$ analogously for the alternative. At each date, the alternative's Hamiltonian cannot exceed $\hat{\mathcal{H}}_t(\tilde{R}_{B,t})$, whereas the candidate attains $\hat{\mathcal{H}}_t(R_{B,t})$. Applying condition (71) and the transformed costate equation yields, for almost every t ,

$$\begin{aligned} D_t \left[\pi_t(\tilde{B}_t) - \tilde{M}_t - \pi_t(B_t) + M_t \right] \\ \leq D_t \hat{\mathcal{H}}'_t(R_{B,t})(\tilde{R}_{B,t} - R_{B,t}) - \mu_t(\dot{\tilde{R}}_{B,t} - \dot{R}_{B,t}) \\ = -\frac{d}{dt} \left[\mu_t(\tilde{R}_{B,t} - R_{B,t}) \right]. \end{aligned}$$

Integrating from zero to T and using the common initial state gives

$$\tilde{J}_T - J_T \leq -\mu_T(\tilde{R}_{B,T} - R_{B,T}) \leq \mu_T R_{B,T},$$

where the second inequality uses $\mu_T > 0$ and $\tilde{R}_{B,T} \geq 0$. Since

$$\psi(B_t) R_{B,t} = -(\bar{B} - B_t) \log \left(1 - \frac{B_t}{\bar{B}} \right) \leq B_t,$$

the developer's transversality condition (25) gives

$$0 \leq \mu_T R_{B,T} = D_T q_T \psi(B_T) R_{B,T} \leq D_T q_T B_T \rightarrow 0.$$

The candidate's present value of net profit is finite by assumption. Hence

$$\limsup_{T \rightarrow \infty} \tilde{J}_T \leq \lim_{T \rightarrow \infty} J_T < \infty,$$

so no feasible alternative yields a higher present value of net profit. Global concavity is sufficient for condition (71), but is not required. $\square$

Lemma 3 (Sufficient conditions for Hamiltonian concavity). *Suppose $\bar{B} < \infty$. Consider a feasible research trajectory with $M_t > 0$ satisfying the research condition (22). The following conditions ensure that the maximized Hamiltonian in equation (75) is concave in transformed AI efficiency R_B . Let $X_t^*(b)$ denote the optimal AI-service choice in problem (18) at efficiency b , and let $\varepsilon_{B,t}(b) = \partial \ln X_t^*(b) / \partial \ln b$ denote its elasticity with respect to AI efficiency. Suppose that, for every $t \geq 0$ and $b \in [B_0, \bar{B})$,*

$$\psi(b)[\varepsilon_{B,t}(b) - 2] \leq \frac{b}{\bar{B}}, \quad (77)$$

and

$$\frac{B_0}{\bar{B}} \geq \max \left\{ 0, \frac{2\eta - 1}{\eta} \right\}. \quad (78)$$

The maximized Hamiltonian is therefore concave on $R_B \in [R_B(B_0), \infty)$ at every date, so the inequality (71) required by Lemma 2 holds.

Condition (77) ensures concavity of operating profit in R_B ; condition (78) ensures concavity of the optimized research term. The latter holds automatically if $\eta \leq 1/2$. For any fixed $0 < \eta < 1$, it also holds when B_0 is sufficiently close to $\bar{B}$. These curvature tests are sufficient, not necessary: they can fail even when the Hamiltonian has the required supporting tangent at the candidate state.

Proof of Lemma 3. Fix a date t . Lemma 1 ensures that $X_t^*(b)$ is unique and smooth. Let $s_{X,t}(b)$ and $e_{X,t}(b)$ be the corresponding elasticities defined in equations (10) and (15). Implicit differentiation of the monopoly condition (19) gives

$$\varepsilon_{B,t}(b) \equiv \frac{\partial \ln X_t^*(b)}{\partial \ln b} = \frac{1 - e_{X,t}(b)}{e_{X,t}(b)[1 - e_{X,t}(b)] + (\alpha - \sigma^{-1})(1 - \sigma^{-1})s_{X,t}(b)[1 - s_{X,t}(b)]}. \quad (79)$$

The denominator is strictly positive by equation (70). The envelope condition (20) gives $\pi'_t(b) = X_t^*(b)/b^2$, so

$$\pi''_t(b) = \frac{X_t^*(b)}{b^3} [\varepsilon_{B,t}(b) - 2]. \quad (80)$$

The transformation (73) satisfies $B'(R_B) = \psi(b)$ and $B''(R_B) = -\psi(b)/\bar{B}$, where $b = B(R_B)$. The chain rule therefore gives

$$\frac{d^2 \pi_t(B(R_B))}{dR_B^2} = \frac{X_t^*(b)\psi(b)}{b^3} \left[\psi(b)(\varepsilon_{B,t}(b) - 2) - \frac{b}{\bar{B}} \right]. \quad (81)$$

Condition (77) therefore makes operating profit concave in the transformed state for every $b \in [B_0, \bar{B})$.

The research term in (75) is a positive multiple of $B(R_B)^{\eta/(1-\eta)}$. The second derivative of this power with respect to R_B is

$$\frac{d^2 B(R_B)^{\eta/(1-\eta)}}{dR_B^2} = \frac{\eta}{1-\eta} b^{\eta/(1-\eta)-2} \psi(b) \left[\frac{2\eta-1}{1-\eta} \psi(b) - \frac{b}{\bar{B}} \right]. \quad (82)$$

The bracket is nonpositive exactly when $b/\bar{B} \geq (2\eta-1)/\eta$. Since $b \geq B_0$, condition (78) makes the research term concave as well. The maximized Hamiltonian is therefore concave, so its tangent at the candidate state is a global upper bound. As shown in the proof of Lemma 2, this is condition (71). □

A.2 Equilibria with bounded AI efficiency

Proof of Proposition 1. I first derive candidate long-run allocations from the static optimum in Lemma 1. I then normalize the dynamic equations and use the stable-manifold and inverse-function theorems to construct convergent trajectories from nearby initial stocks. Finally, I verify feasibility, market clearing, both transversality conditions, and global agent optimality. The result is local in initial stocks, but the constructed equilibria exist for all future time.

Step 1. Feasible efficiency and candidate long-run values.

Equation (72) ensures $B_t < \bar{B}$ at every finite date. I impose $B = \bar{B}$ only in the normalized boundary equations used to identify candidate limits.

Write $\theta = (1-\alpha)/\alpha$. For $\sigma \neq 1$, let $\varphi = (\sigma-1)/\sigma$ and define

$$m(s) = 1 - \frac{1-s}{\sigma} - \alpha s, \quad \mathcal{H}_\sigma(s) = \left[(1-\alpha)s m(s) \left(\frac{\omega_X}{s} \right)^{1/\varphi} \right]^\theta.$$

Here $m(s) = 1 - e_X$ is the marginal-revenue factor, not research compute. The static monopoly optimum has $m(s_X) > 0$ by Lemma 1. The CES share, monopoly condition, and production function, equations (28d), (28i), and (28c), imply

$$\frac{X}{AN} = \left[\frac{\omega_L s_X}{\omega_X (1-s_X)} \right]^{1/\varphi}, \quad \frac{U}{Y} = (1-\alpha)s_X m(s_X), \quad (83)$$

$$\frac{Y}{K} = B^\theta \mathcal{H}_\sigma(s_X). \quad (84)$$

The identities $Z = X(\omega_X/s_X)^{1/\varphi}$ and $X = B(1-\alpha)s_X m(s_X)Y$ give the output–capital ratio in (84).

If $0 < \sigma < 1$, positive marginal revenue requires

$$s > s_{\min} \equiv \frac{1-\sigma}{1-\alpha\sigma}.$$

On $(s_{\min}, 1)$, both $m(s)$ and $s^{-1/(\sigma-1)}$ increase strictly with s . Since

$$\mathcal{H}_\sigma(s)^{1/\theta} = (1 - \alpha)\omega_X^{1/\varphi} m(s)s^{-1/(\sigma-1)},$$

the share map $\mathcal{H}_\sigma(s)$ increases from zero to $\mathcal{H}_\sigma(1) = [(1 - \alpha)^2 \omega_X^{1/\varphi}]^\theta$. Thus the equation

$$\frac{\rho + \gamma + \delta}{\alpha} = \bar{B}^\theta \mathcal{H}_\sigma(\bar{s}_X) \quad (85)$$

has exactly one interior solution if and only if $\bar{B} > \mathcal{B}(\sigma)$.

If $\sigma > 1$, $\mathcal{H}_\sigma(s)$ decreases strictly with s on $(0, 1)$. To verify the sign, put $a = \sigma^{-1} - \alpha$, so $m(s) = \varphi + as$. The logarithmic derivative of $\mathcal{H}_\sigma(s)$ with respect to s has the sign of $a(\sigma - 2)s - \varphi$. If $a(\sigma - 2) \leq 0$, then $a(\sigma - 2)s - \varphi < 0$. If $\sigma > 2$ and $a > 0$, $a(\sigma - 2) < (\sigma - 2)/\sigma < \varphi$; if $1 < \sigma < 2$ and $a < 0$, $a(\sigma - 2) < (1 - \sigma^{-1})(2 - \sigma) < \varphi$. The map therefore decreases from infinity to $\mathcal{H}_\sigma(1)$. Equation (85) has exactly one interior solution if and only if $\bar{B} < \mathcal{B}(\sigma)$. Since $\mathcal{R}(B) = \alpha B^\theta \mathcal{H}_\sigma(1) - \delta$, these calculations also derive (35)–(36). The interior-share condition is therefore $\mathcal{R}(\bar{B}) > \rho + \gamma$ for $\sigma < 1$, but $\mathcal{R}(\bar{B}) < \rho + \gamma$ for $\sigma > 1$.

For either interior-share case, let x, y, u, k, c denote X, Y, U, K, C divided by AN , respectively. Construct their candidate limiting values sequentially:

$$\begin{aligned} \bar{x} &= \left[\frac{\omega_L \bar{s}_X}{\omega_X (1 - \bar{s}_X)} \right]^{1/\varphi}, & \bar{u} &= \frac{\bar{x}}{\bar{B}}, \\ \bar{y} &= \frac{\bar{u}}{(1 - \alpha)\bar{s}_X m(\bar{s}_X)}, & \bar{k} &= \frac{\alpha \bar{y}}{\rho + \gamma + \delta}, \\ \bar{c} &= \bar{y} - \bar{u} - (\delta + n + \gamma)\bar{k}, & \bar{r} &= \rho + \gamma. \end{aligned} \quad (86)$$

At $\sigma = 1$, define $\beta = (1 - \alpha)\omega_X$. The exact static solution is

$$y = \left[(\beta^2 B)^\beta k^\alpha \right]^{1/(1-\beta)}, \quad u = \beta^2 y, \quad x = Bu.$$

The equation $y/k = (\rho + \gamma + \delta)/\alpha$ has exactly one positive solution for every finite $\bar{B} > 0$:

$$\bar{k} = \left[(\beta^2 \bar{B})^\beta \left(\frac{\rho + \gamma + \delta}{\alpha} \right)^{\beta-1} \right]^{1/(1-\alpha-\beta)}. \quad (87)$$

Set $\bar{y} = [(\rho + \gamma + \delta)/\alpha]\bar{k}$, $\bar{u} = \beta^2 \bar{y}$, and $\bar{x} = \bar{B} \bar{u}$, with $\bar{c}, \bar{r}$ as in (86).

Consumption positivity does not require an additional restriction. Positive marginal revenue implies $0 < m(\bar{s}_X) < 1$. Since $0 < \bar{s}_X < 1$, in all three cases

$$\frac{\bar{u}}{\bar{y}} = (1 - \alpha)\bar{s}_X m(\bar{s}_X) < 1 - \alpha, \quad \frac{(\delta + n + \gamma)\bar{k}}{\bar{y}} = \frac{\alpha(\delta + n + \gamma)}{\rho + \gamma + \delta} < \alpha.$$

It follows that $\bar{c}/\bar{y} > 0$.

Step 2. The three positive-labor-share cases.

The cases in Proposition 1(i), 1(ii), and 1(iii) share one normalization. Divide capital, consumption, and the AI-efficiency shadow price by effective labor, and multiply the remaining efficiency gap by effective labor:

$$k = \frac{K}{AN}, \quad c = \frac{C}{AN}, \quad d = (\bar{B} - B)AN, \quad \tilde{q} = \frac{q}{AN}, \quad \tau = (AN)^{-1}. \quad (88)$$

Dividing by AN removes effective-labor growth. Multiplying the shrinking gap by AN allows the scaled gap d to converge to a positive value: $\dot{d}/d = n + \gamma - \dot{B}/(\bar{B} - B)$. The variable d measures the scaled gap, not its rate of closure. The constants $\bar{k}, \bar{d}$ are the limiting values of k, d ; they are determined by the model, not additional parameters. Including τ as a state removes explicit calendar time from the normalized equations. Since $\dot{\tau} = -(n + \gamma)\tau$, its limit is zero. These coordinates leave the model unchanged: $K = k/\tau$ and $B = \bar{B} - d\tau$ at every finite date.

Given k, B , solve the unique static monopoly problem for y, x, u , and set $r = \alpha y/k - \delta$. Define $a = \dot{B}/(\bar{B} - B)$, the proportional rate of closure of the AI-efficiency gap. The research condition gives

$$B = \bar{B} - d\tau, \quad M = \left[\tilde{q} d \chi \eta \frac{B^\eta}{\bar{B}} \right]^{1/(1-\eta)}, \quad a = \chi \frac{B^\eta M^\eta}{\bar{B}}.$$

Substituting (28k) and (29) gives the exact dynamic equations

$$\frac{\dot{k}}{\bar{k}} = \frac{y - c - u - M\tau}{k} - \delta - n - \gamma, \quad (89a)$$

$$\frac{\dot{c}}{\bar{c}} = r - \rho - \gamma, \quad (89b)$$

$$\frac{\dot{d}}{\bar{d}} = n + \gamma - a, \quad (89c)$$

$$\frac{\dot{\tilde{q}}}{\bar{\tilde{q}}} = r - \frac{x}{\tilde{q} B^2} - \eta \frac{a d \tau}{B} + a - n - \gamma, \quad (89d)$$

$$\dot{\tau} = -(n + \gamma)\tau. \quad (89e)$$

Complete the candidate long-run allocation by setting

$$\bar{M} = \left[\frac{(n + \gamma) \bar{B}^{1-\eta}}{\chi} \right]^{1/\eta}, \quad \bar{\tilde{q}} = \frac{\bar{x}}{\bar{r} \bar{B}^2}, \quad \bar{d} = \frac{\bar{M}}{\eta(n + \gamma) \bar{\tilde{q}}}. \quad (90)$$

All are positive. Direct substitution shows that $(\bar{k}, \bar{c}, \bar{d}, \bar{\tilde{q}}, 0)$ is a fixed point of (89). This boundary point is a device for constructing paths with $\tau > 0$ and $B < \bar{B}$, not an admissible date-zero state itself.

I next verify saddle-path stability to determine whether initial consumption and the shadow price can be selected so that nearby initial stocks lead to convergence.

Write $\mu = \eta/(1 - \eta)$, $r_k = \partial r / \partial \log k$, and $a_k = \partial[(y - u)/k] / \partial \log k + \bar{c}/\bar{k}$, evaluated at the

fixed point. In $(\log k, \log c, \log d, \log \tilde{q}, \tau)$ coordinates the Jacobian has two diagonal blocks: one for capital and consumption, and one for the scaled efficiency gap and normalized shadow price,

$$J_R = \begin{pmatrix} a_k & -\bar{c}/\bar{k} \\ r_k & 0 \end{pmatrix}, \quad J_B = \begin{pmatrix} -(n+\gamma)\mu & -(n+\gamma)\mu \\ (n+\gamma)\mu & \bar{r} + (n+\gamma)\mu \end{pmatrix}, \quad (91)$$

and the additional eigenvalue $-(n+\gamma)$. There can be forcing from τ to both blocks and from (k, c) to $(d, \tilde{q})$, but not in the reverse directions at the fixed point.

To check $r_k < 0$, let $D(s_X)$ be the positive curvature expression in (70). Implicit differentiation of the monopoly condition gives

$$\frac{\partial \log X}{\partial \log K} = \frac{\alpha(1 - e_X)}{D(s_X)}, \quad r_k = (r + \delta)(1 - \alpha) \left[\frac{\alpha s_X(1 - e_X)}{D(s_X)} - 1 \right].$$

Moreover,

$$D(s_X) - \alpha s_X(1 - e_X) = (1 - s_X) \left\{ \frac{1 - e_X}{\sigma} + \left(\alpha - \frac{1}{\sigma} \right) \left(1 - \frac{1}{\sigma} \right) s_X \right\} > 0.$$

The expression in braces is positive when $1/\sigma \geq 1$ or $1/\sigma \leq \alpha$. For $\alpha < 1/\sigma < 1$, the expression is linear in s_X , with positive values $(1/\sigma)(1 - 1/\sigma)$ at $s_X = 0$ and $1/\sigma^2 + \alpha(1 - 2/\sigma)$ at $s_X = 1$. The value at $s_X = 1$ is positive if $1/\sigma \leq 1/2$; otherwise $\alpha < 1/\sigma$ implies it exceeds $(1/\sigma)(1 - 1/\sigma) > 0$. Thus $r_k < 0$ in every interior-share case.

Both blocks have negative determinants:

$$\det J_R = (\bar{c}/\bar{k})r_k < 0, \quad \det J_B = -(n+\gamma)\mu\bar{r} < 0.$$

There are therefore three stable and two unstable eigenvalues. The stable direction in J_R has a nonzero k component, and the stable direction in J_B has a nonzero d component. Counting stable directions is not enough: their projection must also allow the three predetermined coordinates to vary independently. To verify this, consider a vector in the stable subspace whose state components (k, d, τ) are all zero. Its linearized τ path is zero. The unforced capital–consumption block must then follow its stable direction; its zero initial k component forces its c component to be zero as well. With that forcing absent, the same argument in the research block forces $\tilde{q} = 0$. The projection of the three-dimensional stable subspace onto the three predetermined coordinates is therefore injective and hence invertible. This argument also covers repeated stable eigenvalues.

Lemma 1 makes the static solution smooth. The normalized vector field extends smoothly through $\tau = 0$, since $B = \bar{B} - d\tau$ remains positive nearby. The stable-manifold theorem therefore gives a local manifold whose trajectories converge to the fixed point. The invertible projection just established lets the inverse-function theorem express this manifold as a local graph selecting $(c, \tilde{q})$ from (k, d, τ) ; see Dyatlov (2018). Choose a small neighborhood of the full fixed point and then a smaller initial-state neighborhood $\mathcal{U}_L$ of $(\bar{k}, \bar{d}, 0)$. The size of this neighborhood depends on the

parameters and the case under consideration. In original initial stocks the restriction is

$$(k_0, d_0, \tau_0) = \left(\frac{K_0}{A_0 N_0}, (\bar{B} - B_0) A_0 N_0, \frac{1}{A_0 N_0} \right) \in \mathcal{U}_L. \quad (92)$$

Closeness is required in all three coordinates; B_0 close to $\bar{B}$ alone is not sufficient. Every physical initial state satisfying (92) has a solution on this graph that stays in the full neighborhood for all future time and converges to the fixed point. It is unique among solutions with those properties. Positivity of $\tau, d, c, \tilde{q}, k$ and the original RSI technology ensure $0 < B_t < \bar{B}$ and $M_t > 0$ at every finite date. Reconstruction gives the original initial stocks, the equilibrium equations, and the limiting rates in (37)–(39). Indeed, the normalized vector field tends to zero at its fixed point, so the time derivatives of the smooth ratios determined by the static solution tend to zero as well. This step, not convergence of ratios alone, establishes convergence of the growth rates. Optimality and transversality are verified below.

Step 3. The AI-dominated case.

Let $\sigma > 1$ and $\bar{B} > \mathcal{B}(\sigma)$. Here $\varphi = (\sigma - 1)/\sigma \in (0, 1)$. I seek an equilibrium in which capital grows faster than effective labor. Since $K/(AN)$ then diverges, I use capital rather than effective labor to normalize consumption, the efficiency gap, and the shadow price. The two sets of coordinates are:

	Labor bottleneck (Step 2)	AI-dominated (Step 3)
Relative stocks	$k = K/(AN)$	$h = (AN/K)^\varphi$
Consumption	$c = C/(AN)$	$c = C/K$
Efficiency gap	$d = (\bar{B} - B)AN$	$d = (\bar{B} - B)K$
Shadow price	$\tilde{q} = q/(AN)$	$v = q/K$

In both cases $\tau = (AN)^{-1}$. The exponent φ makes h enter the CES sum linearly, so the zero-labor-share limit can be studied at $h = 0$. Within this construction, y, x, u denote $Y/K, X/K, U/K$, respectively. The constant $\bar{d}$ now denotes the limit of $(\bar{B} - B)K$, not of $(\bar{B} - B)AN$; no original state or control is redefined. Then

$$B = \bar{B} - d\tau h^{1/\varphi}, \quad M = [vd\chi\eta B^\eta/\bar{B}]^{1/(1-\eta)}, \quad a = \chi B^\eta M^\eta/\bar{B}, \quad r = \alpha y - \delta.$$

Capital's growth rate is $g_K = y - c - u - M\tau h^{1/\varphi} - \delta$. The exact equations are

$$\dot{h} = \varphi h(n + \gamma - g_K), \quad (93a)$$

$$\dot{d}/d = g_K - a, \quad (93b)$$

$$\dot{c}/c = n + r - \rho - g_K, \quad (93c)$$

$$\dot{v}/v = r - x/(vB^2) - \eta ad\tau h^{1/\varphi}/B + a - g_K, \quad (93d)$$

$$\dot{\tau} = -(n + \gamma)\tau. \quad (93e)$$

The CES term divided by capital is $[\omega_L h + \omega_X x^\varphi]^{1/\varphi}$. At $h = 0$ the monopoly equation has the strictly positive curvature $D(1) = \alpha(1 - \alpha)$. The static monopoly solution is therefore smooth in (h, B) near $(0, \bar{B})$. To apply the stable-manifold theorem at $h = 0$, I extend the vector field to an open neighborhood of this boundary. For the remaining terms, replacing $h^{1/\varphi}$ by $\max\{h, 0\}^{1/\varphi}$ outside the physical domain gives a C^1 extension because $1/\varphi > 1$. This meets the regularity requirement of the C^1 stable-manifold theorem; see [Feldman \(2021, p. 7\)](#). The extension is only a mathematical device and does not change any equation on $h \geq 0$.

The limiting interest and aggregate growth rates are

$$\bar{r} = \mathcal{R}(\bar{B}), \quad \bar{g} = n + \bar{r} - \rho > n + \gamma. \quad (94)$$

At $h = \tau = 0$, the terminal constants are

$$\begin{aligned} \bar{y} &= \frac{\mathcal{R}(\bar{B}) + \delta}{\alpha}, & \bar{u} &= (1 - \alpha)^2 \bar{y}, & \bar{x} &= \bar{B} \bar{u}, \\ \bar{r} &= \mathcal{R}(\bar{B}), & \bar{g} &= n + \bar{r} - \rho, & \bar{c} &= \alpha(1 - \alpha) \bar{y} + \rho - n > 0, \\ \bar{v} &= \bar{x} / (\bar{r} \bar{B}^2), & \bar{M} &= [\bar{g} \bar{B}^{1-\eta} / \chi]^{1/\eta}, & \bar{d} &= \bar{M} / (\eta \bar{g} \bar{v}). \end{aligned} \quad (95)$$

The threshold implies $\bar{g} > n + \gamma > 0$ and $\bar{r} > \rho + \gamma > 0$, so all constants are positive. Substitution verifies the fixed point $(0, \bar{d}, \bar{c}, \bar{v}, 0)$.

The h and τ eigenvalues are $\varphi(n + \gamma - \bar{g}) < 0$ and $-(n + \gamma) < 0$. The first equals $\varphi[\rho + \gamma - \mathcal{R}(\bar{B})]$. It becomes zero at $\mathcal{R}(\bar{B}) = \rho + \gamma$, which is why this stable-manifold argument does not cover the threshold. The consumption-ratio eigenvalue is $\bar{c} > 0$. The remaining block in $(\log d, \log v)$ is

$$\begin{pmatrix} -\bar{g}\mu & -\bar{g}\mu \\ \bar{g}\mu & \bar{r} + \bar{g}\mu \end{pmatrix}, \quad \det = -\bar{g}\mu\bar{r} < 0.$$

The linearized system is triangular in the order $(h, \tau), c, (d, v)$. A stable vector with zero state components (h, d, τ) has zero h, τ paths; its consumption component must be zero because the unforced consumption equation is unstable. The research block then has zero d component and hence zero v component, as in the previous construction. Thus the three-dimensional stable subspace projects invertibly onto (h, d, τ) . The same graph argument gives a neighborhood $\mathcal{U}_A$ of $(0, \bar{d}, 0)$ such that the initial-state condition is

$$\left[\left(\frac{A_0 N_0}{K_0} \right)^\varphi, (\bar{B} - B_0) K_0, \frac{1}{A_0 N_0} \right] \in \mathcal{U}_A. \quad (96)$$

The graph determines a convergent trajectory for every admissible initial state in this neighborhood, with a unique nearby convergent choice of C_0, q_0 and positive quantities at every finite date.

The normalized vector field tends to zero, so the time derivatives of the smooth positive ratios Y/K , X/K , and C/K tend to zero. Consequently $g_Y, g_X, g_C, g_K \rightarrow \bar{g}$ and $s_X \rightarrow 1$. Log differentiation of the CES share in equation (28d) and the wage equation (28g) gives the exact

identity

$$g_w = g_Y - n + \frac{\sigma - 1}{\sigma} s_X(n + \gamma - g_X).$$

Taking limits yields

$$g_w \rightarrow \bar{g} - n + \frac{\sigma - 1}{\sigma} (n + \gamma - \bar{g}) = \gamma + \frac{\bar{r} - \rho - \gamma}{\sigma}.$$

This proves the growth and distribution limits in Proposition 1(iv).

Step 4. Optimality and transversality.

In either construction let $\bar{g}$ denote the limiting aggregate growth rate: $n + \gamma$ in the positive labor-share case and $n + \bar{r} - \rho$ in the AI-dominated case. The constructions imply

$$a \rightarrow \bar{g}, \quad M \rightarrow \left[\frac{\bar{g} \bar{B}^{1-\eta}}{\chi} \right]^{1/\eta} > 0, \quad g_q \rightarrow \bar{g}, \quad g_B \rightarrow 0, \quad M/Y \rightarrow 0.$$

Both transversality expressions in Definition 1 have limiting logarithmic growth

$$-\rho + n + g_K - g_C \rightarrow n - \rho < 0, \quad -r + g_q + g_B \rightarrow \bar{g} - \bar{r} = n - \rho < 0. \quad (97)$$

Both transversality expressions therefore converge to zero, rather than merely remaining bounded.

Household utility is finite because $\log(C/N)$ grows at most linearly and $\rho > n$, as imposed in the household problem. Log utility is concave, and the household budget is linear at given prices and distributed profits. More explicitly, let $\lambda_t = e^{-\rho t} N_t / C_t$, so $\dot{\lambda}_t = -r_t \lambda_t$ by (5). Concavity of log utility and equal initial capital give, for any admissible alternative $(\tilde{C}, \tilde{K})$,

$$\int_0^T e^{-\rho t} N_t \log \frac{\tilde{C}_t}{C_t} dt \leq \int_0^T \lambda_t (\tilde{C}_t - C_t) dt = -\lambda_T (\tilde{K}_T - K_T) \leq \lambda_T K_T \rightarrow 0.$$

Here $\tilde{K}_T \geq 0$ and the last limit is the household TVC.

For developer optimality, let $S_t = A_t N_t$ in the positive-share construction and $S_t = K_t$ in the AI-dominated construction, and let $D_t = \exp(-\int_0^t r_s ds)$. Homogeneity of the static problem and compactness of the normalized state neighborhood give a constant C_π such that $0 \leq \pi_t(b) \leq C_\pi S_t$ for every date and every $b \in [B_0, \bar{B}]$. Since $d \log(D_t S_t) / dt \rightarrow \bar{g} - \bar{r} = n - \rho < 0$, $\int_0^\infty D_t S_t dt < \infty$. Thus discounted operating profit is uniformly bounded above across all alternative research policies. The candidate's present value of net profit is finite, because its research compute is bounded and $r_t \rightarrow \bar{r} > 0$. An alternative with infinite discounted research expenditure has payoff $-\infty$ and cannot improve on it.

It remains to verify the curvature conditions of Lemma 3, (77) and (78) for every date and every alternative AI-efficiency level in $[B_0, \bar{B})$, not just on the selected path. First fix a compact normalized-state neighborhood and an interval of alternative AI-efficiency levels $[b_*, \bar{B}]$, with $0 < b_* < \bar{B}$. In the positive labor-share construction, $K/(AN)$ stays in a compact neighborhood of $\bar{k} > 0$. Smoothness of the static monopoly solution bounds $\varepsilon_{B,t}(b)$ uniformly for these states and for AI-efficiency levels near $\bar{B}$. In the AI-dominated construction, h stays near zero, and the same

uniform bound follows from the smooth static limit $\varepsilon_{B,t}(b) \rightarrow 1/\alpha$ as $h \rightarrow 0$. Thus in either case there is a finite constant E with $\varepsilon_{B,t}(b) - 2 \leq E$ throughout that state domain. This bound is chosen before shrinking the initial-state neighborhood. Because $\mu = \eta/(1 - \eta)$ is finite for each $0 < \eta < 1$, choose $b_{**} \in [b_*, \bar{B})$ sufficiently close to $\bar{B}$ that

$$(1 - b_{**}/\bar{B}) \max\{E, \mu - 1, 0\} < b_{**}/\bar{B}.$$

Shrink the open neighborhood $\mathcal{U}_L$ or $\mathcal{U}_A$ so that $B_0 > b_{**}$. This is possible because $B_0 = \bar{B} - d_0\tau_0$ in the first construction and $B_0 = \bar{B} - d_0\tau_0 h_0^{1/\varphi}$ in the second. Every feasible research policy has $b \geq B_0$, so the two curvature conditions hold throughout the entire alternative-state domain at every date. Lemma 3 therefore gives the support inequality (71). Lemma 2 and the verified TVC then establish global developer optimality without imposing $\eta \leq 1/2$. The curvature conditions follow from the primitives and the chosen initial-state neighborhood.

Final-good optimality follows from concavity and constant returns. The pointwise monopoly optimum is global by Lemma 1. All quantities remain finite at every finite date, and market clearing and the original initial conditions hold by reconstruction.

For $\sigma > 1$, at $\bar{B} = \mathcal{B}(\sigma)$ the positive-labor-share limiting point reaches the boundary and a stable eigenvalue in the boundary normalization becomes zero; this proof does not apply. For $\sigma < 1$ and $\bar{B} \leq \mathcal{B}(\sigma)$, it does not rule out equilibria with different long-run ratios.

Step 5. Initial-state sets and the four conclusions.

The preimage of each open neighborhood under its continuous initial-state map, restricted to $A_0, N_0, K_0 > 0$ and $0 < B_0 < \bar{B}$, is open in the original initial stocks. It is nonempty. In the positive-labor-share cases, choose k_0, d_0 close to $\bar{k}, \bar{d}$ and $\tau_0 > 0$ sufficiently small. Then $A_0 N_0 = 1/\tau_0$, $K_0 = k_0/\tau_0$, and $B_0 = \bar{B} - d_0\tau_0$ are admissible. In the AI-dominated case, choose d_0 near $\bar{d}$ and $h_0, \tau_0 > 0$ sufficiently small. The inverse relations are $A_0 N_0 = 1/\tau_0$, $K_0 = 1/(\tau_0 h_0^{1/\varphi})$, and $B_0 = \bar{B} - d_0\tau_0 h_0^{1/\varphi}$. Any positive factorization of $A_0 N_0$ supplies A_0, N_0 . These choices also satisfy the strict efficiency restriction in Step 4.

For each initial state in the resulting set, the local graph selects $C_0, q_0 > 0$. These jump values are unique among the trajectories that remain near the indicated normalized limit and converge to it. The argument does not assert global uniqueness.

Steps 1–2 establish the first three cases, and Step 3 establishes the AI-dominated case. Together with the optimality and transversality checks in Step 4, these constructions satisfy Definition 1. $\square$

Proof of Proposition 2. With all parameters other than σ fixed, define

$$H(\sigma) = \mathcal{R}(\bar{B}) + \delta, \quad \kappa = -\frac{1-\alpha}{\alpha} \log \omega_X > 0, \quad D = \rho + \gamma + \delta.$$

Equation (35) gives

$$H'(\sigma) = \frac{\kappa H(\sigma)}{(\sigma-1)^2} > 0,$$

$$H_\infty \equiv \lim_{\sigma \rightarrow \infty} H(\sigma) = \alpha [(1-\alpha)^2 \bar{B} \omega_X]^{(1-\alpha)/\alpha} < \infty.$$

The AI-dominated domain is $H(\sigma) > D$. Since this domain is nonempty and H is strictly increasing, $H_\infty > D$. By equation (41), limiting wage growth is $\bar{g}_w(\sigma) = \gamma + [H(\sigma) - D]/\sigma$, so

$$\bar{g}'_w(\sigma) = \frac{D - H(\sigma) [1 - \kappa\sigma/(\sigma-1)^2]}{\sigma^2}.$$

The factor $1 - \kappa\sigma/(\sigma-1)^2$ increases strictly from $-\infty$ to 1 as σ increases from 1 to infinity. While this factor is nonpositive, $\bar{g}'_w(\sigma) > 0$. Once this factor is positive, $H(\sigma)[1 - \kappa\sigma/(\sigma-1)^2]$ increases strictly with σ from zero to $H_\infty > D$. The product therefore equals D at exactly one finite $\sigma^* > 1$. At that point the factor is less than one, so $H(\sigma^*) > D$: the critical point lies inside the AI-dominated domain. Limiting wage growth increases with σ below σ^* and decreases above it within that domain. Hence σ^* is its unique maximizer. All other regimes characterized in Proposition 1 have long-run wage growth equal to γ , whereas $\bar{g}_w(\sigma^*) > \gamma$. Hence σ^* uniquely maximizes long-run wage growth across these regimes. $\square$

Proposition 9 (An AI bottleneck under complementarity). *Suppose $\sigma < 1$ and $\bar{B} < \mathcal{B}(\sigma)$. In any equilibrium, long-run average output-per-worker growth is bounded strictly below the rate of labor-augmenting technological progress:*

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \log \left(\frac{Y_t/N_t}{Y_0/N_0} \right) \leq \mathcal{R}(\bar{B}) - \rho < \gamma.$$

AI production services become scarce relative to effective labor, and labor's income share vanishes:

$$\frac{X}{AL} \rightarrow 0, \quad s_X \rightarrow 1, \quad \frac{wL}{Y} \rightarrow 0.$$

Proof of Proposition 9. Consider any equilibrium with $0 < \sigma < 1$ and $0 < B_0 < \bar{B} < \mathcal{B}(\sigma)$. The increasing share map in equation (84), established in Step 1 of the proof of Proposition 1, and $B_t \leq \bar{B}$ imply

$$r_t \leq \mathcal{R}(B_t) \leq \mathcal{R}(\bar{B}) < \rho + \gamma, \quad \frac{Y_t}{K_t} \leq \frac{\mathcal{R}(\bar{B}) + \delta}{\alpha}. \quad (98)$$

The present value of the developer's net profit from any date onward is nonnegative: the developer

can stop research and earn nonnegative operating profit. Wage income is also nonnegative. Let $D_{t,s} = \exp(-\int_t^s r_u du)$. Integrating the household budget (4), using the household transversality condition (6) and the Euler equation (5), gives

$$\frac{C_t}{\rho - n} = K_t + \int_t^\infty D_{t,s}(w_s N_s + \Pi_s) ds.$$

The integral is nonnegative by the preceding argument, yielding the first inequality below. Integrating the Euler equation with the return bound gives the second:

$$C_t \geq (\rho - n)K_t, \quad \frac{C_t}{N_t} \leq \frac{C_0}{N_0} e^{[\mathcal{R}(\bar{B}) - \rho]t}. \quad (99)$$

Together with the output–capital bound in (98), these consumption inequalities yield

$$\frac{Y_t}{N_t} \leq \frac{\mathcal{R}(\bar{B}) + \delta}{\alpha(\rho - n)} \frac{C_0}{N_0} e^{[\mathcal{R}(\bar{B}) - \rho]t}, \quad \limsup_{t \rightarrow \infty} \frac{1}{t} \log \left(\frac{Y_t/N_t}{Y_0/N_0} \right) \leq \mathcal{R}(\bar{B}) - \rho < \gamma. \quad (100)$$

Consequently $Y/(AL) \rightarrow 0$. Since equation (32) gives $X/Y \leq \bar{B}(1 - \alpha)^2$ under complementarity, $X/(AL) \rightarrow 0$: AI production services become scarce relative to effective labor. Equation (10) then implies $s_X \rightarrow 1$ and hence $wL/Y \rightarrow 0$. The growth bound concerns the long-run average, not convergence of the instantaneous growth rate. Neither convergence of C/Y nor $B_t \rightarrow \bar{B}$ is assumed. The growth bound and vanishing labor share are necessary properties of any equilibrium in this region, not a proof of equilibrium existence. $\square$

A.3 Competitive equilibrium and comparison with monopoly

Proof of Proposition 3. Free entry and the common unit cost $1/B$ imply $p_X = 1/B$. Because improvements are immediately available to every supplier and research receives no separate payment, a supplier's net profit is $-M$. Its optimal research expenditure is zero. The RSI technology (17) then gives $B = B_0$, establishing equation (45). The remaining argument constructs and verifies the allocation at this fixed efficiency.

Net production and normalized dynamics. Write $k = K/(AL)$, $c = C/(AL)$, $u = U/(AL)$, and $y = Y/(AL)$. At $p_X = 1/B_0$, final-good firms' demand condition (13) implies $\partial y/\partial u = 1$. Their choice of AI services therefore maximizes output net of compute at a given k . For the proof, denote this net output per unit of effective labor by

$$f(k) = \max_{u \geq 0} \left\{ k^\alpha \left[\omega_L + \omega_X (B_0 u)^{(\sigma-1)/\sigma} \right]^{\sigma(1-\alpha)/(\sigma-1)} - u \right\}. \quad (101)$$

At each fixed $k > 0$, the marginal product of compute, $\partial y/\partial u$, decreases strictly from infinity to zero as u rises from zero to infinity. The derivative of net output with respect to u is $\partial y/\partial u - 1$, so there is a unique positive maximizer. Its first-order condition is $u = (1 - \alpha)s_X y$. Differentiating

this condition with respect to k , using (15), gives

$$\frac{d \log u}{d \log k} = \frac{\alpha}{e_X}, \quad f'(k) = \alpha \frac{y}{k}, \quad \frac{k f''(k)}{f'(k)} = -\frac{(1-\alpha)(1-s_X)}{\sigma e_X} < 0.$$

Thus f is smooth, increasing, and strictly concave, with $f(0) = 0$ and $f'(0+) = \infty$. As $k \rightarrow \infty$, the optimal u tends to infinity and $s_X \rightarrow 1$. The pricing condition gives $X/Y \rightarrow B_0(1-\alpha)$, while (33) gives $Z/X \rightarrow \omega_X^{\sigma/(\sigma-1)}$. It follows that

$$\lim_{k \rightarrow \infty} f'(k) = \lim_{k \rightarrow \infty} \frac{f(k)}{k} = \mathcal{R}^{\text{mc}}(B_0) + \delta, \quad f(k) - k f'(k) = (1-\alpha)(1-s_X)y > 0. \quad (102)$$

The limit of $f'(k)$ also yields equation (44).

Set $d = \rho - n > 0$ and $G(k) = f(k) - (\delta + n + \gamma)k$. The household's objective, up to a positive factor and an additive constant, is $\int_0^\infty e^{-dt} \log c_t dt$. The resource constraint (26) with $M = 0$, the Euler equation (5), and the transversality condition (6) become

$$\dot{k} = G(k) - c, \quad \frac{\dot{c}}{c} = G'(k) - d, \quad \lim_{t \rightarrow \infty} e^{-dt} \frac{k_t}{c_t} = 0. \quad (103)$$

An identity useful for constructing the trajectory is

$$\frac{d}{dt} \log \frac{c}{k} = \frac{c}{k} - d - \frac{f(k) - k f'(k)}{k}. \quad (104)$$

Efficiency below the threshold. If $\mathcal{R}^{\text{mc}}(B_0) < \rho + \gamma$, there is a unique $\bar{k} > 0$ satisfying $G'(\bar{k}) = d$. The stationary consumption level is $\bar{c} = G(\bar{k}) > d\bar{k} > 0$, by (102). The Jacobian at this point is

$$\begin{pmatrix} d & -1 \\ \bar{c} f''(\bar{k}) & 0 \end{pmatrix},$$

whose determinant is negative. The stable curve through $(\bar{k}, \bar{c})$ has positive slope, with $\dot{k}, \dot{c} > 0$ on its left branch and $\dot{k}, \dot{c} < 0$ on its right branch.

Local convergence is not enough to establish existence from every initial capital stock. I therefore show that the two branches of this stable curve together cover every $k > 0$. On the left branch, $dk < c < G(k)$; on the right, $c > \max\{dk, G(k)\}$. To verify these restrictions, at $c = dk$ equation (104) points toward $c/k < d$, from which the path cannot return to $\bar{c}/\bar{k} > d$. At $c = G(k)$, the time derivative of $\dot{k} = G(k) - c$ is $-\dot{c}$. Below $\bar{k}$, crossing this curve makes capital decrease while consumption increases, precluding convergence to $\bar{k}$. Above $\bar{k}$, it makes capital increase while consumption decreases, again precluding convergence. The stable branches cannot make either crossing.

Continuing the left branch backward therefore takes k to zero: otherwise its bounded, monotone coordinates would approach a positive stationary point different from $(\bar{k}, \bar{c})$, which does not exist. The right branch extends to arbitrarily large k . It cannot approach another finite stationary point

or diverge in consumption at a finite positive k : along this branch $dc/dk = c(G'(k) - d)/(G(k) - c)$ remains bounded when c becomes large on a compact interval of k . For each $k_0 > 0$, the stable curve consequently supplies a positive trajectory converging to $(\bar{k}, \bar{c})$ and satisfying the transversality condition in (103).

Efficiency at or above the threshold. If $\mathcal{R}^{\text{mc}}(B_0) \geq \rho + \gamma$, strict concavity implies $G'(k) > d$ and $G(k) > dk$ at every finite $k > 0$. For any $k_0 > 0$, consider initial consumption in $[dk_0, G(k_0)]$ and the region

$$k > 0, \quad dk < c < G(k).$$

I choose initial consumption so that the path stays between these boundaries. At the lower boundary, (104) points strictly out of the region: consumption falls relative to capital. At the upper boundary, $\dot{k} = 0$ and $\dot{c} > 0$, so consumption rises above $G(k)$. Inside, both k and c increase. There is no finite-time escape to infinity because concavity bounds $G(k)$ above by an affine function.

The endpoint choices of initial consumption leave through opposite boundaries. A small change in initial consumption preserves a finite-time exit through either boundary, by continuous dependence of the solution on its initial value. Thus the choices that first exit through the lower boundary and those that first exit through the upper boundary form two nonempty, disjoint sets that are open within the interval. They cannot cover the whole interval $[dk_0, G(k_0)]$, because an interval is connected. At least one initial consumption choice therefore generates a path that never leaves the region.

Along this trajectory $k \rightarrow \infty$. A finite limit would also bound c and require a positive stationary point, but $G'(k) > d$ excludes one. Moreover, $d < c/k < G(k)/k$, and the upper bound satisfies

$$\frac{G(k)}{k} \rightarrow \mathcal{R}^{\text{mc}}(B_0) - n - \gamma.$$

If $\mathcal{R}^{\text{mc}}(B_0) = \rho + \gamma$, these bounds imply $c/k \rightarrow d$. If $\mathcal{R}^{\text{mc}}(B_0) > \rho + \gamma$, the same limit follows from (104): $(f - kf')/k \rightarrow 0$, and any persistent excess of c/k over d would make c/k increase until it exceeded its upper bound $G(k)/k$. More precisely, for any $\varepsilon > 0$ and sufficiently large t , reaching $c/k \geq d + \varepsilon$ implies $d \log(c/k)/dt \geq \varepsilon/2$ thereafter until that contradiction occurs. Thus

$$\frac{c}{k} \rightarrow d, \quad \frac{\dot{k}}{k} \rightarrow \mathcal{R}^{\text{mc}}(B_0) - \rho - \gamma. \quad (105)$$

The transversality condition holds because $c/k > d$. At $\mathcal{R}^{\text{mc}}(B_0) = \rho + \gamma$, capital per effective worker diverges but its growth rate tends to zero.

Optimality, uniqueness, and decentralization. The preceding constructions satisfy global optimality, not only the Euler equation. Let $\lambda_t = e^{-dt}/c_t$ be discounted marginal utility along a constructed path; then $\dot{\lambda} = -G'(k)\lambda$. For any alternative feasible path $(\tilde{k}, \tilde{c})$ with the same k_0 , concavity gives

$$\int_0^T e^{-dt} (\log \tilde{c} - \log c) dt \leq -\lambda_T (\tilde{k}_T - k_T) \leq \lambda_T k_T \rightarrow 0.$$

The constructed path has finite utility: below the efficiency threshold, c converges to a positive constant; at or above the threshold, c is positive, increasing, and at most exponentially growing. Thus the inequality verifies optimality in the fixed- B_0 allocation problem. Strict concavity of log utility and concavity of the resource constraint imply uniqueness of consumption; the state equation then gives uniqueness of capital. This verification does not allow a planner to improve B : it concerns the allocation with the technology fixed by the private research choices.

The maximizing u in (101) determines $U = (AL)u$. This compute use, the factor prices in (11) and (12), and $p_X = 1/B_0$ decentralize this allocation. In particular, $wN = (AL)[f(k) - kf'(k)]$, so the household budget coincides with the resource constraint. The household's Euler and transversality conditions verify household optimality at these prices. Strict concavity also implies that any other competitive equilibrium would solve the same fixed-technology allocation problem and hence have the same aggregate allocation.

Growth rates and income shares. Below the threshold, convergence to $(\bar{k}, \bar{c})$ gives $r \rightarrow \rho + \gamma$, $g_{Y/N}, g_w \rightarrow \gamma$, and a positive limiting labor share, because optimal u and s_X have finite interior limits. At or above the threshold, $k, u \rightarrow \infty$ and $s_X \rightarrow 1$. Differentiating the static allocation gives

$$\frac{d \log y}{d \log k} = 1 - \frac{(1 - \alpha)(1 - s_X)}{\sigma e_X} \rightarrow 1, \quad \frac{d \log(w/A)}{d \log k} = \frac{\alpha}{\sigma e_X} \rightarrow \frac{1}{\sigma}.$$

Combining these elasticities with (105) proves the growth limits in part (ii). Equation (102) gives the interest-rate limit, and $wL/Y = (1 - \alpha)(1 - s_X) \rightarrow 0$ gives the labor-share limit.

At the boundary $B_0 = \mathcal{B}^{\text{mc}}(\sigma)$, the same calculations give $g_{Y/N}, g_w \rightarrow \gamma$, $r \rightarrow \rho + \gamma$, and $wL/Y \rightarrow 0$. The labor share therefore vanishes without a permanent growth premium. $\square$

Proof of Proposition 4. Fix $\sigma > 1$. The limiting returns in equations (35) and (44), derived above, satisfy

$$\mathcal{R}^{\text{mc}}(B) = \mathcal{R}(B/(1 - \alpha)).$$

Both returns increase strictly with B . Since each threshold is the efficiency at which the corresponding return equals $\rho + \gamma$,

$$\mathcal{R}^{\text{mc}}((1 - \alpha)\mathcal{B}(\sigma)) = \mathcal{R}(\mathcal{B}(\sigma)) = \rho + \gamma.$$

Uniqueness of the competitive efficiency threshold gives $\mathcal{B}^{\text{mc}}(\sigma) = (1 - \alpha)\mathcal{B}(\sigma) < \mathcal{B}(\sigma)$, establishing equation (47). $\square$

Proof of Corollary 1. Proposition 4 ensures that the interval in equation (48) is nonempty. Fix $\bar{B}$ in this interval. The labor-bottleneck construction of Proposition 1(iii) uses initial values satisfying equation (92), with $B_0 = \bar{B} - d_0\tau_0$, d_0 near $\bar{d} > 0$, and $\tau_0 = 1/(A_0N_0) > 0$ arbitrarily small. Since $\bar{B} > \mathcal{B}^{\text{mc}}(\sigma)$, I can restrict this construction to $B_0 > \mathcal{B}^{\text{mc}}(\sigma)$ while preserving a nonempty open set of initial states.

From each state in this set, the constructed monopoly equilibrium has $B \rightarrow \bar{B}$ and $g_{Y/N}, g_w \rightarrow \gamma$. Proposition 3 supplies a competitive equilibrium from the same state. Since $B_0 > \mathcal{B}^{\text{mc}}(\sigma)$, this equilibrium is AI-dominated, with $B = B_0$ and both limiting growth rates strictly above γ . This proves the comparison without requiring monopoly equilibrium existence from arbitrary initial states or global uniqueness under monopoly.

The accompanying comparison of AI services follows from the CES share identity in equation (83) and $L = N$. Under monopoly, s_X converges to an interior value, so $X/(AL)$ converges to a positive constant. Under competition, $s_X \rightarrow 1$; since $\sigma > 1$, the same identity gives $X/(AL) \rightarrow \infty$. The common paths of A and L therefore imply that AI services eventually exceed their monopoly level under competition, without establishing their ordering at every date. $\square$

Proof of Corollary 2. Equations (35) and (44) imply

$$\mathcal{R}(\bar{B}) = \mathcal{R}^{\text{mc}}((1 - \alpha)\bar{B}).$$

Since $\mathcal{R}^{\text{mc}}$ is strictly increasing in AI efficiency, it follows that

$$\mathcal{R}(\bar{B}) \gtrless \mathcal{R}^{\text{mc}}(B_0) \iff (1 - \alpha)\bar{B} \gtrless B_0,$$

with equality on one side if and only if equality holds on the other. Under the two AI-dominated equilibria, these returns are the limiting interest rates. Equations (40), (41), and (46) express limiting output-per-worker and wage growth as strictly increasing functions of the limiting interest rate, with slopes one and $1/\sigma$, respectively. Since the parameters are common to both economies, the same ordering applies to both growth rates, including the equality case. $\square$

A.4 Equilibria without an efficiency bound

Proof of Proposition 5. Dropping the positive AI term from the CES sum gives an upper bound for Z , because the outer exponent $\sigma/(\sigma - 1)$ is negative. This gives the production bounds in (51) and (52). Define $k = K/(AL)$. The resource constraint (29a), nonnegative consumption and compute expenditure, and $L = N$ imply

$$\dot{k} \leq \omega_L^{\sigma(1-\alpha)/(\sigma-1)} k^\alpha - (\delta + n + \gamma)k. \quad (106)$$

Since $\delta + n + \gamma > 0$ and $0 < \alpha < 1$, the right-hand side is negative whenever k exceeds a finite positive level. In particular,

$$k_t \leq \max \left\{ k_0, \left[\frac{\omega_L^{\sigma(1-\alpha)/(\sigma-1)}}{\delta + n + \gamma} \right]^{1/(1-\alpha)} \right\}.$$

Equation (52) then bounds $Y/(AL)$ uniformly. Since $A_t = A_0 e^{\gamma t}$ and $L = N$, Y/N is bounded above by a constant times $e^{\gamma t}$. Taking logarithms, dividing by t , and taking the upper limit gives (53). $\square$

Proof of Proposition 6. I first bound consumption and discounted research spending, then show that capital per unit of effective labor stays away from zero. These bounds establish that research per unit of effective labor vanishes, allowing convergence to the limiting Ramsey allocation and its growth rates.

All auxiliary variables below are confined to this proof. Write $x = X/(AL)$, $k = K/(AL)$, $c = C/(AL)$, $y = Y/(AL)$, $u = U/(AL)$, $m = M/(AL)$, $d = \rho - n > 0$, and $\nu = \delta + n + \gamma$. No convergence of a growth rate or allocation ratio is assumed.

Step 1. Static optimality gives a limiting production function. Put $\varphi = (\sigma - 1)/\sigma < 0$ and $f(x) = [\omega_L + \omega_X x^\varphi]^{(1-\alpha)/\varphi}$, so $y = k^\alpha f(x)$. The monopoly condition (19) requires $e_X = (1 - s_X)/\sigma + \alpha s_X < 1$. Thus

$$s_X > s_\infty := \frac{1 - \sigma}{1 - \alpha\sigma}, \quad 0 < x < x_\infty := \left[\frac{\omega_L s_\infty}{\omega_X (1 - s_\infty)} \right]^{1/\varphi}. \quad (107)$$

Define $f_\infty = f(x_\infty) > 0$. In particular, $y \leq f_\infty k^\alpha$ and $0 < u \leq x_\infty/B$.

The monopoly condition can be written

$$k^\alpha v(x) = 1/B, \quad v(x) = \frac{(1 - \alpha)s_X f(x)(1 - e_X)}{x}.$$

On $(0, x_\infty)$, $v(x)$ decreases strictly with x , from infinity to zero. Indeed,

$$\frac{de_X}{d \log x} = (\alpha - \sigma^{-1})\varphi s_X (1 - s_X) > 0, \quad -\frac{d \log v}{d \log x} = e_X + \frac{1}{1 - e_X} \frac{de_X}{d \log x} > \alpha s_X.$$

Consequently x increases with B and k , while

$$\frac{\partial \log y}{\partial \log k} = \alpha + \frac{\alpha(1 - \alpha)s_X}{e_X + (1 - e_X)^{-1} de_X / d \log x} < 1.$$

Thus y/k decreases with k and increases with B . For k in any compact interval bounded away from zero, as $B \rightarrow \infty$,

$$x \rightarrow x_\infty, \quad y \rightarrow f_\infty k^\alpha, \quad u \rightarrow 0$$

uniformly. These limits concern the monopoly allocation at given k, B , not the equilibrium path. Together with the monotonicity of y/k , they imply that, for any prescribed finite return, sufficiently small k and sufficiently large B make r exceed that return, uniformly over those small values of k .

Step 2. Optimality bounds consumption per unit of effective labor. Proposition 5 bounds k and y above by finite constants $k_{\max}, y_{\max}$. The household Euler equation and $r \geq -\delta$ give $\dot{c}/c \geq -(\delta + \rho + \gamma)$. Integrating the resource constraint over $[t, t+1]$ therefore gives

$$c_t \int_0^1 e^{-(\delta+\rho+\gamma)v} dv \leq \int_t^{t+1} c_s ds \leq k_{\max} + y_{\max}.$$

Hence c has a finite uniform upper bound $c_{\max}$.

Write $D_{t,s} = \exp(-\int_t^s r_v dv)$. Euler implies $\int_t^\infty D_{t,s} C_s ds = C_t/d$. Integrating the household budget (4) and using its TVC (6) shows that the discounted integral of labor income plus developer net profit converges. Since labor income is nonnegative, the discounted integral of developer net profit has a limit that is either finite or minus infinity. The latter is impossible: the developer can stop research and obtain nonnegative operating profits. By continuation optimality, the developer's continuation value $V_t = \int_t^\infty D_{t,s} \Pi_s ds$ is therefore finite and nonnegative. It follows that

$$\frac{C_t}{d} = K_t + V_t + \int_t^\infty D_{t,s} w_s L_s ds, \quad c_t \geq dk_t, \quad \int_t^\infty D_{t,s} w_s L_s ds \leq \frac{C_t}{d}. \quad (108)$$

Here V_t is an auxiliary continuation value, not an additional traded asset or a change to the household budget.

Step 3. Discounted research spending vanishes relative to effective labor. I first bound the present value of future effective labor relative to its current level, without assuming that the interest rate converges. Since $U \leq x_\infty AL/B$, this bound will let me use the developer's optimality conditions to bound discounted research spending. Fix a positive rate a . Step 1 permits a small $k_a > 0$ and a sufficiently late date such that $r - (n + \gamma) \geq a$ whenever $k \leq k_a$. On $[k_a, k_{\max}]$, the wage equation (12) and the static monopoly allocation give a uniform positive lower bound ℓ_a on w/A . Since $r \geq -\delta$, there is a finite constant $H_a > 0$ such that

$$a \leq r - (n + \gamma) + H_a w/A$$

at every sufficiently late date. Multiplication by $D_{t,s} A_s L_s$, integration, and (108) give

$$\int_t^\infty D_{t,s} A_s L_s ds \leq \frac{A_t L_t + H_a C_t/d}{a} \leq J A_t L_t \quad (109)$$

for a fixed finite J . The omitted terminal contribution from discounted effective labor is nonpositive.

The research condition (22) and costate equation (23) imply

$$\frac{d(qB)}{dt} = rqB - U + \frac{1-\eta}{\eta} M. \quad (110)$$

Since B is nondecreasing, $U_s \leq x_\infty A_s L_s / B_t$ for $s \geq t$. Integrating (110) first over finite intervals and then using the developer TVC (25) gives

$$q_t B_t + \frac{1-\eta}{\eta} \int_t^\infty D_{t,s} M_s ds = \int_t^\infty D_{t,s} U_s ds \leq \frac{x_\infty J}{B_t} A_t L_t. \quad (111)$$

The upper bound is finite by (109). Since $q_t B_t \geq 0$, the discounted research integral is finite as well. That integral, divided by $A_t L_t$, tends to zero because $B_t \rightarrow \infty$. This argument requires $0 < \eta < 1$, but neither $\eta \leq 1/2$ nor $\eta < \alpha$.

Step 4. Capital per unit of effective labor stays away from zero. At sufficiently late dates, a low k makes the return high enough for c to grow. The upper bound $c_{\max}$ therefore limits how long k can remain low. I also need to rule out repeated returns to low k at increasingly late dates. Let $F(k) = f_\infty k^\alpha - \nu k$. Choose $\varepsilon > 0$ small enough that F is positive and increasing on $(0, \varepsilon]$. Step 1 also permits a sufficiently late date and $a > 0$ such that $\dot{c}/c = r - \rho - \gamma \geq a$ whenever $k \leq \varepsilon$. For any $j \leq \varepsilon$, the equilibrium path cannot enter $\{k \leq j, c \geq F(j)\}$: there $\dot{k} \leq F(k) - c \leq 0$, whereas c continues to grow at least at rate a , contradicting its upper bound.

If k stayed below ε forever, c would likewise exceed $c_{\max}$. Hence k eventually reaches ε . Suppose arbitrarily late crossings from ε to $\varepsilon/2$ were possible. Let t_0 be the last crossing of ε before the first subsequent crossing t_1 of $\varepsilon/2$. Throughout this interval $k \in [\varepsilon/2, \varepsilon]$ and c increases. Equation (108) gives $c_{t_0} \geq d\varepsilon$, so

$$t_1 - t_0 \leq T_\varepsilon := a^{-1} \log\left(\frac{c_{\max}}{d\varepsilon}\right) < \infty.$$

At t_1 , the path cannot lie in the region excluded above, so $c_{t_1} < F(\varepsilon/2)$. Thus $c_s < F(\varepsilon/2)$ throughout the crossing. The uniform convergence in Step 1, applied on the crossing interval, yields numbers $\epsilon_{t_0} \rightarrow 0$ such that

$$\dot{k}_s = y_s - u_s - \nu k_s - c_s - m_s \geq -\epsilon_{t_0} - m_s.$$

On this interval r is bounded above by a fixed positive R_ε . Since $n + \gamma > 0$, equation (111) then implies

$$\frac{\varepsilon}{2} \leq T_\varepsilon \epsilon_{t_0} + \int_{t_0}^{t_1} m_s ds \leq T_\varepsilon \epsilon_{t_0} + e^{R_\varepsilon T_\varepsilon} \frac{\eta x_\infty J}{(1-\eta)B_{t_0}} \rightarrow 0,$$

The fall in k would be $\varepsilon/2$, but the maximum fall allowed by the resource constraint tends to zero. This contradiction rules out such late crossings. Therefore k is eventually bounded away from zero, and so is $c \geq dk$. In particular, r is now bounded above.

Step 5. Research per unit of effective labor vanishes and Ramsey dynamics determine the limit. The discounted-spending bound from Step 3 does not by itself imply $m_t \rightarrow 0$: m could remain large over increasingly short intervals. The following bound on the growth rate of aggregate research expenditure M rules out that possibility. Differentiating the research condition and using the costate equation gives

$$(1-\eta)g_M = r - \frac{\eta \chi U B^{\eta-1}}{M^{1-\eta}} \leq r. \quad (112)$$

Choose a positive R above all eventual values of r and put $H = R/(1 - \eta)$. Then $M_s \geq M_t e^{-H(t-s)}$ on $[t - 1, t]$ at sufficiently late dates. Consequently

$$\frac{1}{A_{t-1}L_{t-1}} \int_{t-1}^t D_{t-1,s} M_s \, ds \geq e^{n+\gamma-R-H} m_t.$$

The left side tends to zero by (111). Hence $m \rightarrow 0$, and Step 1 already gives $u \rightarrow 0$.

I now show that capital and consumption per unit of effective labor converge to their stationary values. The bounds established above rule out divergence and convergence to zero, but do not by themselves rule out persistent fluctuations. Since $u, m \rightarrow 0$, the normalized equilibrium equations approach, uniformly on the compact positive region containing the path, the Ramsey system

$$\dot{k} = F(k) - c, \quad \dot{c} = c[\alpha f_\infty k^{\alpha-1} - \delta - \rho - \gamma]. \quad (113)$$

This limiting system has one positive stationary point,

$$k_\infty = \left[\frac{\alpha f_\infty}{\rho + \gamma + \delta} \right]^{1/(1-\alpha)}, \quad c_\infty = F(k_\infty) > 0.$$

The Jacobian at (k_∞, c_∞) has negative determinant, so the stationary point is a saddle. The standard Ramsey phase portrait is discussed by [Groth \(2013\)](#).

To connect this stationary point to the original equilibrium path, I examine that path over intervals centered at increasingly late dates, resetting each center to time zero. Bounded derivatives and uniform convergence of the equations imply that, from any sequence of such dates, a subsequence of the shifted paths converges on every bounded time interval to a solution of (113). This follows by taking limits in the integral form of the equations. Each limiting trajectory remains in the same compact positive region and is defined for all positive and negative times relative to its center; such a trajectory is called a complete orbit. I next show that it must be stationary.

The following calculation excludes cycles and nonconstant paths that approach the same stationary point in both time directions. Multiplying both equations of the limiting system by $1/c^2$ gives the divergence

$$\frac{\partial}{\partial k} \frac{F(k) - c}{c^2} + \frac{\partial}{\partial c} \frac{\alpha f_\infty k^{\alpha-1} - \delta - \rho - \gamma}{c} = \frac{\rho - n}{c^2} > 0.$$

Its strictly positive sign rules out both types of paths by the Bendixson–Dulac argument; the second type is a homoclinic loop. With these paths excluded and only one stationary point, the planar limit-set theorem implies that any nonconstant complete orbit confined to the compact positive region would have to approach (k_∞, c_∞) in both time directions. That would be precisely the homoclinic loop already ruled out. Hence every limiting trajectory constructed above is the stationary path. Every accumulation point of the original equilibrium path is therefore (k_∞, c_∞) , which proves $k_t \rightarrow k_\infty$ and $c_t \rightarrow c_\infty$.

It follows that $y \rightarrow y_\infty = f_\infty k_\infty^\alpha > 0$ and

$$r \rightarrow \rho + \gamma, \quad \frac{K}{Y} \rightarrow \frac{\alpha}{\rho + \gamma + \delta}, \quad \frac{C}{Y} \rightarrow 1 - \frac{\alpha(\delta + n + \gamma)}{\rho + \gamma + \delta} > 0. \quad (114)$$

The limit of C/Y is positive because the numerator of its expression is $\rho - n + (1 - \alpha)(\delta + n + \gamma) > 0$. This proves (54). Since $u, m \rightarrow 0$ and $y \rightarrow y_\infty > 0$, it follows that $U/Y = u/y \rightarrow 0$ and $M/Y = m/y \rightarrow 0$, proving (55). The resource constraint and Euler equation then give $\dot{k} \rightarrow 0$ and $g_C \rightarrow n + \gamma$.

Step 6. Output and wage growth converge as well. A convergent level need not have a convergent derivative. The bound on g_M from Step 5 and the RSI technology give

$$B_t^{1-\eta} \geq (1 - \eta)\chi M_t^\eta \int_0^1 e^{-\eta H v} dv.$$

Thus $g_B = \chi M^\eta / B^{1-\eta}$ is bounded above. At x_∞ , the marginal-revenue function v defined in Step 1 satisfies $v(x_\infty) = 0$ and $v'(x_\infty) < 0$. Differentiating $k^\alpha v(x) = 1/B$ yields

$$\dot{x} = -\frac{v(x)}{v'(x)} \left(g_B + \alpha \frac{\dot{k}}{k} \right) \rightarrow 0.$$

Hence $\dot{y} \rightarrow 0$ and $\dot{s}_X \rightarrow 0$. Since $y_\infty > 0$ and $1 - s_\infty > 0$, $g_{Y/N} \rightarrow \gamma$ and, by (12), $g_w \rightarrow \gamma$. The limiting income shares follow by substituting s_∞ into (12) and (13). The result is conditional on equilibrium existence and $B \rightarrow \infty$, not an equilibrium-existence theorem or an assumption of balanced growth. $\square$

Proof of Proposition 7. I first construct and verify the reference BGP, denoted by a superscript $*$, and then establish convergence from nearby initial stocks.

Step 1. Construction of the reference BGP. At unit elasticity, $s_X = \omega_X$ and $e_X = 1 - (1 - \alpha)\omega_X$. The AI-demand and monopoly conditions, equations (13) and (19), give

$$\frac{p_X X}{Y} = (1 - \alpha)\omega_X, \quad \frac{U}{Y} = (1 - \alpha)^2 \omega_X^2. \quad (115)$$

Substituting $X = BU = (1 - \alpha)^2 \omega_X^2 BY$ into production yields

$$Y^{1-(1-\alpha)\omega_X} = K^\alpha (AL)^{(1-\alpha)\omega_L} [(1 - \alpha)^2 \omega_X^2 B]^{(1-\alpha)\omega_X}. \quad (116)$$

On an exact BGP with positive constant K/Y , C/Y , and M/Y , log differentiation of this equation and the RSI technology (50) gives the growth restrictions (60) and (61). Define

$$\Delta \equiv (1 - \alpha)(1 - \eta - \omega_X) > 0. \quad (117)$$

The unique solution to the two growth restrictions has

$$g_Y^* = \frac{(1-\eta)(1-\omega_X)}{1-\eta-\omega_X}(n+\gamma), \quad (118)$$

and g_B^* as in (66). The equality $g_K = g_C = g_Y^*$ follows from constant capital and consumption ratios; equations (12) and (5) then give (63) and (64). Since $n+\gamma > 0$, the two growth restrictions are inconsistent if $\Delta = 0$ and imply $g_B < 0$ if $\Delta < 0$, contrary to the RSI technology. Thus $\eta + \omega_X \geq 1$ excludes this positive finite-rate BGP, not every possible equilibrium trajectory.

Define the candidate ratios $k^* = K/Y$, $u^* = U/Y$, $i^* = (\dot{K} + \delta K)/Y$, $m^* = M/Y$, and $c^* = C/Y$ by

$$k^* = \frac{\alpha}{r^* + \delta}, \quad u^* = (1-\alpha)^2 \omega_X^2, \quad (119)$$

$$i^* = (g_Y^* + \delta)k^*, \quad (120)$$

$$m^* = \frac{u^* \eta g_B^*}{\rho - n + \eta g_Y^*}, \quad (121)$$

$$c^* = 1 - u^* - i^* - m^*. \quad (122)$$

Here k^* is a capital-output ratio, not capital per unit of effective labor, and i^* is gross physical investment as a share of output. The research share comes from the developer's first-order and costate conditions; the consumption share comes from market clearing. These ratios are endogenous, not fixed saving or research rules.

All candidate ratios are positive. The denominator of the research share m^* , $\rho - n + \eta g_Y^*$, is positive. Moreover,

$$i^* = \frac{\alpha(g_Y^* + \delta)}{\rho - n + g_Y^* + \delta} < \alpha, \quad m^* < \frac{u^* \eta}{1 - \eta}.$$

Since $\Delta = (1-\alpha)(1-\eta-\omega_X) > 0$, $\omega_X < 1-\eta$ and $\omega_X < 1$. Consequently,

$$u^* + m^* < \frac{u^*}{1-\eta} < (1-\alpha)\omega_X < 1-\alpha, \quad c^* = 1 - u^* - i^* - m^* > 0.$$

Thus positive consumption follows from the stated conditions; it is not an additional assumption.

For given $A_0, N_0 > 0$, choose the reference initial stocks as

$$B_0^* = \left[A_0 N_0 (k^*)^{\alpha/[(1-\alpha)\omega_L]} (u^*)^{\omega_X/\omega_L} m^* \left(\frac{\chi}{g_B^*} \right)^{1/\eta} \right]^{\eta\omega_L/(1-\eta-\omega_X)}, \quad (123)$$

$$Y_0^* = A_0 N_0 (k^*)^{\alpha/[(1-\alpha)\omega_L]} (u^* B_0^*)^{\omega_X/\omega_L}, \quad K_0^* = k^* Y_0^*. \quad (124)$$

The outer exponent in the expression for B_0^* is positive. These choices solve simultaneously the initial production equation and $g_B^* B_0^* = \chi (B_0^* m^* Y_0^*)^\eta$. Set $U_0^* = u^* Y_0^*$, $M_0^* = m^* Y_0^*$, $C_0^* = c^* Y_0^*$,

$X_0^* = B_0^* U_0^*$, and

$$q_0^* = \frac{M_0^*}{\eta g_B^* B_0^*}. \quad (125)$$

Let $Y_t^* = Y_0^* e^{g_Y^* t}$, $B_t^* = B_0^* e^{g_B^* t}$, $K_t^* = k^* Y_t^*$, $C_t^* = c^* Y_t^*$, $U_t^* = u^* Y_t^*$, $M_t^* = m^* Y_t^*$, and $q_t^* = q_0^* e^{(g_Y^* - g_B^*)t}$. Use the exogenous A_t, N_t and set $L_t = N_t$ and $X_t^* = B_t^* U_t^*$. The static equilibrium equations determine all remaining quantities and prices.

The two growth identities (60) and (61) ensure that the production equation and the RSI technology hold at every date once they hold initially. Equation (125) gives $M = \eta q \dot{B}$, which is the research first-order condition (28k) with $\psi = 1$. The definitions of k^* , r^* , and c^* verify the capital-return, Euler, and resource equations. The costate equation reduces to

$$g_Y^* - g_B^* = r^* - \frac{u^* \eta g_B^*}{m^*} - \eta g_B^*.$$

This holds by equation (121) and $(1 - \eta)g_B^* = \eta g_Y^*$. All allocations are positive and finite at every finite date.

Both transversality conditions are explicit:

$$e^{-\rho t} \frac{N_t K_t^*}{C_t^*} = \frac{N_0 k^*}{c^*} e^{-(\rho-n)t} \longrightarrow 0, \quad (126)$$

$$e^{-r^* t} q_t^* B_t^* = q_0^* B_0^* e^{-(\rho-n)t} \longrightarrow 0. \quad (127)$$

Discounted developer profit is a constant multiple of $e^{-(\rho-n)t}$. Discounted household utility is $N_0 e^{-(\rho-n)t} [\log(C_0^*/N_0) + (g_Y^* - n)t]$. Both improper objective integrals are finite.

Step 2. Global optimality.

The constructed path satisfies the necessary conditions, but these do not yet rule out profitable deviations. For the developer, holding the paths of K, A, L fixed, static optimization gives

$$\begin{aligned} \pi_t(B) &= (1 - \alpha) \omega_X [1 - (1 - \alpha) \omega_X] [K_t^\alpha (A_t L_t)^{(1-\alpha)\omega_L}]^{1/[1-(1-\alpha)\omega_X]} \\ &\quad \times [(1 - \alpha)^2 \omega_X^2 B]^{(1-\alpha)\omega_X/[1-(1-\alpha)\omega_X]}. \end{aligned} \quad (128)$$

To verify developer optimality by concavity, define the invertible state transformation and its current-value costate by

$$\nu = \frac{1 - \eta}{\eta}, \quad S = B^\nu, \quad p = \frac{q}{\nu B^{\nu-1}}. \quad (129)$$

The exponent ν is chosen so that the transformed research technology has constant returns in S and M :

$$\dot{S} = \nu \chi S^{1-\eta} M^\eta,$$

which is jointly concave in S, M for every $0 < \eta < 1$. At given K, A, L , operating profit in

equation (128) is a positive time-dependent coefficient times

$$S^{\frac{(1-\alpha)\omega_X}{\nu[1-(1-\alpha)\omega_X]}}.$$

Since $\eta + \omega_X < 1$ implies $(1 - \alpha)\omega_X < 1 - \eta$, this exponent lies strictly between zero and one. Operating profit is therefore concave in S . With $p > 0$, the transformed Hamiltonian $\pi_t(S^{1/\nu}) - M + p_t\nu\chi S^{1-\eta}M^\eta$ is jointly concave in S, M . The research condition (22) and costate equation (23), with $\psi = 1$, give the transformed problem's control first-order condition and costate equation by the chain rule.

Write $D_t = \exp(-\int_0^t r_s ds)$ and $J_T = \int_0^T D_t[\pi_t(B_t) - M_t] dt$. For any admissible alternative with the same initial B , concavity, the research condition, and the costate equation give, after integration by parts,

$$\tilde{J}_T - J_T \leq -D_T p_T (\tilde{S}_T - S_T) \leq D_T p_T S_T = \frac{D_T q_T B_T}{\nu} \rightarrow 0. \quad (130)$$

The last limit follows from the developer TVC. Since the transformation preserves feasible research paths and their payoffs, no admissible alternative has a greater limiting payoff, or a greater upper limit of truncated payoff, than this finite-valued candidate. The same verification applies to nearby candidates satisfying the uncapped equations and the developer TVC; it does not require exact balanced growth.

Household optimality follows from log-utility concavity, the linear budget, the Euler equation, and the TVC, by the same integration-by-parts argument as in Step 4 of the proof of Proposition 1.

Competitive final-good firms face a concave constant-returns technology, and Lemma 1 establishes global static monopoly optimality. The constructed path satisfies Definition 1 with the uncapped RSI technology (50).

Step 3. Local stability and selection of initial consumption and research. The initial stocks are predetermined. I must select initial consumption and the shadow price so that the resulting path converges to the BGP; the research condition then determines initial research expenditure. For the given exogenous paths of population and labor-augmenting technology, compare each variable with its value on the reference BGP at the same date. Define log deviations

$$\xi_K = \log \frac{K}{K^*}, \quad \xi_B = \log \frac{B}{B^*}, \quad \xi_C = \log \frac{C}{C^*}, \quad \xi_q = \log \frac{q}{q^*}. \quad (131)$$

Time subscripts are suppressed: the reference quantities in the denominators follow the BGP rather than remain fixed at their initial values. Substitution into equations (29a)–(29d), with $\psi = 1$ and $\psi' = 0$, gives a smooth autonomous system $\dot{\xi} = f(\xi)$ with $f(0) = 0$, where $\xi = (\xi_K, \xi_B, \xi_C, \xi_q)'$. Let $J^* = Df(0)$ denote its Jacobian. No constant growth rates or allocation shares are imposed along these transitions.

I next establish two properties needed for this selection. First, the linearized system has exactly two independent directions of convergence and no eigenvalues with zero real part. Writing E^s for

the real invariant subspace associated with the convergent directions, this means

$$\text{spec}(J^*) \cap i\mathbb{R} = \emptyset, \quad \dim E^s = 2. \quad (132)$$

Second, let $V_S \in \mathbb{R}^{4 \times 2}$ be any real basis matrix for E^s , and let V_{KB} consist of its first two rows. I show that

$$\det V_{KB} \neq 0. \quad (133)$$

This nonsingularity allows the stable manifold to select C_0, q_0 for every sufficiently close pair K_0, B_0 , rather than only for stocks on a curve. Both properties follow from the parameter restrictions, rather than being additional assumptions. I establish them by locating the stable eigenvalues from the signs of the characteristic polynomial, then checking that their eigenvectors allow the two stocks to vary independently. Retain ν from (129) and, for this linearization, define

$$\begin{aligned} a_K &= \frac{\alpha}{1 - (1 - \alpha)\omega_X}, & a_T &= \frac{(1 - \alpha)\omega_X\eta}{[1 - (1 - \alpha)\omega_X](1 - \eta)}, \\ d &= \rho - n, & P &= r^* + \delta, & h &= \rho - n + \eta g_Y^*. \end{aligned}$$

The quantities ν, a_K, a_T, d, P, h are positive, and the restriction $\eta + \omega_X < 1$ gives

$$1 - a_K - a_T = \frac{(1 - \alpha)(1 - \eta - \omega_X)}{[1 - (1 - \alpha)\omega_X](1 - \eta)} > 0. \quad (134)$$

For this calculation, return to the transformed state S and its shadow price p from Step 2. This gives the invertible linear change of deviations

$$\hat{\xi} = (\xi_K, \nu\xi_B, \xi_C, \xi_q + (1 - \nu)\xi_B)'$$

The second and fourth coordinates are the log deviations of the transformed state S and its shadow price p . This changes neither the model nor the eigenvalues. In these coordinates, the reduced production equation and the research condition give

$$\log \frac{Y}{Y^*} = a_K \hat{\xi}_K + a_T \hat{\xi}_S, \quad \log \frac{M}{M^*} = \hat{\xi}_S + \frac{\hat{\xi}_p}{1 - \eta}.$$

Differentiating the capital, transformed research, Euler, and transformed costate equations at the BGP, using $U/(qB) = \rho - n + \eta g_Y^*$, gives a matrix similar to J^* :

$$\hat{J} = \begin{pmatrix} A & B & -C & -V \\ 0 & 0 & 0 & D \\ -G & H & 0 & 0 \\ -E & F & 0 & d \end{pmatrix}, \quad (135)$$

where the following letters abbreviate matrix coefficients only:

$$\begin{aligned} \mathbf{A} &= d + (1 - \alpha)\omega_X P, & \mathbf{B} &= \frac{(1 - u^*)a_T - m^*}{k^*}, & \mathbf{C} &= \frac{c^*}{k^*}, \\ \mathbf{V} &= \frac{m^*}{k^*(1 - \eta)}, & \mathbf{D} &= \frac{\eta g_Y^*}{1 - \eta}, & \mathbf{G} &= P(1 - a_K), \\ \mathbf{H} &= Pa_T, & \mathbf{E} &= \mathbf{G} + ha_K, & \mathbf{F} &= \mathbf{H} + h(1 - a_T). \end{aligned}$$

All coefficients other than $\mathbf{B}$ are positive by their definitions. The BGP allocation ratios imply the identities

$$\mathbf{L} \equiv \mathbf{B} - \frac{\mathbf{V}d}{\mathbf{D}} = \frac{(1 - \alpha)\omega_X \eta}{(1 - \eta)k^*} > 0, \quad (136)$$

$$\mathbf{G}\mathbf{F} - \mathbf{H}\mathbf{E} = hP(1 - a_K - a_T) > 0. \quad (137)$$

Set $\mathbf{N} = \mathbf{C}\mathbf{G} + \mathbf{V}\mathbf{E} > 0$ and $\mathbf{Q} = \mathbf{C}\mathbf{H} + \mathbf{V}\mathbf{F} > 0$. Expanding the determinant gives the characteristic polynomial

$$\begin{aligned} p(\ell) &= \det(\ell I - \hat{J}) \\ &= (\ell^2 - \mathbf{A}\ell - \mathbf{N})(\ell^2 - d\ell - \mathbf{D}\mathbf{F}) - \mathbf{D}\mathbf{E}(\mathbf{Q} - \mathbf{L}\ell). \end{aligned} \quad (138)$$

To locate the negative eigenvalues without solving the quartic, first evaluate the polynomial at zero:

$$p(0) = \mathbf{D}\mathbf{C}hP(1 - a_K - a_T) > 0.$$

Let $\mu = (d - \sqrt{d^2 + 4\mathbf{D}\mathbf{F}})/2 < 0$, the negative root of $\ell^2 - d\ell - \mathbf{D}\mathbf{F}$. At $\ell = \mu$, the product of the two quadratic factors in $p(\ell)$ vanishes, leaving $p(\mu) = -\mathbf{D}\mathbf{E}(\mathbf{Q} - \mathbf{L}\mu) < 0$. Since $p(\ell) \rightarrow +\infty$ as $\ell \rightarrow -\infty$, there is one negative root below μ and another between μ and zero. These are the only negative roots: the cubic coefficient of p is $-(\mathbf{A} + d) < 0$, its linear coefficient is $\mathbf{A}\mathbf{D}\mathbf{F} + d\mathbf{N} + \mathbf{L}\mathbf{D}\mathbf{E} > 0$, and its constant coefficient is positive. Thus the coefficient signs of $p(-x)$, in descending powers, are $(+, +, \text{arbitrary}, -, +)$. Descartes' rule of signs gives exactly two sign changes, regardless of the quadratic coefficient. The two negative roots are therefore real, distinct, and simple. Write them as $\ell_1 < \mu < \ell_2 < 0$. The remaining two roots have positive product and positive sum, because $p(0) > 0$ and the sum of all four roots is $\mathbf{A} + d > 0$. If real, both are positive; if complex, their real parts are positive. This proves (132).

The eigenvalue count does not by itself ensure that a convergent path can be selected for every nearby pair of initial stocks. I establish this by showing that the stock components of the stable eigenvectors are linearly independent. Write a stable eigenvector in transformed coordinates as $(x, y, z, u)'$. The component y cannot be zero: the second, fourth, and first rows of the eigenvalue equation for $\hat{J}$ would successively imply $u = x = z = 0$. Normalize $y = 1$. The second and fourth rows then give

$$x(\ell) = \frac{\mathbf{D}\mathbf{F} + d\ell - \ell^2}{\mathbf{D}\mathbf{E}}.$$

The ordering of the roots implies $x(\ell_1) < 0 < x(\ell_2)$. Hence the two projected stable eigenvectors $(x(\ell_1), 1)'$ and $(x(\ell_2), 1)'$ are linearly independent. Returning to (ξ_K, ξ_B) only rescales their second component by $1/\nu > 0$, so (133) follows as well.

These matrix properties now provide a selection rule for the nonlinear system. The exact normalized vector field is continuously differentiable near $\xi = 0$. The spectral property just proved and the stable-manifold theorem supply a two-dimensional local manifold whose trajectories remain near zero and converge exponentially. The derivative of its projection onto (ξ_K, ξ_B) is V_{KB} . The nonsingularity just proved and the inverse-function theorem therefore make this manifold locally the graph of a continuously differentiable function selecting (ξ_C, ξ_q) from the two initial stock deviations. Returning from log deviations to levels is a smooth invertible change of variables, so the same holds for (C_0, q_0) as functions of (K_0, B_0) .

Step 4. Reconstruction of nearby equilibria.

I first choose a neighborhood of initial states so that the selected trajectories stay within the region where the local construction is valid. For some constants $a, H > 0$, trajectories on a sufficiently small stable manifold satisfy

$$\|\xi(t)\| \leq H e^{-at} \|\xi(0)\|, \quad t \geq 0.$$

For any sufficiently small open neighborhood $\mathcal{V}$ of zero, choose a ball contained in $\mathcal{V}$. Continuity of the graph selecting the two jump variables permits an open neighborhood $\mathcal{U}$ of the initial stock deviations for which $H\|\xi(0)\|$ is smaller than that ball's radius. Every initial stock pair in $\mathcal{U}$ therefore has a trajectory staying in $\mathcal{V}$ for all $t \geq 0$ and converging to zero. Exponentiation maps $\mathcal{U}$ to an open neighborhood of (K_0^*, B_0^*) . Shrinking these neighborhoods preserves positivity, feasibility, and existence for every $t \geq 0$. Local uniqueness follows because every trajectory remaining in this local region and converging to zero belongs to the same stable manifold, which is a graph over stocks. This does not assert convergence from arbitrary initial C_0, q_0 or exclude equilibrium trajectories that leave the region.

The statement also permits nearby A_0, N_0 . Equations (123) and (124) make the reference B_0^*, K_0^* smooth functions of these two positive initial values. After normalization by the corresponding BGP, the exact vector field depends only on the parameters and BGP ratios, not on A_0, N_0 . The set of (A_0, N_0, K_0, B_0) whose stock deviations belong to $\mathcal{U}$ is therefore open and contains the reference initial condition. From each such initial condition, the construction converges to the BGP associated with its own A_0, N_0 , not necessarily to the same levels as the original reference path.

Since $f(\xi(t)) \rightarrow 0$, the growth rates of K, B, C, q converge to their BGP values, not only normalized levels. The static equations give the same convergence for the interest rate, wage growth, and allocation ratios. The unit-elastic labor income share remains $(1 - \alpha)\omega_L$ at every date.

Exponential convergence implies that $r_t - r^*$ is integrable. Thus D_t/e^{-r^*t} tends to a finite positive constant, and both TVC expressions equal their BGP counterparts in (126) and (127) times factors tending to finite positive limits. Both TVCs hold. Output and all expenditure flows are

bounded multiples of $e^{g_Y^* t}$, while log consumption per person is a linear function of time plus a bounded deviation. Both objective integrals therefore remain finite. The global concavity verification established above applies at these new price and allocation paths, proving agent optimality. Static optimality and market clearing then establish equilibrium, not merely a solution of the necessary conditions. $\square$

Proof of Proposition 8. I construct a sequence of research trajectories whose discounted net profit grows without bound, although each trajectory has finite expenditure and AI efficiency. Write $\theta = (1 - \alpha)/\alpha$. For a research trajectory $\{M_t\}_{t \in [0, T]}$, discounted net profit on the fixed horizon is

$$J_T(M) = \int_0^T \exp \left\{ - \int_0^t r_s ds \right\} [\pi(B_t) - M_t] dt.$$

Integrability of r bounds the discount factor above and away from zero on $[0, T]$. At the optimal service choice, equations (13) and (19) imply

$$\pi(B) = p_X X - \frac{X}{B} = e_X p_X X = (1 - \alpha) e_X s_X Y.$$

For $\sigma > 1$, marginal revenue is positive at every finite X and approaches zero as $X \rightarrow \infty$, as established in Lemma 1. The monopoly condition therefore implies $X \rightarrow \infty$ as $B \rightarrow \infty$, holding K, A, L fixed. Consequently, $s_X \rightarrow 1$ and $e_X \rightarrow \alpha$. Substituting these limits into equation (68) gives

$$\pi(B) \sim \kappa_X K B^\theta, \quad \kappa_X = \alpha(1 - \alpha) \left[(1 - \alpha)^2 \omega_X^{\sigma/(\sigma-1)} \right]^\theta > 0.$$

The operating-profit asymptotic is uniform over dates in $[0, T]$ because K, A, L are continuous, bounded, and bounded away from zero on $[0, T]$. Thus a common lower bound proportional to B^θ applies at every date once B is sufficiently large.

The research trajectories constructed below are deviations in the developer's optimization problem, not alternative equilibrium allocations. Fix $\Delta \in (0, T)$, choose $M = \mathcal{M}$ on $[0, \Delta]$, and choose $M = 0$ afterward. The uncapped RSI technology gives

$$B_{\mathcal{M}}(t)^{1-\eta} = B_0^{1-\eta} + (1 - \eta) \chi \mathcal{M}^\eta \min\{t, \Delta\}.$$

As $\mathcal{M} \rightarrow \infty$, efficiency after the research burst is asymptotically proportional to $\mathcal{M}^{\eta/(1-\eta)}$. Discounted operating profit over the remaining interval therefore grows in proportion to $\mathcal{M}^{\eta\theta/(1-\eta)}$, while research expenditure grows only linearly in $\mathcal{M}$. Since optimized operating profit is nonnegative, a lower bound on discounted net profit counts only operating profits earned after research stops, while subtracting all research expenditure. For all sufficiently large $\mathcal{M}$,

$$J_T(\mathcal{M}) \geq c_\pi \mathcal{M}^{\eta(1-\alpha)/[\alpha(1-\eta)]} - c_M \mathcal{M}, \tag{139}$$

where $c_\pi, c_M > 0$ are independent of $\mathcal{M}$. The exponent exceeds one because $\eta > \alpha$, so $J_T(\mathcal{M}) \rightarrow$

$+\infty$ as $\mathcal{M} \rightarrow \infty$. Each finite $\mathcal{M}$ keeps AI efficiency and discounted net profit finite on $[0, T]$; it is the sequence of payoffs that is unbounded.

Now consider an infinite-horizon equilibrium candidate $\{\widehat{B}_t, \widehat{M}_t\}$ with finite discounted net profit $\widehat{V}_0$, and let $\widehat{J}_T$ denote its payoff through T . After T , continue the deviation using $\widehat{M}_t$. For sufficiently large $\mathcal{M}$, the deviation has $B_{\mathcal{M}}(T) > \widehat{B}_T$. Because $d(B^{1-\eta})/dt = (1-\eta)\chi M^\eta$, this efficiency advantage persists under the common subsequent expenditure. Operating profit is increasing in efficiency by equation (20), so the continuation payoff is at least $\widehat{V}_0 - \widehat{J}_T$. The full deviation therefore yields

$$J_\infty(\mathcal{M}) \geq J_T(\mathcal{M}) + \widehat{V}_0 - \widehat{J}_T \rightarrow +\infty.$$

No candidate with finite discounted net profit can be optimal. Moreover, every equilibrium must have finite discounted developer profit. Indeed, integrating the household budget (4) and using its Euler equation (5) and transversality condition (6) gives

$$\int_0^\infty e^{-\int_0^t r_s ds} (w_t N_t + \Pi_t) dt = \frac{C_0}{\rho - n} - K_0.$$

Since labor income is nonnegative, discounted developer profit is either finite or minus infinity. A payoff of minus infinity cannot be optimal because stopping research yields nonnegative operating profit. Hence no equilibrium exists under the stated restrictions. This argument does not establish a finite-time singularity. $\square$

B Numerical algorithm and equilibrium audit

B.1 Numerical solution

For the RSI-activation comparisons in Sections 7.2 and 7.3, I compute numerical approximations to equilibrium trajectories by solving a boundary-value problem (BVP). Initial capital K_0 and AI efficiency B_0 are given, whereas initial consumption C_0 and the developer's shadow price q_0 are determined jointly with the transition. I solve the equilibrium differential equations (29) over a finite horizon, using the given stocks as initial conditions and the long-run characterization in Proposition 1 to construct terminal conditions.

For each σ , I first solve the existing-AI economy with efficiency fixed at B_0 and research unavailable. A scalar root search in $\log K_0$ solves Equation (69) using the production and monopoly equations. The resource constraint (26), with $M = 0$ and $\dot{K} = (n + \gamma)K$, gives pre-event consumption. The post-activation problem inherits the stocks but selects consumption, research, and the shadow price anew.

To represent positive variables and keep AI efficiency below its upper bound, I use the computational coordinates

$$\left(\log K, \log \frac{B}{\overline{B} - B}, \log C, \log q \right).$$

The second coordinate expresses AI efficiency relative to the remaining gap below its upper bound;

it can grow without bound while AI efficiency remains bounded. I subtract the long-run growth rate of aggregate capital times t from each coordinate: $n + \gamma$ in the labor-bottleneck regime and $n + \mathcal{R}(\bar{B}) - \rho$ in the AI-dominated regime. The efficiency coordinate has the same asymptotic slope because the remaining gap $\bar{B} - B$ shrinks at that proportional rate. Removing these trends makes the numerical coordinates converge to finite limits without changing the economic variables or calendar time.

The terminal conditions approximate convergence to the long-run equilibrium. In the normalized coordinates of Appendix A.2, I require the terminal deviation from the limiting point to lie in the stable subspace of the linearized dynamics. This gives two restrictions excluding locally divergent directions. Together with the initial stocks K_0, B_0 , they select initial consumption and the developer's shadow price. The conditions do not require the economy to reach its long-run limit at a finite date.

I solve this BVP with `scipy.integrate.solve_bvp` from the open-source Python library SciPy (Virtanen et al., 2020). The solver implements fourth-order collocation with residual control similar to that of Kierzenka and Shampine (2001). At each trial state, I solve the static monopoly problem and recover research expenditure from its first-order condition.

For each of the eight parameter configurations, I start near its analytical limiting point and move the initial stocks to their prescribed values through a sequence of auxiliary boundary-value problems, using each solution as the initial guess for the next. This procedure changes the initial conditions of the numerical problem, not the dates on the economic trajectory.

For the competitive-to-monopoly comparison in Section 7.4, I instead obtain K_0 from $r = \rho + \gamma$ with marginal-cost pricing $p_X = 1/B_0$; at $\sigma = 1.5$, $K_0 = 6.535$. Pre-event consumption follows from the same resource constraint. I then solve the monopoly BVP at these initial stocks, using the RSI-activation trajectories only as numerical initial guesses, not as restrictions on consumption or research. This procedure also applies to the four-elasticity comparisons in the [digital edition](#).

B.2 Accuracy and equilibrium checks

To check whether the computed transition depends on the terminal date or solver accuracy, I extend each computational horizon twice by 500 years and tighten the collocation tolerance from 2×10^{-6} to 10^{-8} and 10^{-9} . The final boundary tolerance is 10^{-11} . All plotted values lie within the solved interval; none is extrapolated beyond it.

I check the four dynamic equations independently of the solver's residual report. Five-point finite differences approximate the time derivatives of the four logarithmic coordinates above; I compare these derivatives with the equations' right-hand sides expressed in the same coordinates. The dynamic residual is the largest absolute discrepancy across coordinates and checked dates, measured per year, with denser checks over the first ten years. For the monopoly and research first-order conditions, I report the absolute logarithmic deviations of $Bp_X(1 - e_X)$ and $q\chi\eta B^\eta M^{\eta-1}\psi(B)$ from one. Acceptance requires dynamic residuals below 10^{-6} and both first-order residuals below 10^{-9} .

I also require the last horizon extension to change the four detrended coordinates by less than 2×10^{-5} over years 0–500, with the same tolerance for changes in $\log C_0$ and $\log q_0$. At the terminal date, the largest coordinate deviation from the analytical limiting point must be below 10^{-4} . This last check uses the regime-specific normalized coordinates of Appendix A.2, with logarithms for coordinates having positive limits and with $\tau = (AN)^{-1}$ scaled by $\bar{d}/\bar{B}$. Here $\bar{d}$ is the limiting value of the normalized efficiency gap d in the corresponding regime. Unlike the horizon comparison, this check measures proximity to the limiting equilibrium, not agreement between two numerical solutions. I also check the pre-event equilibrium conditions and the continuity of production quantities and prices at RSI activation.

Small equation residuals do not establish that the developer prefers the computed research path to every feasible alternative. I therefore check feasibility and the sufficient conditions for global optimality. When the concavity conditions of Lemma 3 fail, I use the support condition in Lemma 2: the tangent at the candidate state must lie above the maximized Hamiltonian at every alternative AI-efficiency level. At each checked date, I minimize the difference between that tangent and the Hamiltonian over alternative efficiency levels, holding the transformed shadow price, capital, and effective labor fixed. An analytical bound covers the alternative states beyond the numerical search range as efficiency approaches $\bar{B}$.

I assess the trajectory beyond the computational horizon separately. For the AI-dominated paths, I check the support bound derived below at the terminal date, as well as proximity to the limiting equilibrium. Along an AI-dominated path converging to that equilibrium, the bound holds sufficiently late. Both transversality expressions also decay asymptotically at rate $n - \rho < 0$. These checks support the numerical approximation over an infinite horizon; they do not constitute a proof of equilibrium existence from arbitrary initial stocks. The competitive-to-monopoly paths satisfy the same residual, horizon-refinement, transversality, and developer-optimality criteria.

Table 3: Numerical accuracy of the illustrative RSI comparisons

χ	σ	Final horizon	Dynamic residual	Horizon change
7.5	0.90	5,428.0	1.18×10^{-9}	3.06×10^{-8}
7.5	1.00	4,944.5	1.38×10^{-9}	3.46×10^{-8}
7.5	1.10	4,828.8	1.66×10^{-9}	3.73×10^{-8}
7.5	1.50	3,013.0	1.69×10^{-9}	1.56×10^{-8}
1.5	0.90	6,047.0	1.02×10^{-9}	4.30×10^{-8}
1.5	1.00	5,563.5	1.11×10^{-9}	3.80×10^{-8}
1.5	1.10	5,447.8	1.19×10^{-9}	4.08×10^{-8}
1.5	1.50	3,632.0	1.23×10^{-9}	1.80×10^{-8}

Horizons are measured in years. Dynamic residuals are the largest absolute errors in the derivatives of the four logarithmic coordinates, across the full-horizon and dense first-decade checks. Horizon change compares the last two solutions in detrended logarithmic coordinates over years 0–500. Reported error maxima are rounded upward. All eight paths pass the equation, event-continuity, developer-optimality and long-run-continuation checks. The $\sigma = 1.50$ cases use the global Hamiltonian-support test and its analytical long-run bound; the other cases pass the concavity test.

B.3 Analytical checks on developer optimality

Two analytical results supplement the numerical optimality checks: global concavity at unit elasticity and a long-run support bound for the AI-dominated paths, which have $\sigma = 1.50$ in these simulations.

At unit elasticity, define $\beta = (1 - \alpha)\omega_X$. The static monopoly solution implies $\pi_t(B) = a_t B^{\beta/(1-\beta)}$ for some $a_t > 0$, conditional on the dated values of capital and effective labor. Here R_B is the transformed AI-efficiency state in equation (73). Because $B(R_B)$ is increasing and concave in R_B , $R_B \mapsto \pi_t(B(R_B))$ is globally concave whenever $\beta \leq 1/2$. The illustrative values $\beta = 0.067$ and $\eta = 0.20$ therefore satisfy both concavity conditions in Lemma 3.

For the AI-dominated paths, I next show why the support condition holds eventually even if global concavity fails at low alternative efficiency levels. For $1 < \sigma \leq 1/\alpha$, the elasticity of optimal AI services with respect to AI efficiency, defined in equation (79), satisfies $\varepsilon_{B,t}(b) \leq 1/\alpha$ for every t and b . Suppressing these arguments, subtracting $\alpha(1 - e_X)$ from the denominator in that equation gives

$$(\sigma^{-1} - \alpha)(1 - s_X) [(1 - \sigma^{-1})(1 - s_X) + (\sigma^{-1} - \alpha)s_X] \geq 0.$$

To separate the alternative efficiency levels covered by concavity from those requiring an additional bound, choose $B_* \in (B_0, \bar{B})$ with $B_*/\bar{B} > \max\{0, (1 - 2\alpha)/(1 - \alpha)\}$. Operating profit is then concave in the transformed state R_B for $B \geq B_*$ by equation (81). If $\eta \leq 1/2$, the research term in (75) is concave there as well. For a candidate with $B_t \geq B_*$, concavity gives the tangent bound at all alternative efficiencies above B_* . Below B_* , the maximized Hamiltonian increases with alternative efficiency b and is therefore bounded above by its value at B_* . The tangent has positive slope by (76), so its lowest value over feasible transformed efficiencies is at $R_{B,0}$. If that value is at least the Hamiltonian at B_* , the tangent also bounds every alternative below B_* . It therefore suffices to check

$$\widehat{\mathcal{H}}_t(R_{B,t}) - \widehat{\mathcal{H}}'_t(R_{B,t})(R_{B,t} - R_{B,0}) \geq \widehat{\mathcal{H}}_t(R_B(B_*)). \quad (140)$$

At the terminal date, I verify $B_t \geq B_*$ and (140) with $B_* = 0.75\bar{B}$.

Along an AI-dominated path converging to its long-run limit, the left-hand side minus the right-hand side of (140), divided by Y_t , converges to

$$\alpha(1 - \alpha) \left[1 - \left(\frac{B_*}{\bar{B}} \right)^{(1-\alpha)/\alpha} \right] > 0.$$

To obtain this limit, use $M/Y \rightarrow 0$, $\psi(B)R_B \rightarrow 0$, and $\pi(B_*)/Y \rightarrow \alpha(1 - \alpha)(B_*/\bar{B})^{(1-\alpha)/\alpha}$. Thus the global support condition holds sufficiently late along such a path even when the stronger curvature condition fails at low alternative AI-efficiency levels. The positive terminal margin and the small terminal-coordinate gap provide numerical evidence that the computed path has approached this region. The finite transition still requires its own support check.

Replication files. I use NumPy for array operations and linear algebra ([Harris et al., 2020](#)) and Matplotlib to generate the figures ([Hunter, 2007](#)). The code and simulation data are available in the [repository](#). The [replication guide](#) records package versions, implementation details, numerical tolerances, and commands for checking and reproducing the results, with links to the additional institutional experiments.

Declaration of AI use

During the preparation of this work, I used OpenAI Codex to assist with mathematical derivations, consistency checks, manuscript editing, and the preparation and testing of computational code. I take full responsibility for the content of the paper.

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